

GE Healthcare

Signa HD, HDx & HDxt 3.0T Block Diagrams & Supplemental Schematics (Release 12.x, 14.x & 15.x)

Technical Publication



5133320
Rev 11

DAMAGE IN TRANSPORTATION

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File a report with

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- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
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- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
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- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
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- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
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- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
1 . .	Oct 28, 2004	Initial Release
2 . . .	May 11, 2005	Overview Tab – Sec 3: Added new acronyms and descriptions. System Tab – Added XW8200 host computer, removed CAN Link for ACGD, added HFD–S CAN Link, and other minor corrections. Operator Workspace Tab – Added XW8200 host computer and other minor corrections. Patient Handling Tab – Made minor corrections. Gradient Tab – Removed ACGD cabinet pages and inserted HFD–S Cabinet page. MGD/RRF/RF Tab – Corrected minor errors and added informational callouts. Added MRC II Tool & Cable and MRC II Tool – 8 Channel Connector pages.
3 . . .	Oct 15, 2005	System Tab – Added J116 and J119 to TAC Cabinet and relocated Terminator. Patient Handling Tab – Added test points and LED's to Magnet Enclosure and PHPS. Added MNS connection information to SSM. MGD/RRF/RF Tab – Added LED info to Exciter and Receiver boards. Corrected RRF Receive Chain connection labels. Added MNS connection info to Driver Module and SSM. Added MC Preamp Bias Filter pinouts. Added LED and Pinout info for 16 Ca'hannel Switch.
4 . . .	May 10, 2006	Major update to manual with the addition of HDx (Release 14.x) product. Changed title page and banners on all pages. Also added 3T94 information throughout this manual. Overview Tab – Sec 3: Added new acronyms and descriptions. System Tab – Added HP9300 computer info, 32 Channel option info, and the new RFS cabinet power distribution using two power strips. Revised all sheet references. OW Tab – Added HP9300 computer info. MGD/RRF/RF Tab – Added HDx (14.x) info as needed which includes VRE modules: ICN's, 16/24 Port Ethernet Switch, Infiniband board and module. Added HDx A, B, and C port connection information for the HDx LPCA. Added MCR III Tool
5 . . .	Oct 20, 2006	Added Environmental Sensor to System/RFS cabinet in System. Operator Workspace, and MGD/RRF/RF Sections (MRIhc19337). MGD/RRF/RF Tab –Added Port 2 to Port 2 connection for 4 ICN configuration. Added 4 ICN configuration without Infiniband switch. Updated Multi–Coil Adapter by adding MC Bias 17–24 for 14.x systems. Updated several pages with additional MC signal information (MRIhc18215).
6 . . .	Dec 20, 2006	SYSTEM Tab – Added 32 Channel configuration without Infiniband Switch. OW Tab – Added Audio IN/OUT signals for PC Host for 14.x system. GRADIENT Tab – Added Pang block configuration info (MRIhc20732)
7	Jun 6, 2007	SYSTEM Tab – Added note for designating components required for 16 channel systems. MGD/RRF/RF Tab –Added notes to designate parts not included with newer 8 Channel systems and for parts that are required for 16 Channel systems.
8 . . .	Oct 26, 2007	SYSTEM Tab – Added CAM Chassis designation to MGD Chassis where applicable. Added CAN LINK block diagram for RFS cabinet with CAM Chassis. MGD/RF/RRF Tab – Added board slot locations for CAM Chassis.
9 . . .	Sep 11, 2008	Changed title of manual and all page headers to <i>Signa HD, HDx & HDxt 3.0T (12.x, 14.x & 15.x) Block Diagrams & Supplemental Schematics</i> . Updates for HDxT (15.x) in all sections. Added information on Sun ICN's.. Added depopulated 16–channel switch for 8–channel installations.
10 . .	Dec 17, 2008	SYSTEM Tab – Added new environmental sensor diagram (ECO 2055310). PATIENT HANDLING Tab – Revised E–Stop circuit voltage from 24V to 12V in Teal PDU (PQR 13207104).
11	Jun 30, 2010	SYSTEM Tab – Added new Z400 computer page (ECO2056587) and added 5337894–3 and –4 ICN's (ECO 2101939). MGD/RRF/RF Tab – Added new ICN's (ECO 2101939).

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* This revision number/letter corresponds to the indicated document's revision control system.

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OVERVIEW

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Standard Conventions
Abbreviations



SYSTEM

System Block Diagrams



OPERATOR WORKSPACE

Operator Workspace Block Diagrams



PATIENT HANDLING

Patient Handling Block Diagrams



GRADIENT

Gradient Block Diagrams (HFD & ACGD)



PDU

PDU Block Diagrams

Note:

Phoenix PDU found in Vendor Manuals only. (SEE Service Methods CDRom)



MGD/RRF/RF

MGD/RRF/RF Block Diagrams
RF Supplemental Schematics



COOLING SYSTEM

Water Chillers Block Diagrams

Note:

For all Chillers and Cooling System variations (SEE Vendor Manuals on MR Service Methods CDRom)



OVERVIEW

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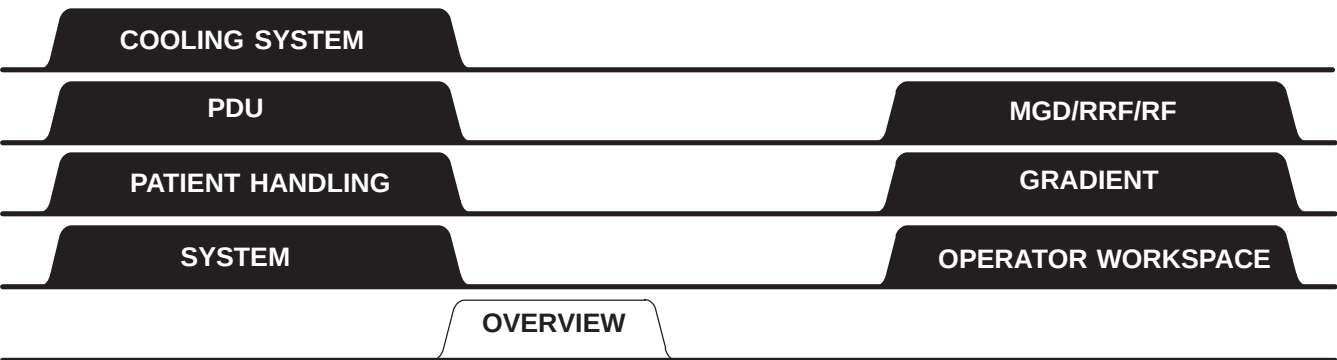
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SECTION 1 – PURPOSE

The purpose of this document is to provide an overview of the hardware which makes up the MR Signa System, and a quick means to troubleshoot major signals within the system.

It is recommended that you page through this manual to become familiar with its content and organization, beginning with System Tab, OVERALL SYSTEM. This section shows how the major subsystems are grouped by cabinet, and the main communication lines between them.

The remaining block diagrams are organized into sections which correspond to major subsystems. Some diagrams show hardware from another subsystem in order to complete an entire signal flow route. For such exceptions, the board/module title of the circuitry shown will generally help identify the applicable subsystem. See Illustration 1–1 for tab organization of this manual.



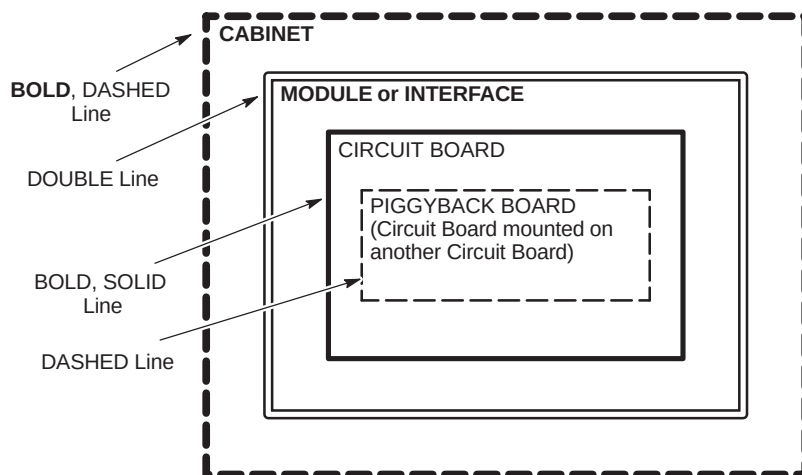
TAB ORGANIZATION FOR THIS DIRECTION
ILLUSTRATION 1

These block diagrams, along with provided diagnostics, are intended to be the primary tools for troubleshooting the MR System from a system or subsystem level on down to the specific module or board level. Since some of the RF subsystem block diagrams are presently insufficient for troubleshooting to the FRU (Field Replaceable Unit) level, schematics have been provided to supplement those block diagrams. When using the diagnostics, the block diagrams can provide a better understanding of what circuitry the diagnostics are testing, and how to further isolate a problem if required.

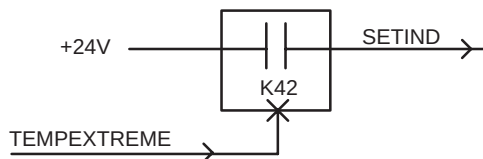
SECTION 2 – STANDARD CONVENTIONS

The following standard conventions are used within the block diagrams:

- The type of line used to enclose a major box on a block diagram indicates the following:



- The component designator for cabinets, modules, circuit boards, etc. is enclosed in "()" after or below their name on the block diagram.
- Logic levels: A "0" indicates a logic "low" level (typ. <0.8VDC), a "1" indicates a logic "high" level (typ. >2.0VDC).
- An "*" is used to identify an "active low" signal (e.g. "INT*" signal when at a logic "0" indicates that an interrupt is present; a logic "1" level indicates that no interrupt is present).
- A "#/" indicates the number of wires in a cable (e.g. $\frac{9}{1}$ indicates 9 wires).
- Signal flow is generally from left to right on a block diagram. Arrows are used to identify exceptions to this rule and to aid in identifying input/output signal lines on the diagram.
- The "x" symbol (e.g. $\rightarrow x$) shown on boxes in a block diagram are used to indicate that the function within the box is enabled by the "active" signal on this input line.



For this example, when signal TEMPEXTREME goes active ("1"), relay K42 will close (become enabled) and the signal SETIND now becomes active to set a temperature indicator.

SECTION 3 – ABBREVIATIONS

Note

This list has been compiled from a wide variety of sources. Not all of these abbreviations will apply to your system. For an electronic copy of over 1000 acronyms, see *Direction 2124201–2*.

A

AC = Alternating Current
 ACGD = Advanced Concept Gradient Driver
 ACK = Acknowledge
 ACQ = Acquisition
 A/D = Analog-to-Digital
 ADC = Analog to Digital Converter
 ADD or ADDR = Address
 ADJ = Adjust
 ADV = Advance
 AGC = Automatic Gain Control
 AGP = Applications Gateway Processor (Board)
 AI = Applications Interface, Artificial Intelligence
 ALGN = Align
 ALU = Arithmetic Logic Unit
 AM = Amplitude Modulation
 AMP = ampere, amplifier
 A/N = Alpha-numeric
 ANSI = American National Standards Institute
 ANT = Antenna
 AP = Array Processor
 APM = Analog Power Monitor
 APS = Auto Pre-Scan or Acquisition Processing System (Board)
 APSGT = Auto Pre-Scan Gain Control
 ARCH = Archive/Remove Process
 ASC = Automated Support Center or Amplifier Support Controller
 ASCII = American Standard Code for Information Interchange
 ASM or ASSY = Assembly
 ASM = Analog Service Module
 ATTEN = Attenuator
 AUTO = Automatic
 AUX = Auxiliary

B

B = magnetic field
 BATT = Battery
 BB = Buffered Data or Broadband
 BC = Body Coil

BD = Board
 BDS = Buffered Data Strobe
 BLK = Black
 BNC = BNC Connector
 BRM = Body Resonance Module
 BRT = Brightness
 BUF = Buffer
 B/W = BandWidth

C

C = centigrade, Celsius
 CAB = Cabinet
 CAL = Calibration
 CALC = Calculate
 CAN = Controller Area Network
 CATTEN = Course Attenuation
 CB = Circuit Breaker
 CCC = CAN Core Controller card
 CCW = Counter Clockwise
 CD-ROM = Compact Disk Read-Only Memory
 CE = Chip Enable
 CF = Center Frequency
 CFA = Center Frequency Adjust
 CHAN or CHNL = Channel
 CHK = Check
 CKT = Circuit
 CLR = Clear
 CMD = Command
 CMMR = Common Mode Rejection Ratio
 CMOS = Complimentary Symmetry Metal Oxide Semiconductor
 CMP = Compare or Compressor
 CNT = Contrast
 COAX = Coaxial
 COF = Cut Off Frequency
 CONFIG = Configuration
 CONT = Continuous
 CONV = Converter
 CPD = Communication Pin Driver
 CPU = Central Processing Unit

CRT = Cathode-Ray Tube
 CTL = Clock Time Control
 CTRL = Control
 CV = Configuration Variable
 CW = Clockwise or Controller Wave
 CYL = Cylinder

D

D/A = Digital-to-Analog
 DAC = Digital to Analog Converter
 dB = Decibel
 dBm = Decibels above or below one Milliwatt
 DC = Direct Current
 DCB = Driver Control Board
 DCKSTP = Dock Stop
 DD = Dynamic Disable
 DDS = Direct Digital Synthesizer
 DDSB = Dynamic Disable Switch Box
 DEC = Decimal
 DEG = Degree(s)
 DEMUX = Demultiplexer
 DEV = device
 DIA = Diagnostic Imaging Accessories
 DIAG = Diagnostic
 DIFF = Differential
 DIG = Digit
 Dig B0 = Digital B0 Adjustment
 DIP = Dual In-line Package
 DIR = Direction
 DIS = Disable
 DM = Driver Module
 DMA = Direct Memory Access
 DMAC = Direct Memory Access Controller
 DMM = Digital Multi-Meter
 DPR or DPRAM = Dual Port Random Access Memory
 DQA = Daily Quality Assurance
 DRAM = Dynamic Random Access Memory
 DRC = Dynamic Ram Controller
 DRF = Digital Receive Filter (Board)
 DRVR = Driver
 DSP = Digital Signal Processor
 DVM = Digital Volt Meter
 DWG = Drawing

E

ECC = Extended Cursor Control or Error Checking and Correction
 ECG = Electro-Cardiogram
 EEPROM = Electrically Erasable Programmable Read Only Memory
 EFB = Envelop Feedback

EPI = Echo Planer Imaging
 e.g. = for example
 EM or EMERG = Emergency
 EMI = Electro-Magnetic Interference
 EN or ENA = Enable
 ENV = Envelope
 EOS = End Of Sequence
 EOT = End Of Travel
 EOW = End Of Waveform
 EPO = Emergency Power Off
 EPROM = Erasable Programmable Read Only Memory
 ERU = Emergency Rundown Unit
 EXCITE = Expanding Applications with Multi-channel Imaging Technology
 EXER = Exerciser

F

f = farad
 F = Fahrenheit, Fuse (designator)
 FBD = Functional Block Diagram
 FBK = Feedback
 FC = footcandle
 Fc = Center Frequency
 FDBK = Feedback
 FDO = Future Delivery Order
 FE = Field Engineer
 FET = Field-Effect Transistor
 FF or FIFO = First In – First Out
 FFD = Full Field Distortion
 FFT = Fast Fourier Transform
 FID = Free Induction Decay
 FIDAT = FIFO In Data
 FIFO or FF = First In – First Out
 FILT = Filter
 FM = Frequency Modulated
 FMI = Field Modification Instruction
 FOV = Field Of View
 FREQ = Frequency
 FRNT = Front
 FRU = Field Replaceable Unit
 ft = foot
 FT = Feet
 FTP = File Transfer Protocol

G

G = gauss
 G/A = Gradient Amplifier
 GEN = Generator
 GND = Ground
 GOC AA = Global Operator Console Audio Assembly
 GRAD = Gradient
 GRN = Green
 GRY = Grey
 GP = Gradient Processor
 GSW = Gradient Switch

H

H = henry
 H = Hexadecimal
 HART = Hardware Tracker
 HC = Head Coil
 HD = Head or High Definition
 He = Helium
 He-ALARM = Helium Alarm
 HEX = Hexadecimal
 He-WARNING = Helium Warning
 HFA = High Fidelity Amplifier
 HFD = High Fidelity Driver
 HFD-S = High Fidelity Driver – Special
 HORIZ or HORZ = Horizontal
 HPC = Hardware and Protocol Control
 HPFS = Host Platform Software
 hr = hour
 HSN = High Speed Network
 HSS (D) = High Speed Serial Data Link
 HTR = Heater
 HV = High Voltage
 HV PSU = High Voltage Power Supply Unit
 HVAC = Heating, Ventilation, & Air Conditioning
 Hz = hertz

I

I = Current or Phase or Real
 IAMP = Amplifier Current (Gradient)
 ICD = Installation Certification Document
 ICL = Coil Overcurrent (Gradient)
 ICN = Image Compute Node
 ID = inner diameter
 ID = Identity, Identifier
 i.e. = that is to say
 IEC = International Electrical Code
 IEEE = Institute of Electrical & Electronics Engineers
 I/F = Interface
 in. = inch
 INFO = Information

INS, INST, or INSTR = Instruction
 INT or INTR = Interrupt
 I/O = Input/Output
 IP = Image Processor or Plate Current (RF amp) or
 Instruction Processor or Internet Protocol
 IPA = Intermediate Power Amplifier
 IPC = Interprocess Communication
 IQ = Image Quality
 IR = Image Reject
 IRF = Interface and Remote Functions (Board)
 IRF I/O = Interface and Remote Functions
 Input/Output (Board)
 IRQ = Interrupt Request
 ISA = Interrupt Start Address
 ISI = Inter-Sequence Interrupt
 ISO = Isolator or Isolation
 IT-MGD = Image Transfer-MDG (Board)
 IT-Host = Image Transfer-Host (Board)
 IVI = Interactive Vascular Imaging

J

J = Jack Connector (designator)
 JP = Jumper

K

K = Kelvin (unit of temperature) or Kilo (Thousand)
 KB = one thousand bytes of memory
 KCal = kiloCalorie
 kg = kilogram
 kHz = kilohertz
 KVA = kilo Volt Amperes
 kW = Kilowatt

L

L or LFT = Left
 LAN = Local Area Network
 lb = pound
 LBK = Loopback
 LCA = Logic Cell Arrays
 LED = Light Emitting Diode
 LFC = Load From Cold
 LHe = Liquid Helium
 LIN = Linear
 LM or LMK = Landmark
 LMP = Lamp
 LNK = Link
 LPCA = Low Profile Carriage Assembly
 LO = Local Oscillator
 LOEP = List of Effective Pages
 LONG = Longitudinal
 LVLE = Low Voltage, Low Energy

M

m = meter
 M = mega – one million
 mA = millamps
 MAG = Magnet
 MB = megabyte
 MC = Multi-Coil
 MCD = Multi-Coil switch Driver
 MCR = Multicoil Receive (Tool or Diagnostic)
 MDS = Multidrop Serial interface
 MEM = Memory
 Mfg. = Manufacturing
 MFU = Multi-Format Unit
 MGD = Multi-Generational Data acquisition (Chassis)
 MGMNT = Management
 MHz = megahertz
 mm = millimeter
 MOD = Modifier
 MON = Monitor
 MOSFET = Metal Oxide Semiconductor Field Effect Transistor
 MOV = Metal Oxide Varistor
 MPS = Manual Pre-Scan
 MR = Magnetic Resonance
 MSG = Message
 MTR = Motor
 MUX = Multiplexer
 mV = millivolts
 MW = megawatt
 mW = milliwatt

N

N = Negative
 N2 = Nitrogen
 n.c. = normally closed
 NEC = National Electrical Code
 Ni-Cad = Nickel Cadmium
 n.o. = normally open
 NO. = Number
 NOPROC = No Processing
 NOREC = No Reconstruction
 nS = Nanosecond

O

O2 = oxygen
 OC = Operator's Console
 OD = outside diameter
 OM = Oxygen Monitor
 OPI = Oblique Plane Imaging
 OPTO = Optical
 ORG = Orange

OT = Overtemp
 OV = Over voltage
 OVR = Over
 OVRLD or OL = Overload
 oz = ounce
 OW = Operator Workspace

P

P = Positive
 PA = Patient or Power Amplifier
 PAC = Physiological Acquisition Controller
 PAL = Programmable Array Logic or Patient Alignment Lights
 PAN = Primary Archive Node
 PAT = Patient
 PCI = Programmable Communications Interface or Peripheral Component Interconnect
 PCT = Pulse Controller Task
 PDI = Product Delivery Instruction
 PDU = Power Distribution Unit
 PEN = Penetration
 PHPS = Patient Handling Power Supply
 PIN = Positive-Intrinsic-Negative
 PIO = Programmed Input/Output
 PLL = Phase Lock Loop
 PIXEL = Picture Element
 PM = Phase Modulation, or Planned Maintenance
 PMC = PCI Mezzanine Connector
 P/N = Positive/Negative
 PNL = Panel
 POS = Position
 POT = Potentiometer
 PP = Penetration Panel
 PPC = Power PC
 P-P = Peak to Peak
 ppm = parts per million
 PREAMP = Preamplifier
 PROC = Process, Processor
 PROG = Program, -ed, -able
 PROM = Programmable Read Only Memory or Program
 PS = Power Supply
 PSD = Pulse Sequence Database
 PSG = Pulse Sequence Generator
 PSU = Power Supply Unit
 PSS = Power Supply System Board
 PT = Patient Transport or Patient Table

PUR = Purple
 PVC = Polyvinylchloride
 PW = Pulse Width
 PWA = Printed Wiring Assembly
 PWB = printed wiring board
 PWR = Power

Q

Q = Quadrature, Quality Factor or Imaginary
 Qty = Quantity
 QUAD = Quadrature or Quadrant

R

R or RT = Right
 RAM = Random Access Memory
 RCV = Receive
 RD = Read
 RDY = Ready
 RECON = Reconstruction
 REF = Reference or Reflected
 REG = Register or Regulator
 RET or RTN = Return
 REV = Revision or Reverse
 RF = Radio Frequency
 RFAMP = Radio Frequency Amplifier
 RFA = RF Amplifier
 RFI = Radio Frequency Interference or RF Interface
 (Module)
 RFS = RF System Cabinet
 RGB = Red Green Blue
 RLA or RLY = Relay
 rms = root mean square
 ROI = Region of Interest
 ROM = Read Only Memory
 RPC = Remote Control Panel
 RRF = Remote RF (Chassis)
 RRF-DIF = Remote RF-Digital Interface board
 RSR = Register
 RST = Reset
 R/W = Read/Write
 RX = Receive
 RXD = Receive Data

S

S = Switch (designator)
 SACK = Slave Acknowledge
 SAG = Sagittal
 SAR = Specific Absorption Rate
 S/C = Superconducting
 SCA = Scan Control Assembly
 SCP = Scan Control Processor (Board)
 SCR = Silicon Controlled Rectifier

SCSI = Small Computer System Interface
 SEC = second
 SEG = Segment
 SEL = Select
 SEPI = Spin Echo Planer Imaging
 SGA = Switchable Gradient Amplifier
 SGI = Silicon Graphics Inc.
 SIG = Signal
 SII = "S-2" (type of magnet)
 SIII = "S-3" (type of magnet)
 SIMMS = Single In-line Memory Module
 SIP = Single In-line Package
 SL = Slice
 SMC = Status Monitoring and Control
 SMPTE = Society of Motion Picture and Television
 Engineers
 SN = Service Note
 SNR = Signal-to-Noise Ratio
 SPI = Serial Peripheral Interface
 SPT = System Performance Test
 SPU = Signal Processing Unit
 SRAM = Static Random Access Memory
 SRF = Solid State RF Amplifier SGA or Scalable RF
 (Cabinet)
 SRF/TRF = Sequencer Related Function / Trigger and
 Rotation Function (Board)
 SRI = Scan Room Interface
 SRV = Service
 SSM = System Support Module
 SSP = Sequence Synchronous Protocol
 SST = Small Sample Test
 STAB = Stability
 STAT = Status
 STD = Standard
 STD DEV = Standard Deviation
 STIF Bd = SRF/TRF InterFace Board
 SUB-D = Subminiature D (connector)
 SW = Switch or Software
 S/W = Software
 SYNC = Synchronous
 SYS = System
 SYSID = System Identification

T

T = tesla
TAC = Twin Accessory Cabinet
TBD = To Be Determined
TBL = Table
TEMP = Temperature
TFU = Thyristor Firing Unit
TG = Transmit Gain
TGWC = TwinSpeed Gradient Water Cooler
TNS = Transient Noise Suppressor
TP = Test Point (designator)
TR or T/R = Transmit–Receive
TRMAP = Transmit/Receive Map
TRM = Twin Coil Resonator Module
TS = Terminal Strip
TSCC = TwinSpeed Cooling Cabinet
TST = Test
TTL = Transistor to Transistor Logic
TX = Transmit
TXD = Transmit Data

U

UART = Universal Asynchronous Receiver–Transmitter
uf = microfarad
UFI = Ultra Fast Imaging
UNBLK = Unblank
uP = Microprocessor
UPLMT = Up Limit
UPM = Universal Power Monitor
USART = Universal Synchronous/Asynchronous
Receiver–Transmitter
uS or usec = microsecond
UTNS = Universal Transient Noise Suppressor (Board)
UV = Under voltage

V

VAC = Volts Alternating Current
VAR = Variable
VBUS = Bus Voltage
VCR = Video Cassette Recorder
VCNTRL = Control Voltage for 8645 Techrons
VDC = Volts Direct Current
VERT = Vertical
Vp–p = Volts Peak–to–Peak
VRE = Volume Recon Engine
VRMS = Volts, Root Mean Square
VTR = Video Tape Recorder

W

WARP = Waveform And Rotation Processor
WC = Water Chiller
WR = Write
WHT = White
WIM = Workspace Interface Module
WND = Window
WORM = Write Once Read Many
WRT = Write

X

XCVR = Transceiver
XFMR = Transformer
XMT or XMIT = Transmit

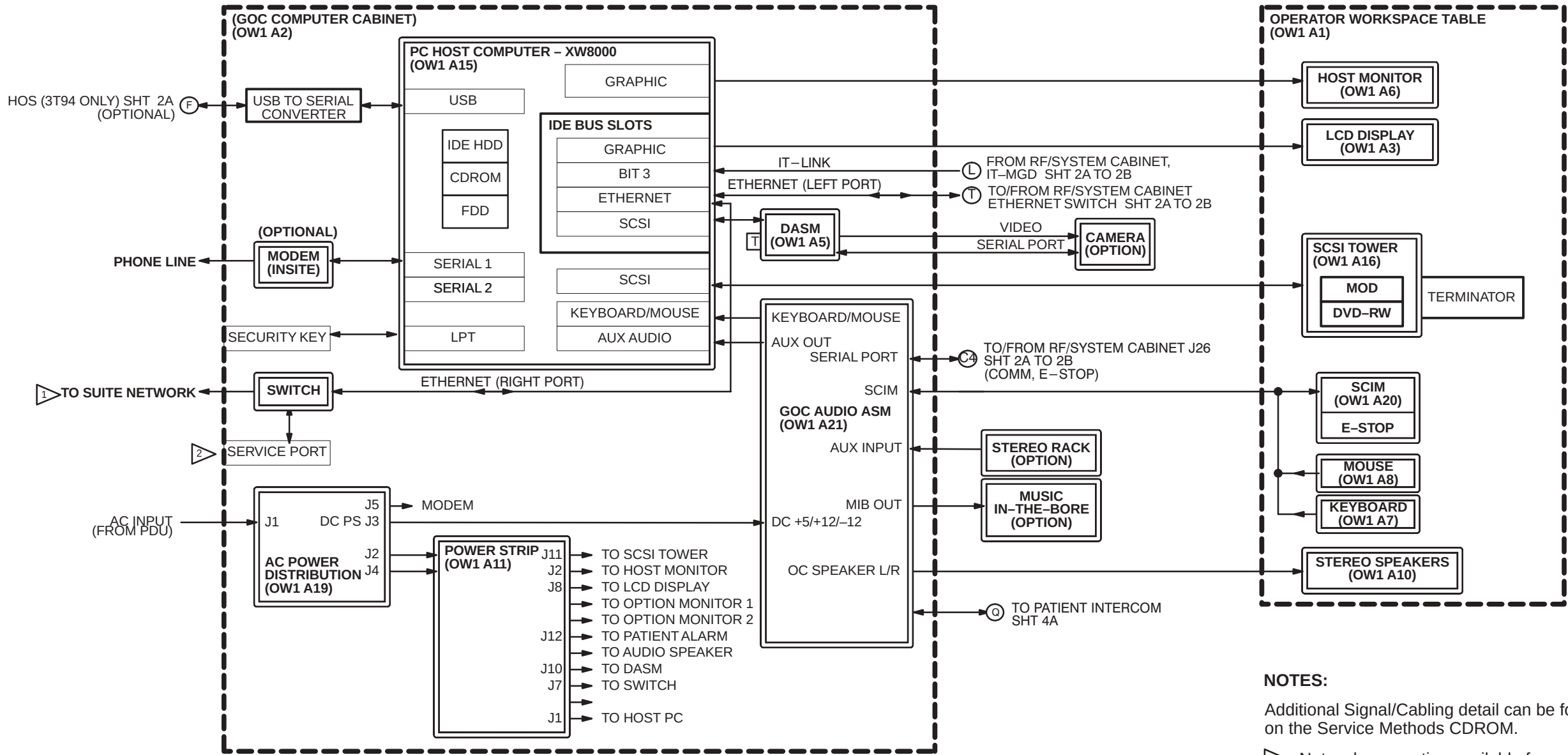
Y

YEL = Yellow

SYSTEM

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NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

➤ Network connection available for external suite connection

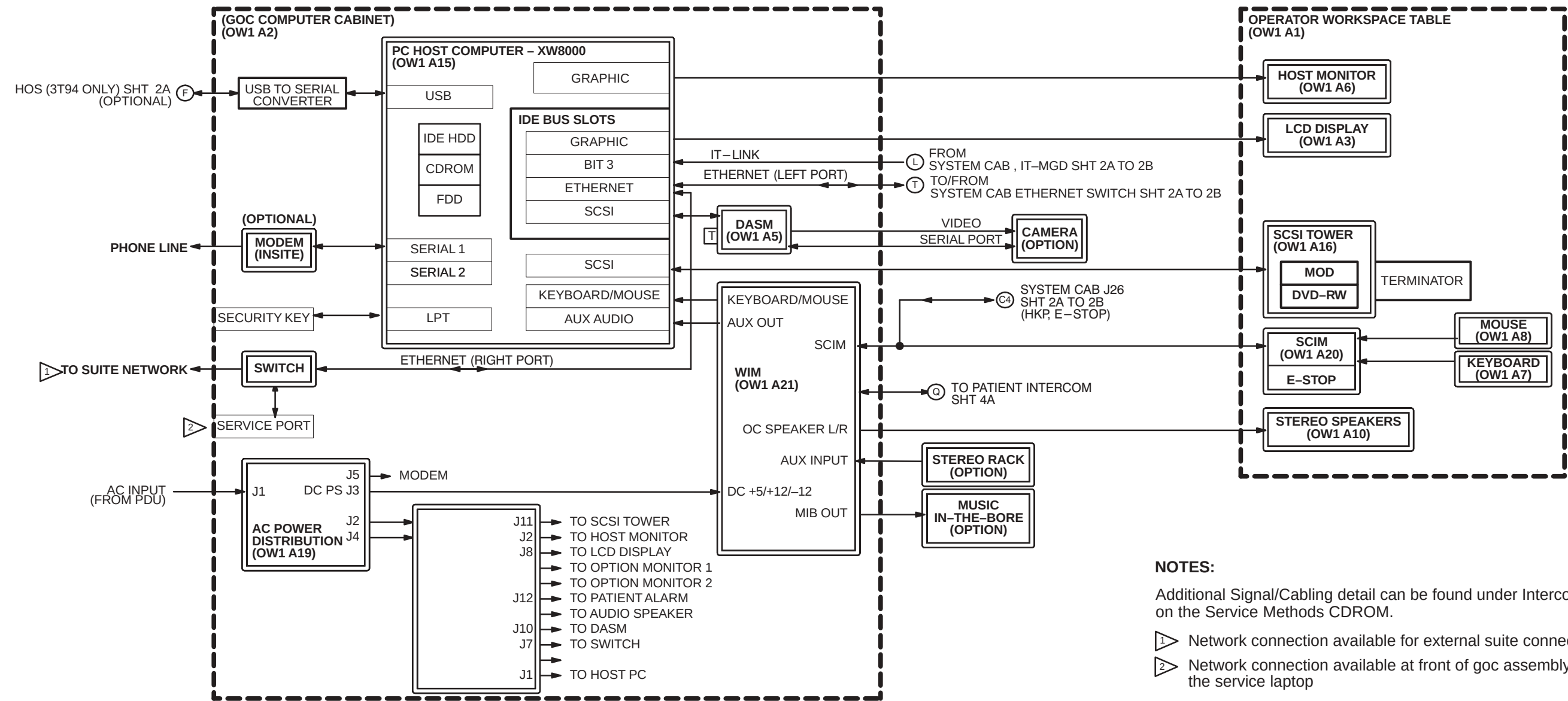
➤ Network connection available at front of goc assembly for the service laptop

Refer to the following functional block diagram for additional information:

NOTE:

1) PATIENT HANDLING

2) MGD/RRF/RF



NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

- 1) Network connection available for external suite connection
- 2) Network connection available at front of goc assembly for the service laptop

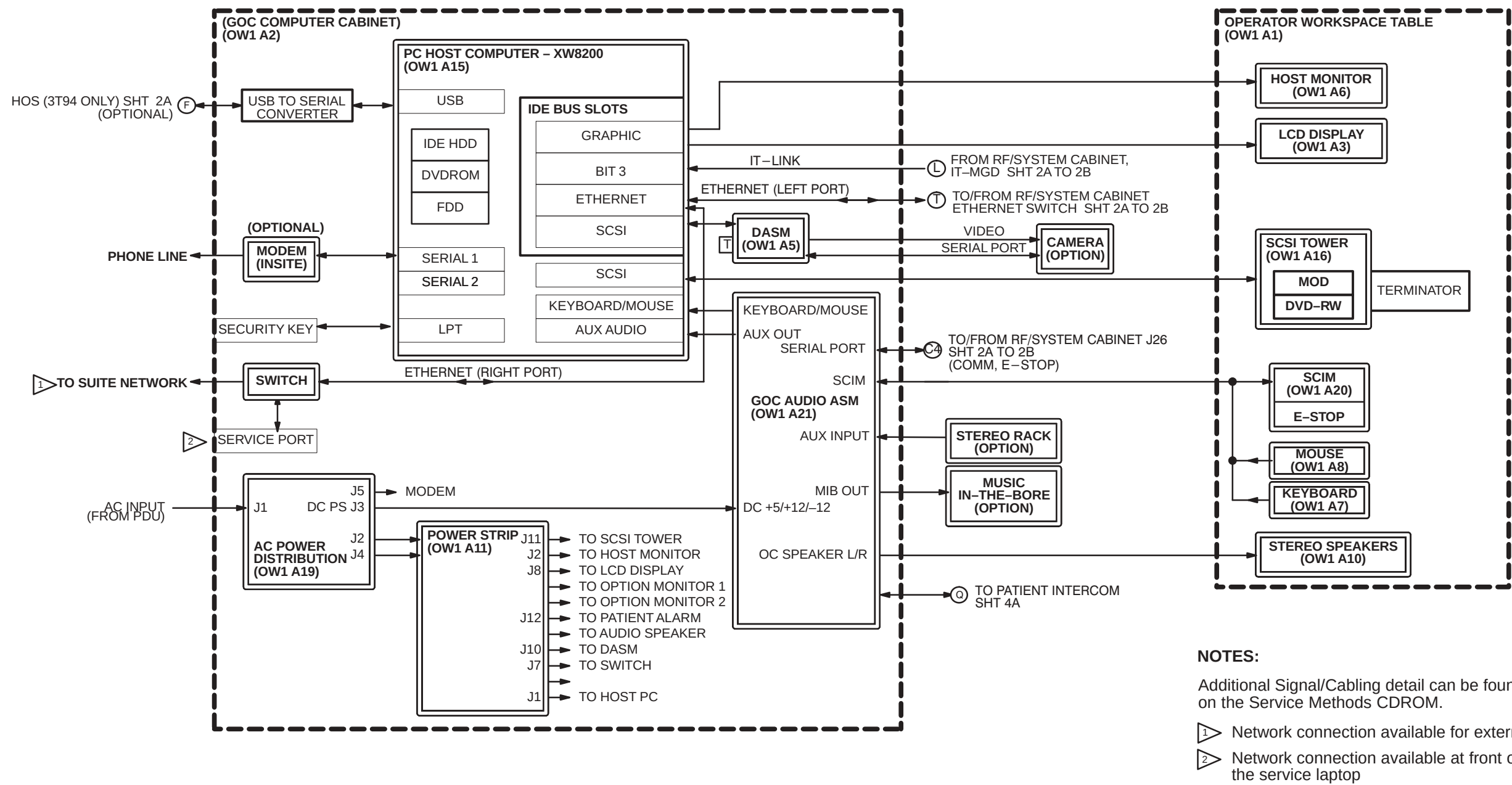
Refer to the following functional block diagram for additional information:

NOTE:

- 1) PATIENT HANDLING
- 2) MGD/RRF/RF

G3 & 3T94 EXCITE HD (12.x) OVERALL SYSTEM
EXCITE to HD Upgrades
(OPERATOR WORKSPACE CABINET/TABLE)

SHEET 1B OF 6



G3 & 3T94 EXCITE HD (12.x) OVERALL SYSTEM
Forward Production or Pre-EXCITE Upgrades
(OPERATOR WORKSPACE CABINET/TABLE)
SHEET 1C OF 6

NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

1) Network connection available for external suite connection

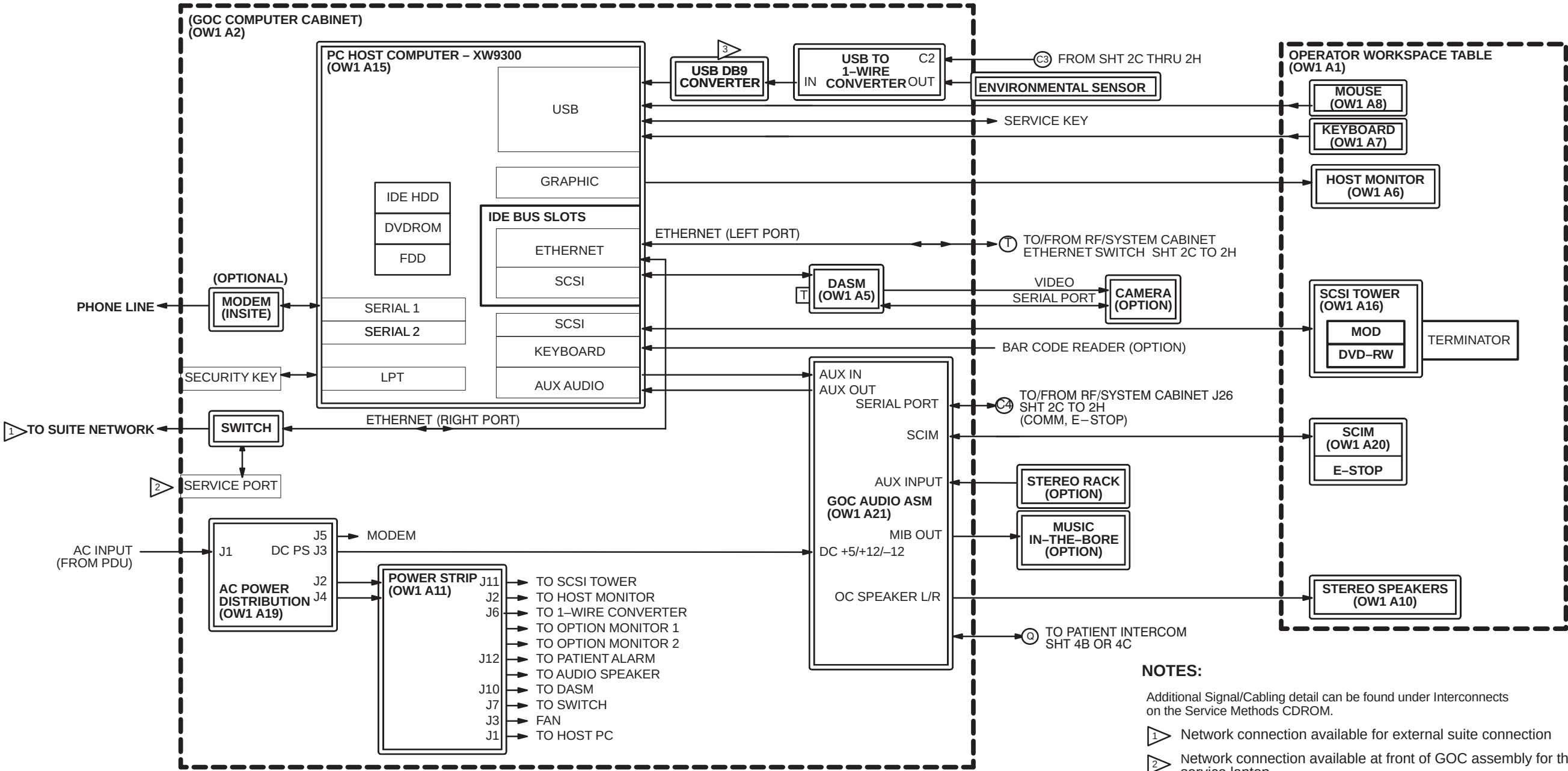
2) Network connection available at front of goc assembly for the service laptop

Refer to the following functional block diagram for additional information:

NOTE:

1) PATIENT HANDLING

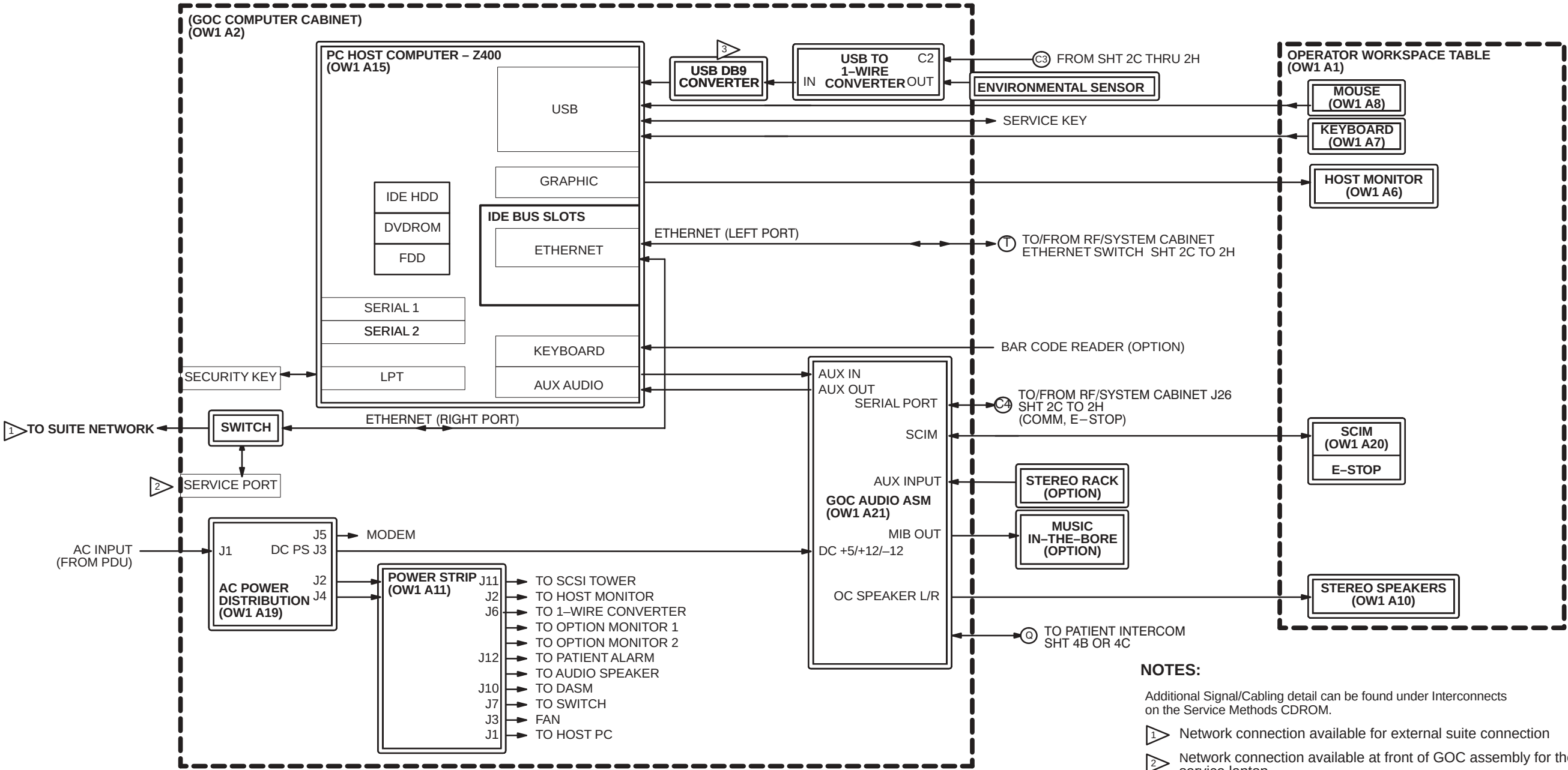
2) MGD/RRF/RF



- NOTES:**
- Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.
- 1 Network connection available for external suite connection
 - 2 Network connection available at front of GOC assembly for the service laptop
 - 3 Beginning August 2008, design changed to 1-wire to USB converter. Environmental Sensors Block Diagram utilizing 1-wire 5147136-2
-
- Refer to the following functional block diagram for additional information:

NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF

G3 & 3T94 HDx (14.x) & HDxt (15.x) OVERALL SYSTEM
Forward Production, EXCITE, or Pre-EXCITE to HDx Upgrades
(OPERATOR WORKSPACE CABINET w/9300 & TABLE)



NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

1) Network connection available for external suite connection

2) Network connection available at front of GOC assembly for the service laptop

3) Beginning August 2008, design changed to 1-wire to USB converter. Environmental Sensors Block Diagram utilizing 1-wire 5147136-2

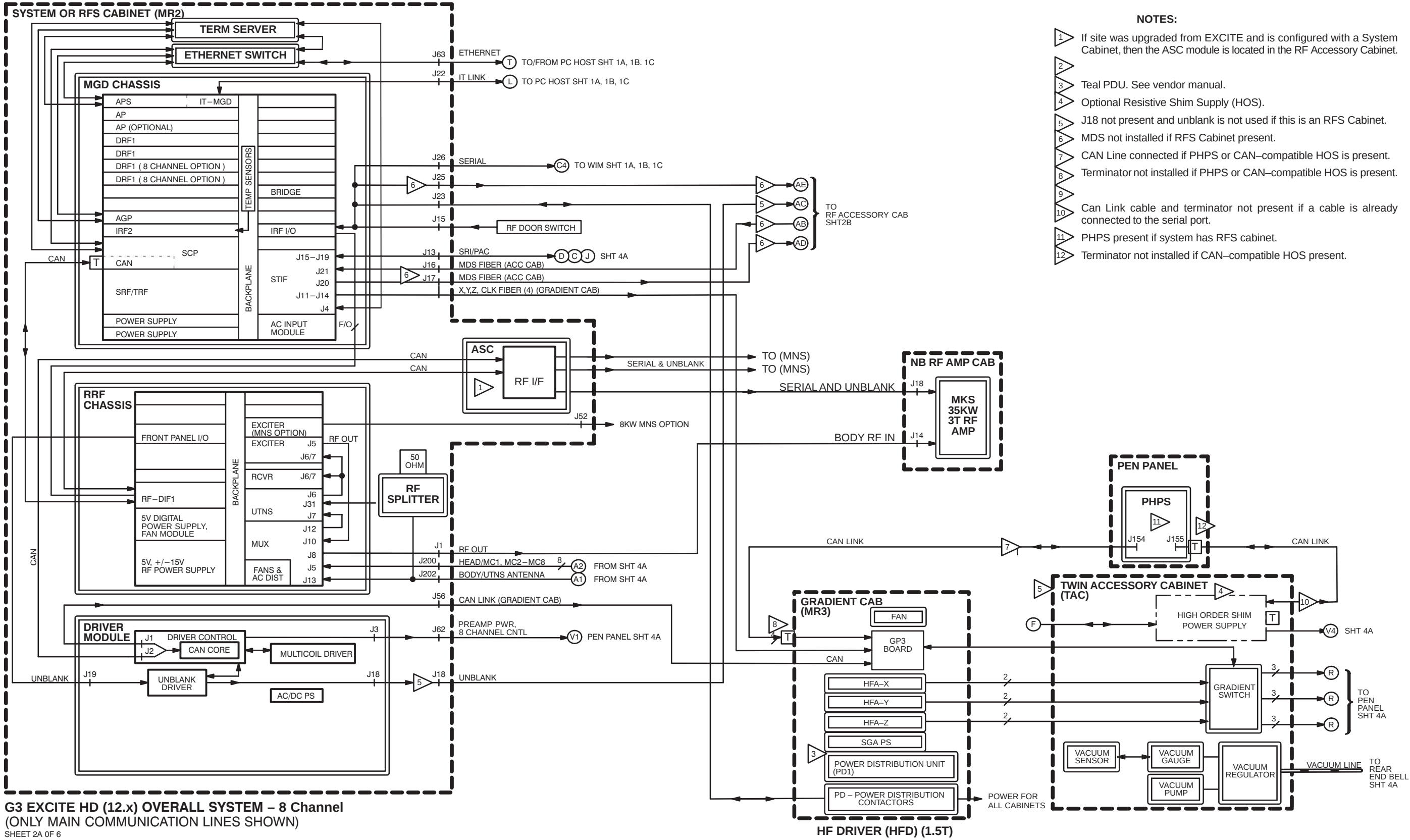
The functional block diagram shows the Host PC USB Port connected to a USB to 1-wire Converter. This converter is connected to Run 1321, which leads to the GOC Sensor. Run 1322 connects the GOC Sensor to the Sys Cab Sensor.

Refer to the following functional block diagram for additional information:

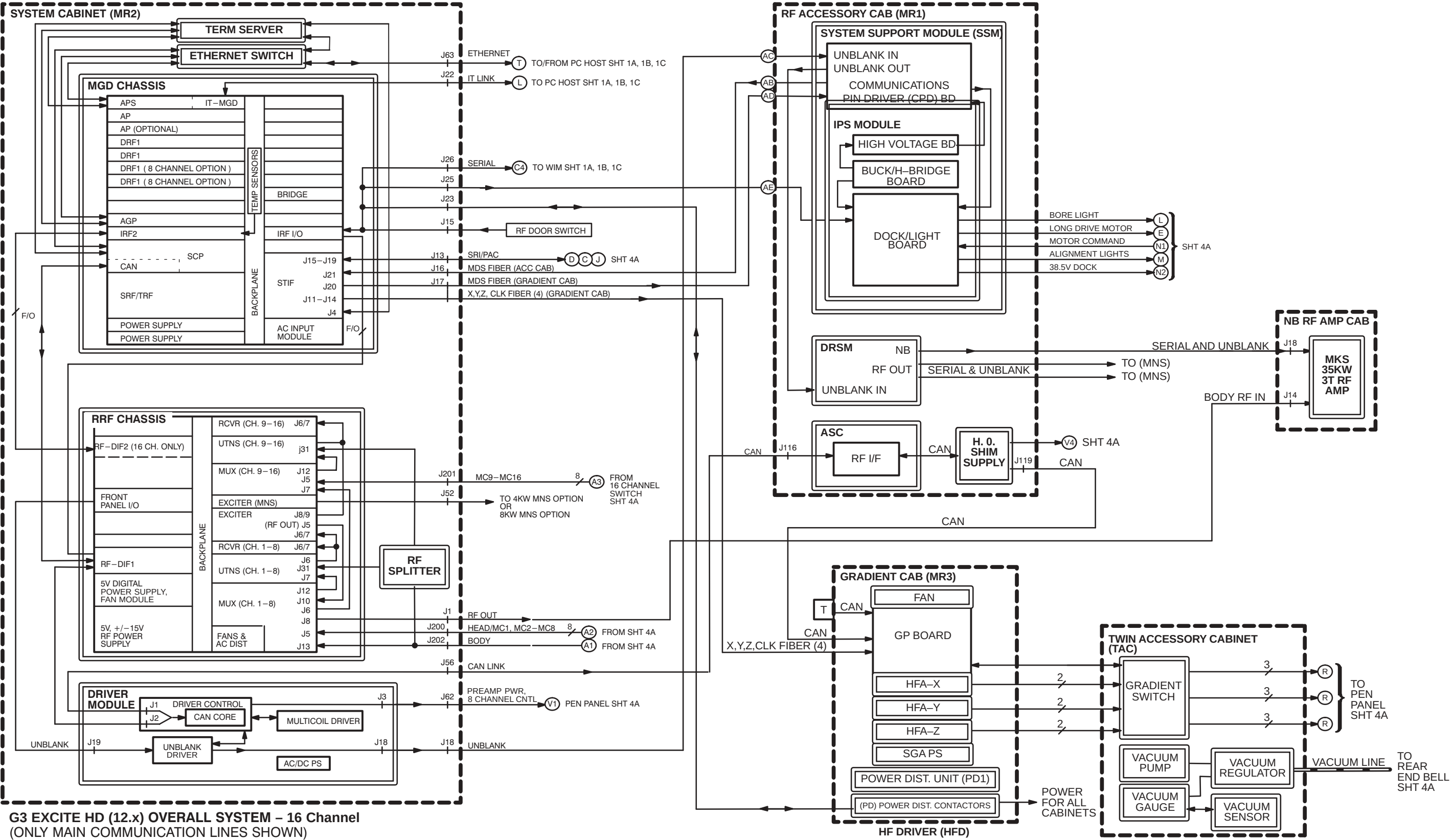
NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF

G3 & 3T94 HDx (14.x) & HDxt (15.x) OVERALL SYSTEM
Forward Production, EXCITE, or Pre-EXCITE to HDx Upgrades
(OPERATOR WORKSPACE CABINET w/Z400 & TABLE)

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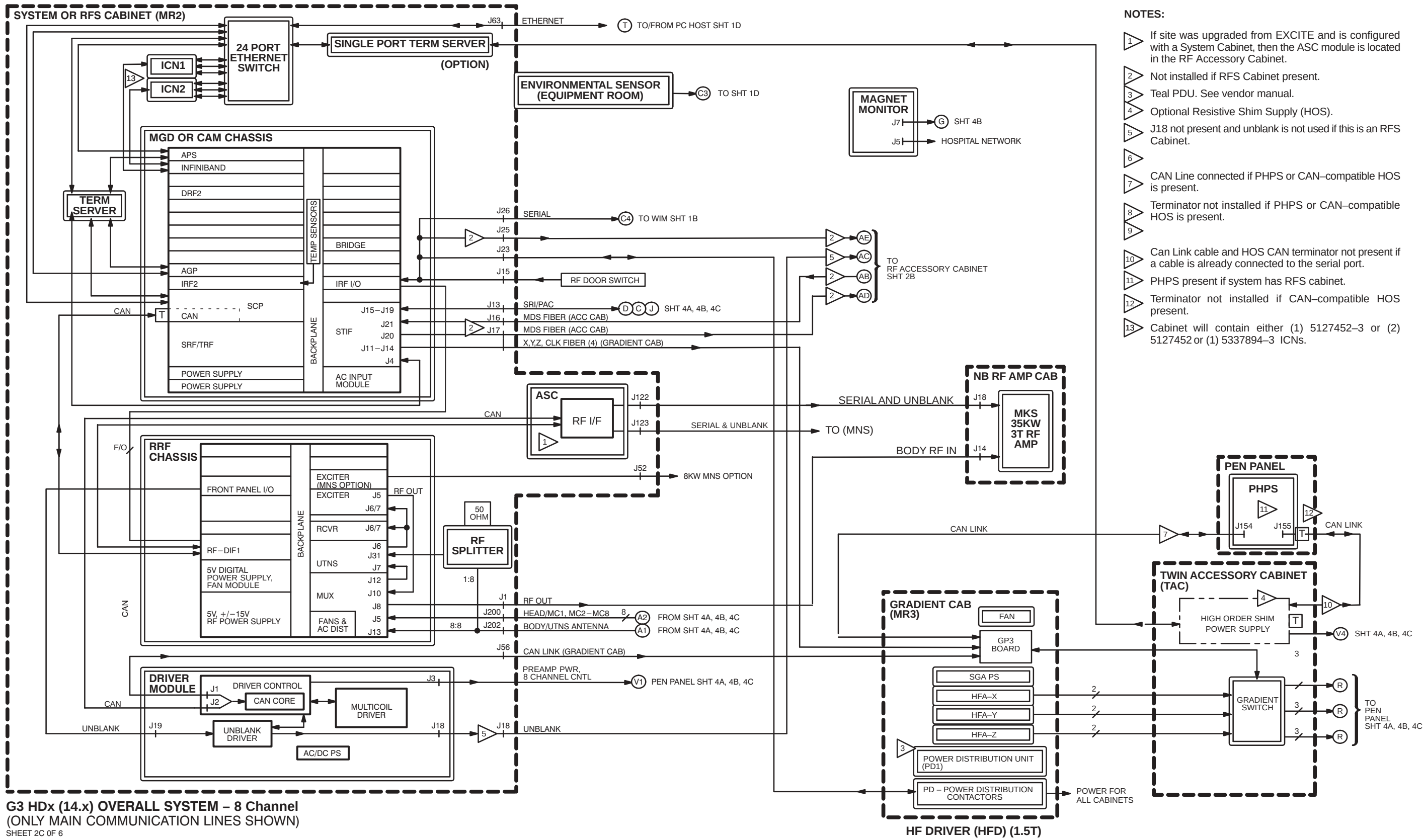


G3 EXCITE HD (12.x) OVERALL SYSTEM - 8 Channel
(ONLY MAIN COMMUNICATION LINES SHOWN)

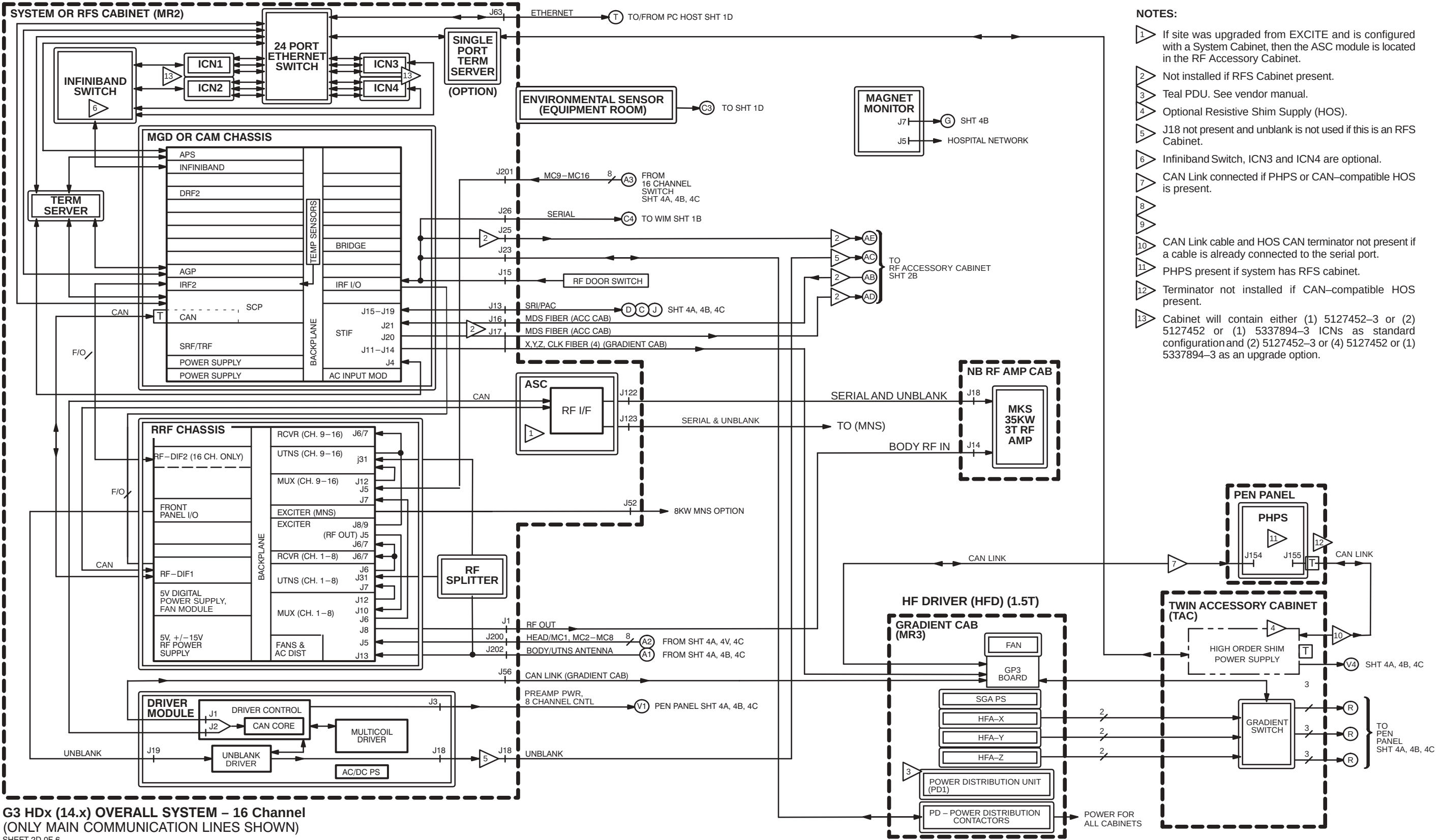


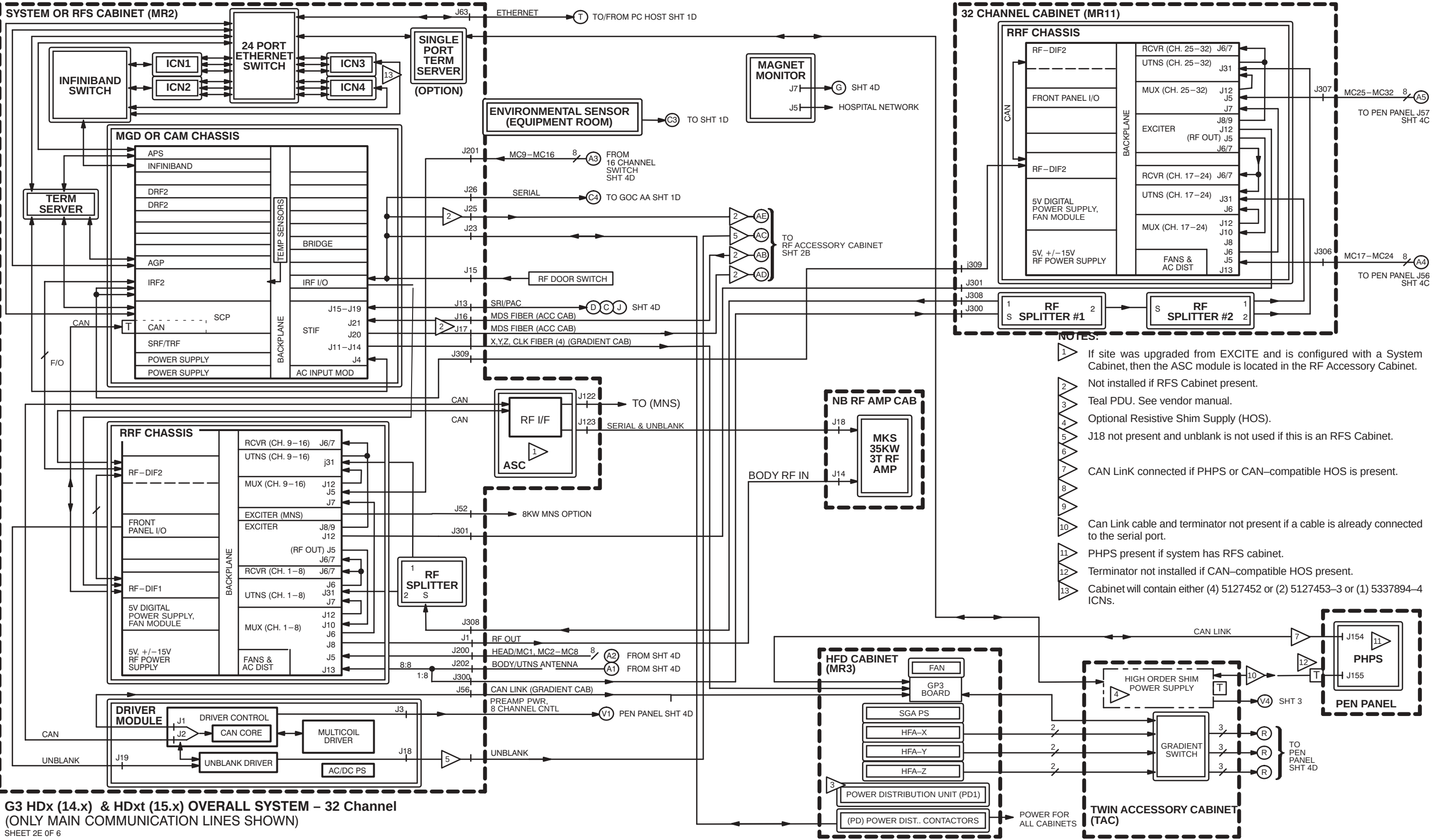
G3 EXCITE HD (12.x) OVERALL SYSTEM - 16 Channel
(ONLY MAIN COMMUNICATION LINES SHOWN)

SHEET 2B OF 6



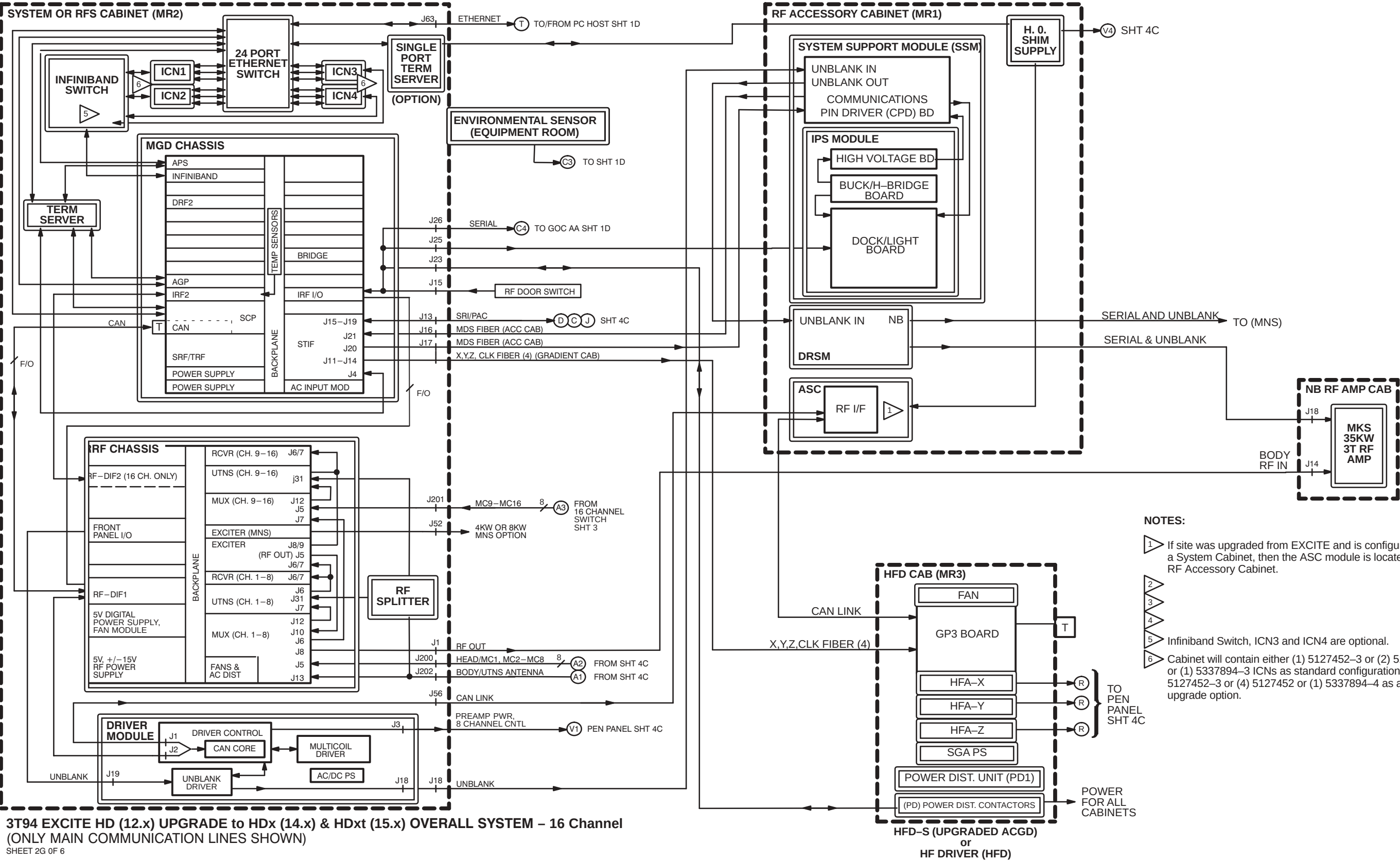
G3 HDx (14.x) OVERALL SYSTEM - 8 Channel
(ONLY MAIN COMMUNICATION LINES SHOWN)
SHEET 2C OF 6





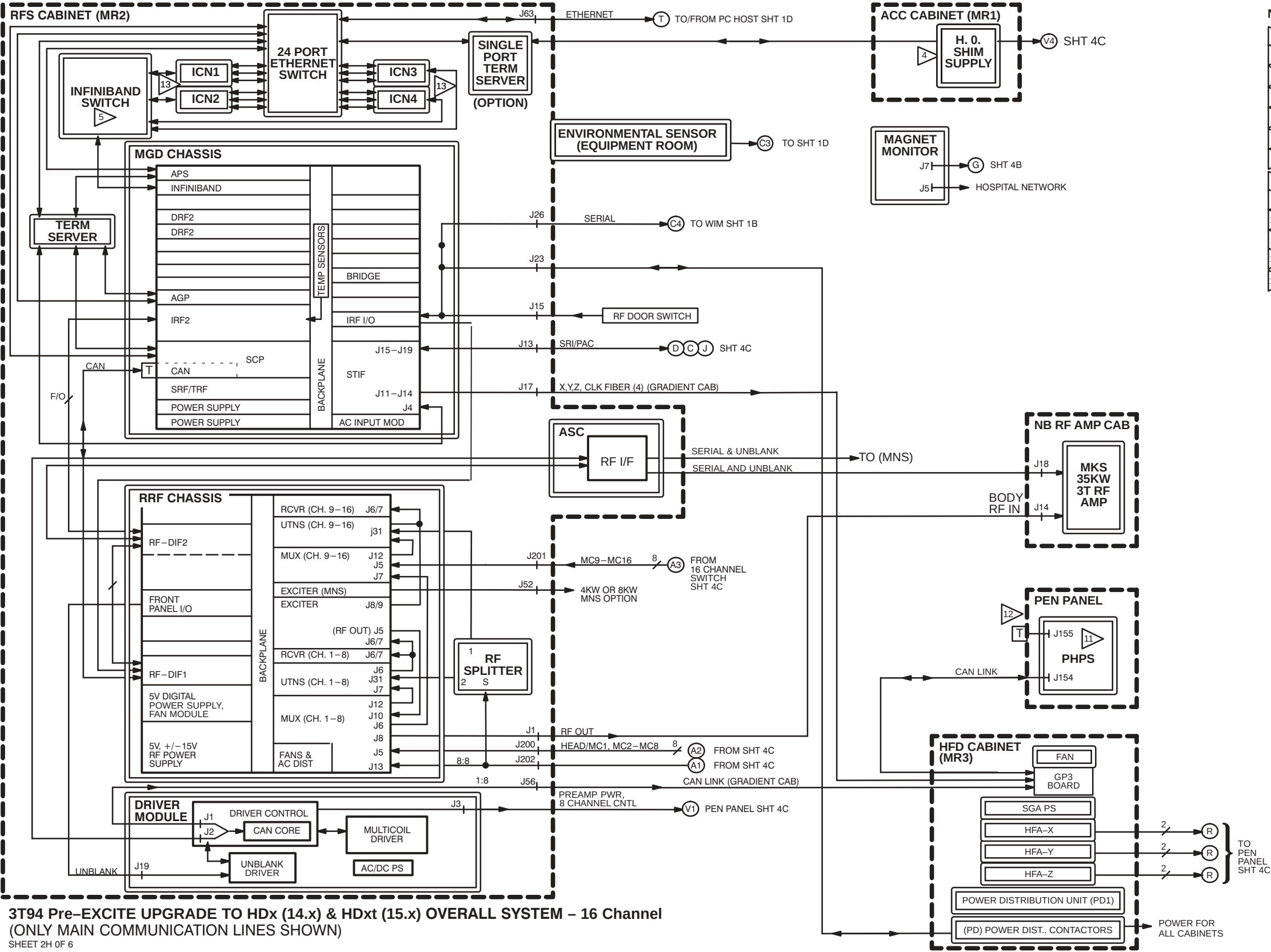
G3 HDx (14.x) & HDxt (15.x) OVERALL SYSTEM – 32 Channel
(ONLY MAIN COMMUNICATION LINES SHOWN)
SHEET 2E OF 6





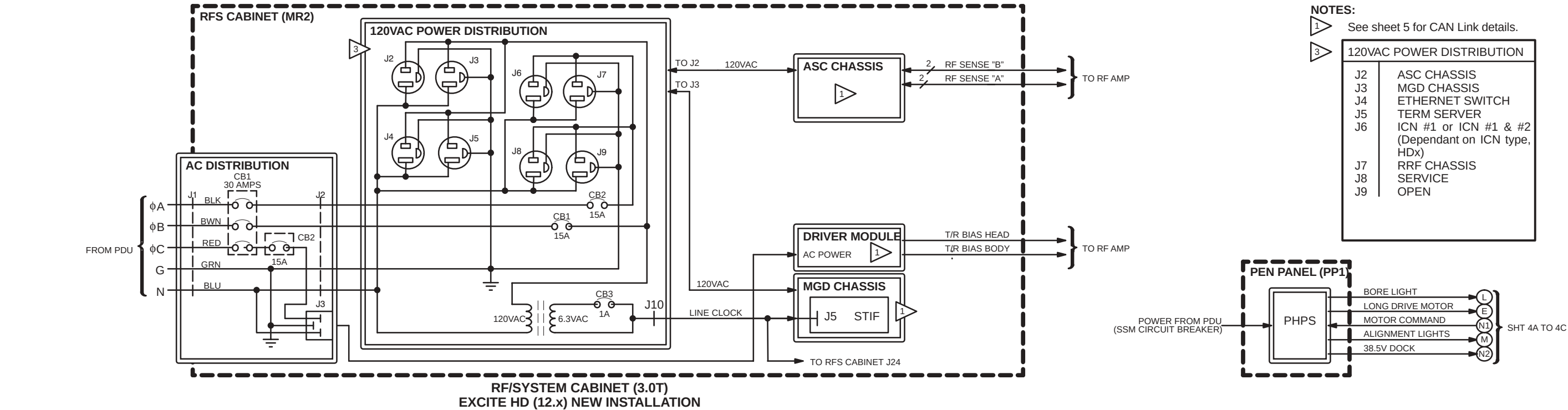
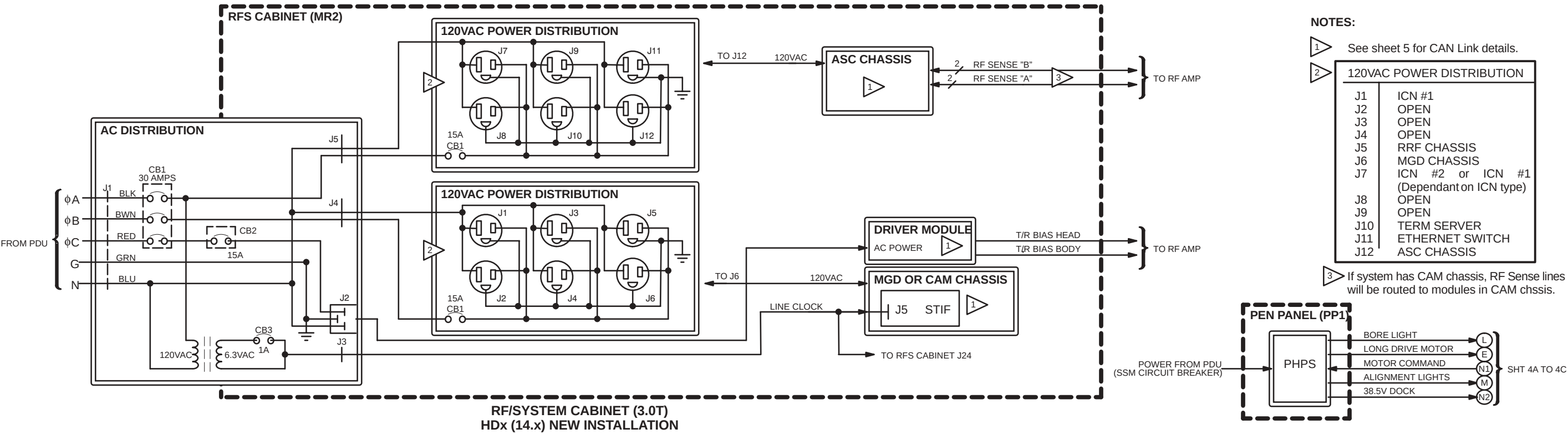
3T94 EXCITE HD (12.x) UPGRADE to HDx (14.x) & HDxt (15.x) OVERALL SYSTEM - 16 Channel
(ONLY MAIN COMMUNICATION LINES SHOWN)

SHEET 2G OF 6

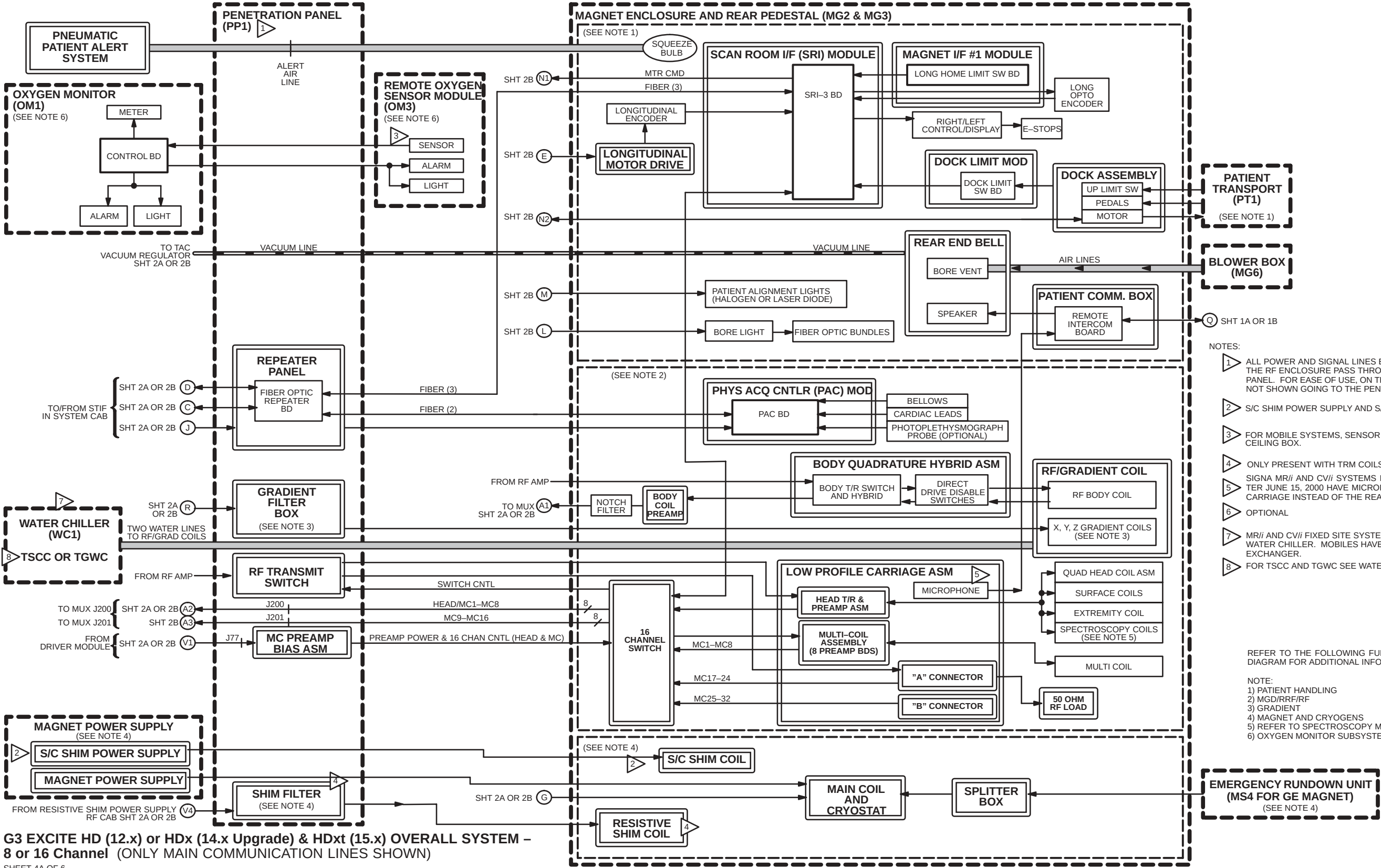


NOTES:

- 1 MDS not installed if RFS Cabinet present.
- 2
- 3
- 4 Optional Resistive Shim Supply (HOS).
- 5 Infiniband Switch, ICN3 and ICN4 are optional.
- 6
- 7
- 8
- 9
- 10
- 11 PHPS present if system has RFS cabinet.
- 12 Terminator not installed if CAN-compatible HOS present.
- 13 Cabinet will contain either (1) 5127452-3 or (2) 5127452 or (1) 5337894-3 ICNs as standard configuration and (2) 5127452-3 or (4) 5127452 or (1) 5337894-3 as an upgrade option.



G3 & 3T94 EXCITE HD (12.x) and HDx (14.x) & HDxt (15.x) OVERALL SYSTEM – New Installation & Upgrade
(ONLY MAIN COMMUNICATION LINES SHOWN)



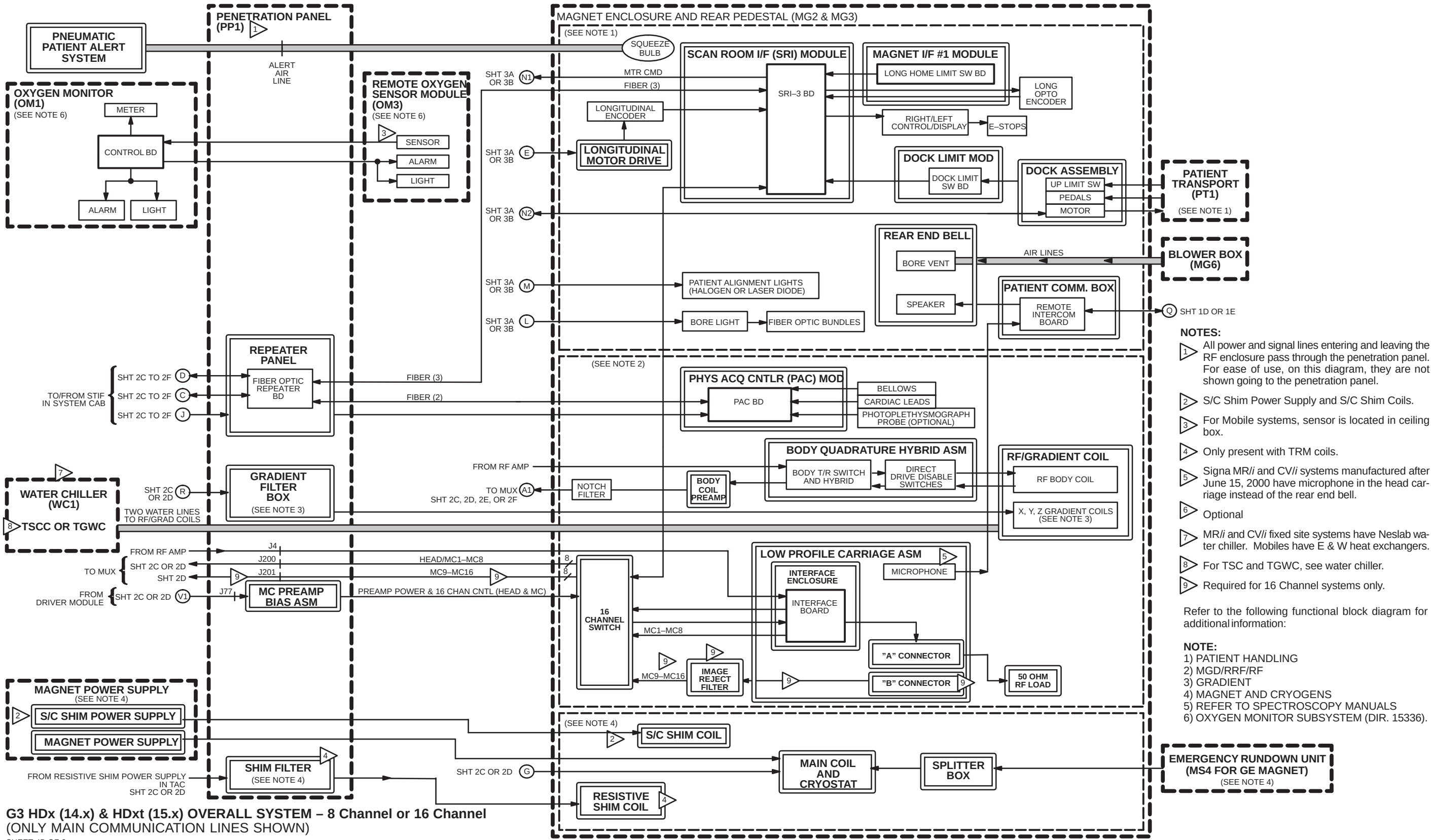
G3 EXCITE HD (12.x) or HDx (14.x Upgrade) & HDxt (15.x) OVERALL SYSTEM –
8 or 16 Channel (ONLY MAIN COMMUNICATION LINES SHOWN)

SHEET 4A OF 6

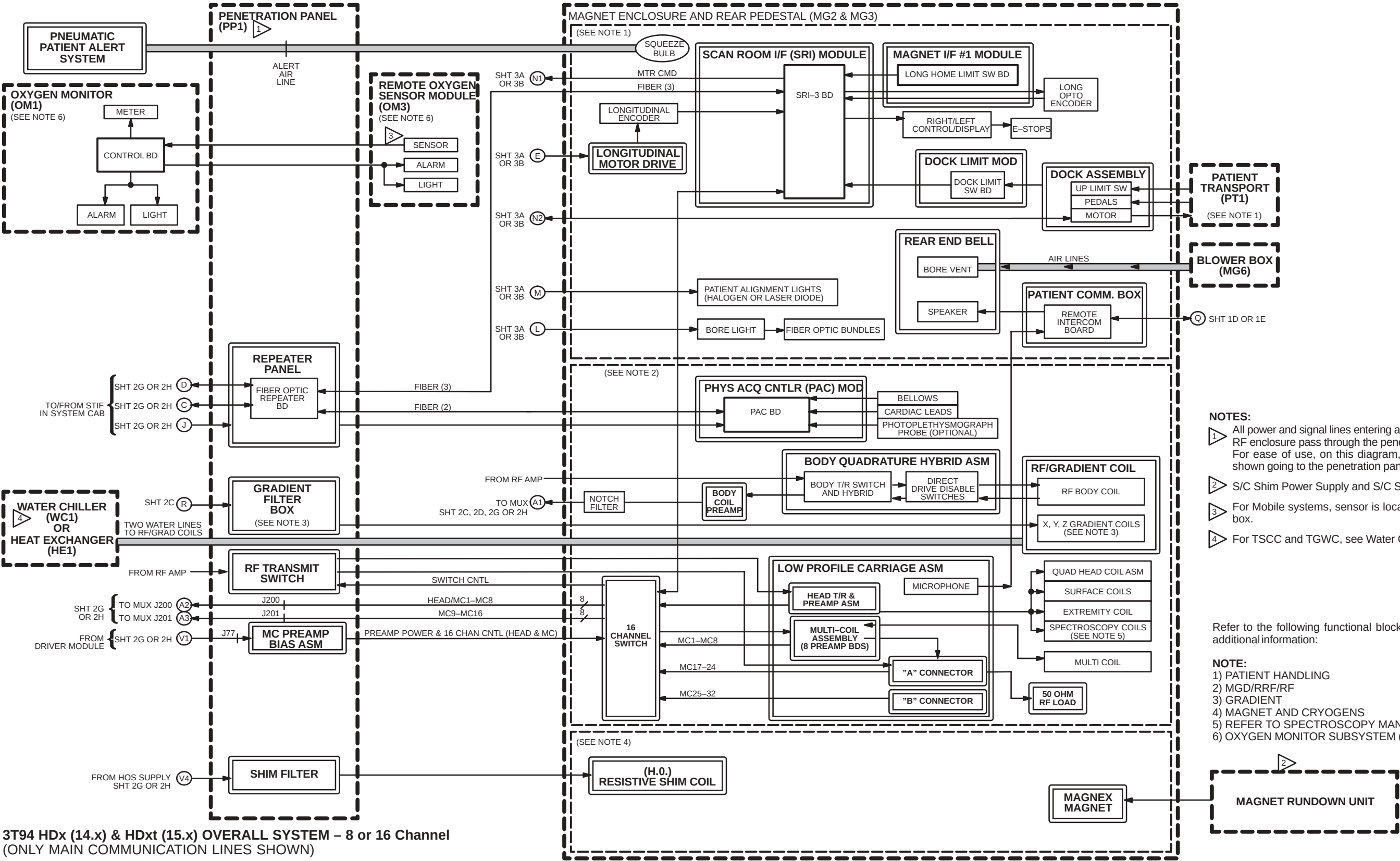
- NOTES:
- 1 ALL POWER AND SIGNAL LINES ENTERING AND LEAVING THE RF ENCLOSURE PASS THROUGH THE PENETRATION PANEL. FOR EASE OF USE, ON THIS DIAGRAM, THEY ARE NOT SHOWN GOING TO THE PENETRATION PANEL.
 - 2 S/C SHIM POWER SUPPLY AND S/C SHIM COILS.
 - 3 FOR MOBILE SYSTEMS, SENSOR IS LOCATED IN CEILING BOX.
 - 4 ONLY PRESENT WITH TRM COILS.
 - 5 SIGNA MR// AND CV// SYSTEMS MANUFACTURED AFTER JUNE 15, 2000 HAVE MICROPHONE IN THE HEAD CARRIAGE INSTEAD OF THE REAR END BELL.
 - 6 OPTIONAL
 - 7 MR// AND CV// FIXED SITE SYSTEMS HAVE NESLAB WATER CHILLER. MOBILES HAVE E & W HEAT EXCHANGER.
 - 8 FOR TSCC AND TGWC SEE WATER CHILLER.

REFER TO THE FOLLOWING FUNCTIONAL BLOCK
DIAGRAM FOR ADDITIONAL INFORMATION:

- NOTE:
- 1) PATIENT HANDLING
 - 2) MGD/RRF/RF
 - 3) GRADIENT
 - 4) MAGNET AND CRYOGENS
 - 5) REFER TO SPECTROSCOPY MANUALS
 - 6) OXYGEN MONITOR SUBSYSTEM (DIR. 15336).



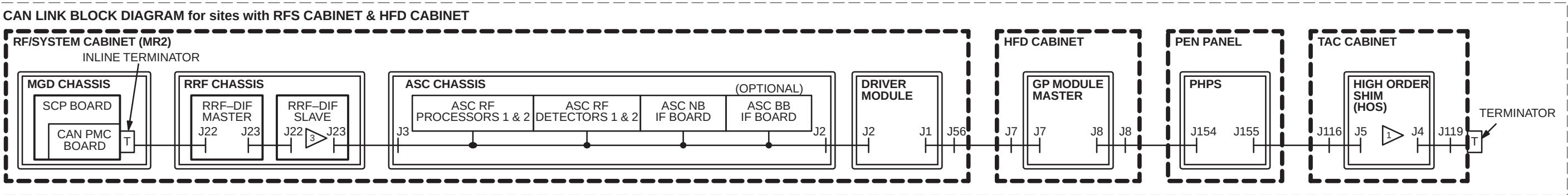
G3 HDx (14.x) & HDxt (15.x) OVERALL SYSTEM – 8 Channel or 16 Channel
(ONLY MAIN COMMUNICATION LINES SHOWN)
SHEET 4B OF 6



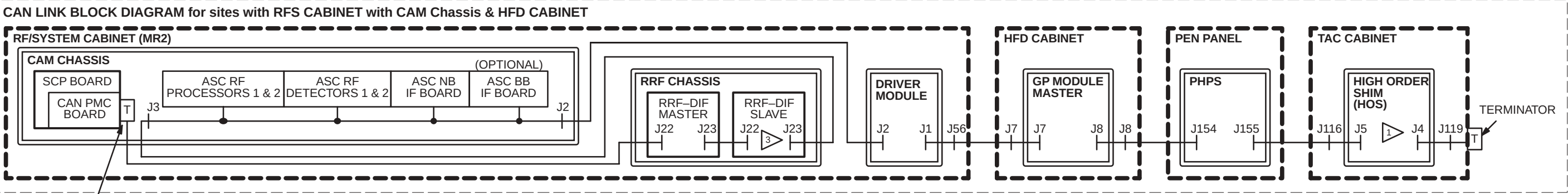
- NOTES:**
- 1 All power and signal lines entering and leaving the RF enclosure pass through the penetration panel. For ease of use, on this diagram, they are not shown going to the penetration panel.
 - 2 S/C Shim Power Supply and S/C Shim Coils.
 - 3 For Mobile systems, sensor is located in ceiling box.
 - 4 For TSCC and TGWC, see Water Chiller.

- Refer to the following functional block diagram for additional information:
- NOTE:**
- 1) PATIENT HANDLING
 - 2) MGD/RRF/RF
 - 3) GRADIENT
 - 4) MAGNET AND CRYOGENS
 - 5) REFER TO SPECTROSCOPY MANUALS
 - 6) OXYGEN MONITOR SUBSYSTEM (DIR. 15336).

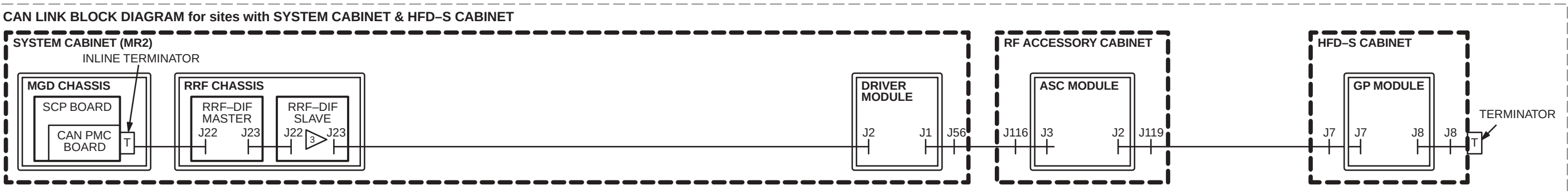




OR



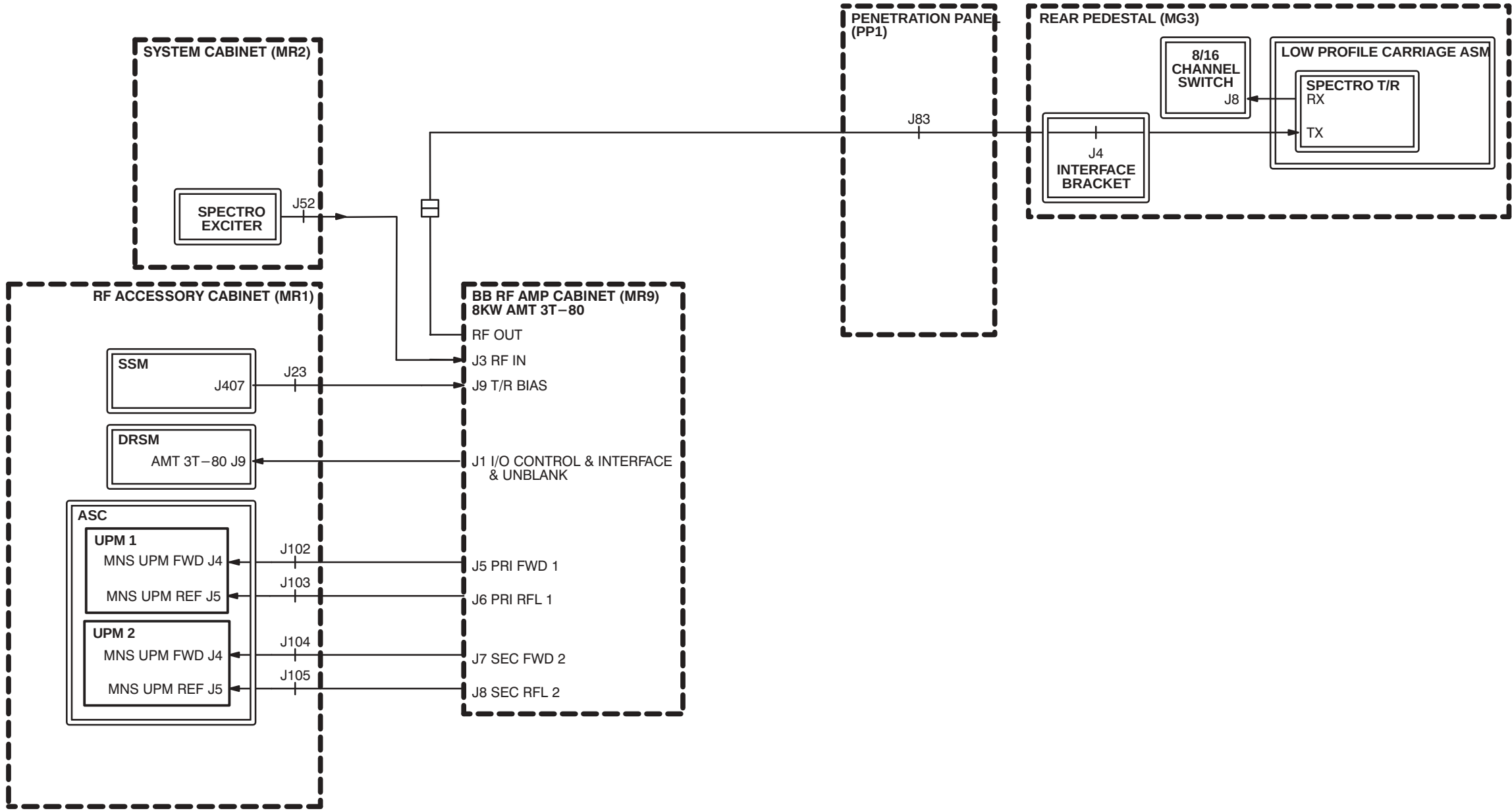
OR



- NOTES:
- 1

HOS is optional. Terminator must be connected to J155 on PHPS if HOS not present.
- 2
- 3

Only present in systems equipped with 16 or more Channels. If system is 8 Channels or less, then RF-DIF Master J23 connects to next unit in link as shown.

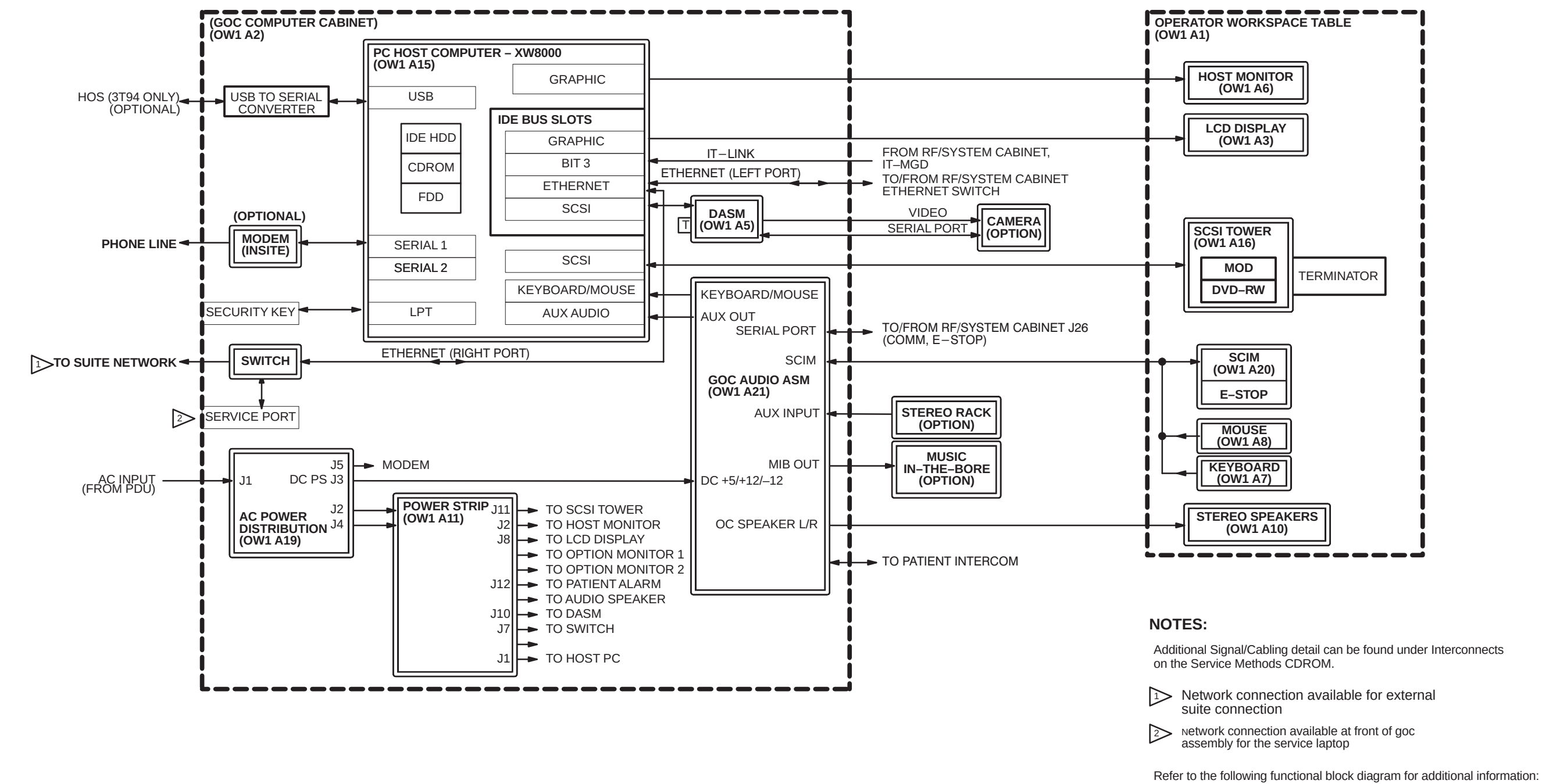


G3 EXCITE HD (12.x) and HDx (14.x) & HDxt (15.x) OVERALL SYSTEM – 8KW MNS OPTION
(ONLY MAIN COMMUNICATION LINES SHOWN)
SHEET 5 OF 6

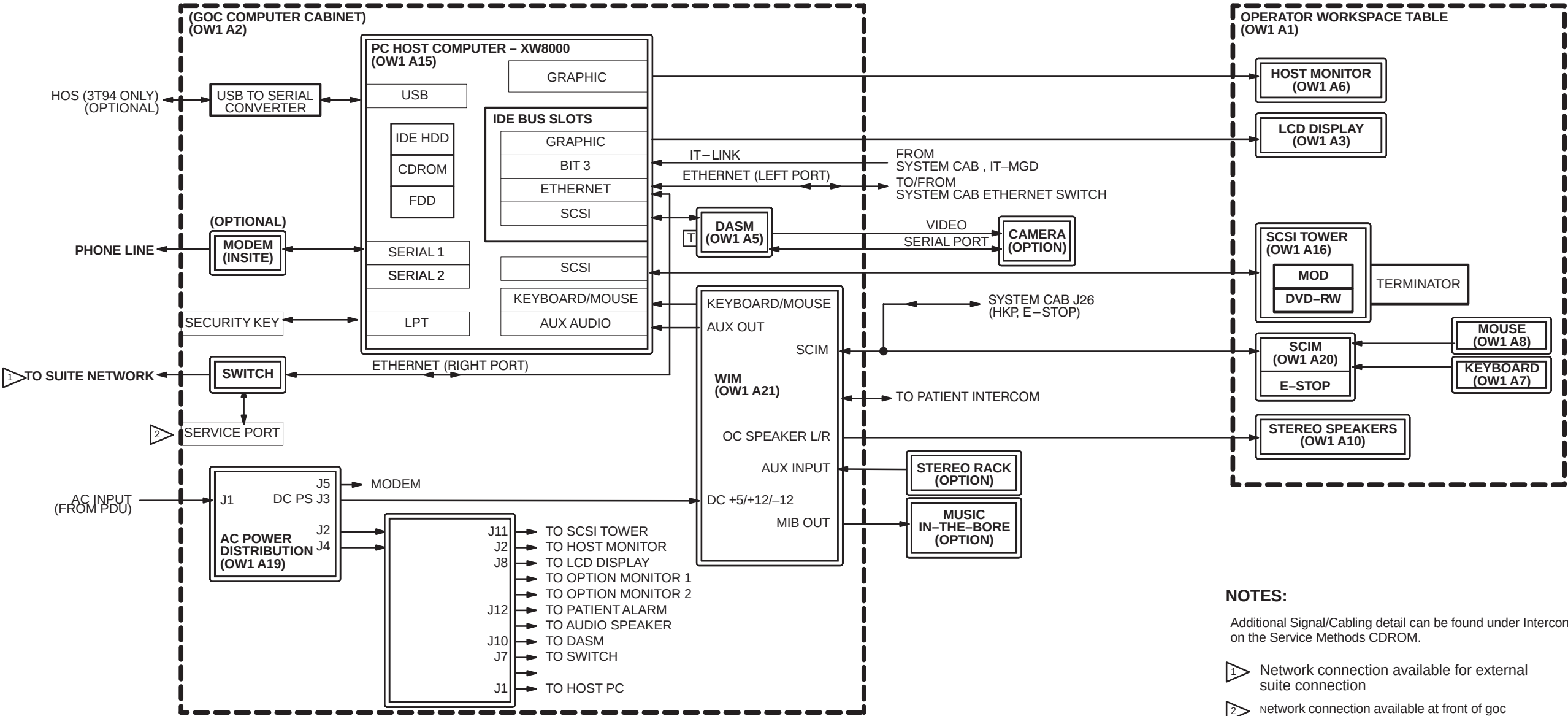
OPERATOR WORKSPACE

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G3 & 3T94 EXCITE HD (12.x) OVERALL SYSTEM
Forward Production or Pre-EXCITE Upgrades
(OPERATOR WORKSPACE CABINET w/8000 & GOC AA and TABLE)
SHEET 1A OF 5



NOTES:

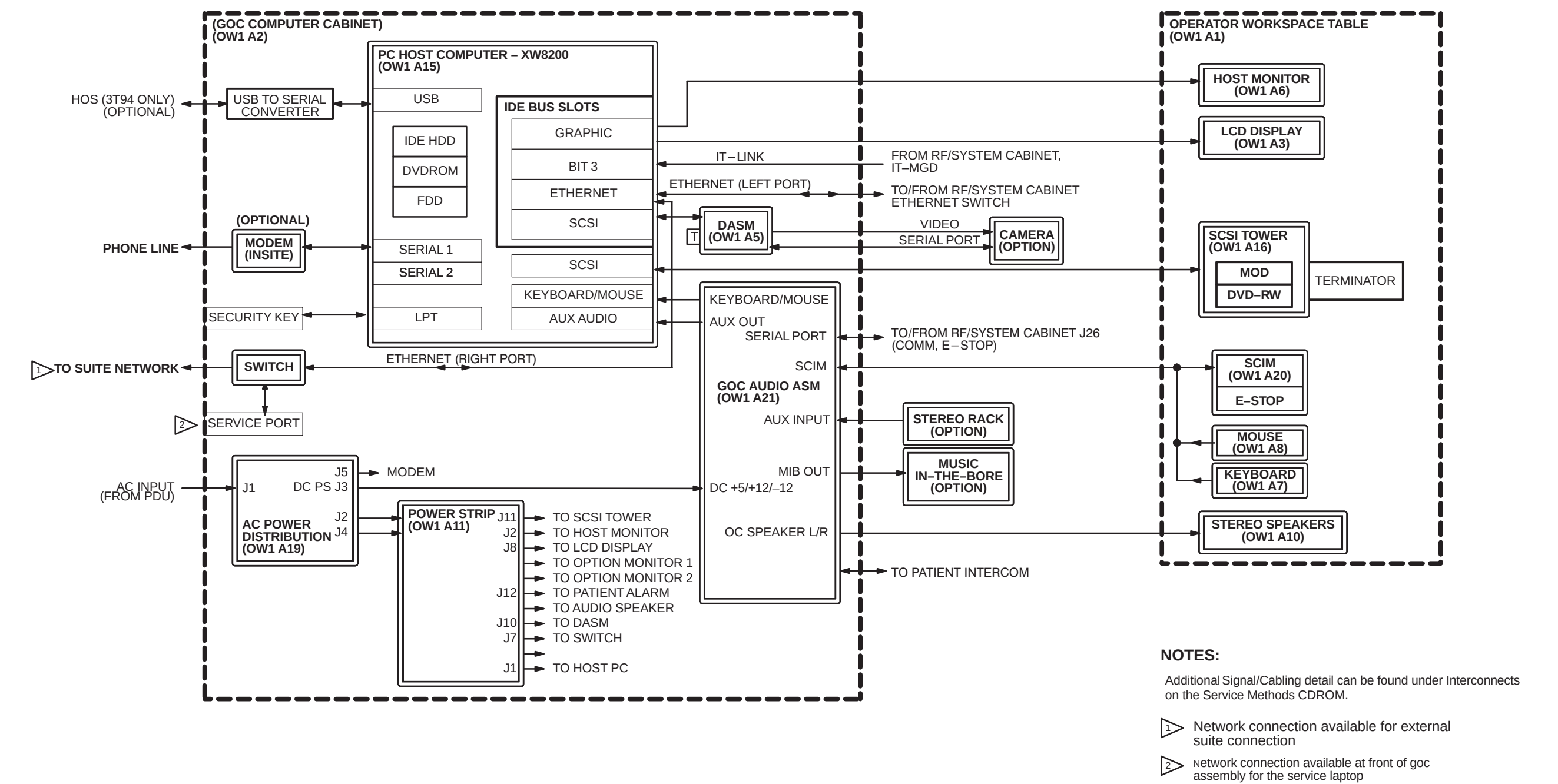
Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

1 Network connection available for external suite connection

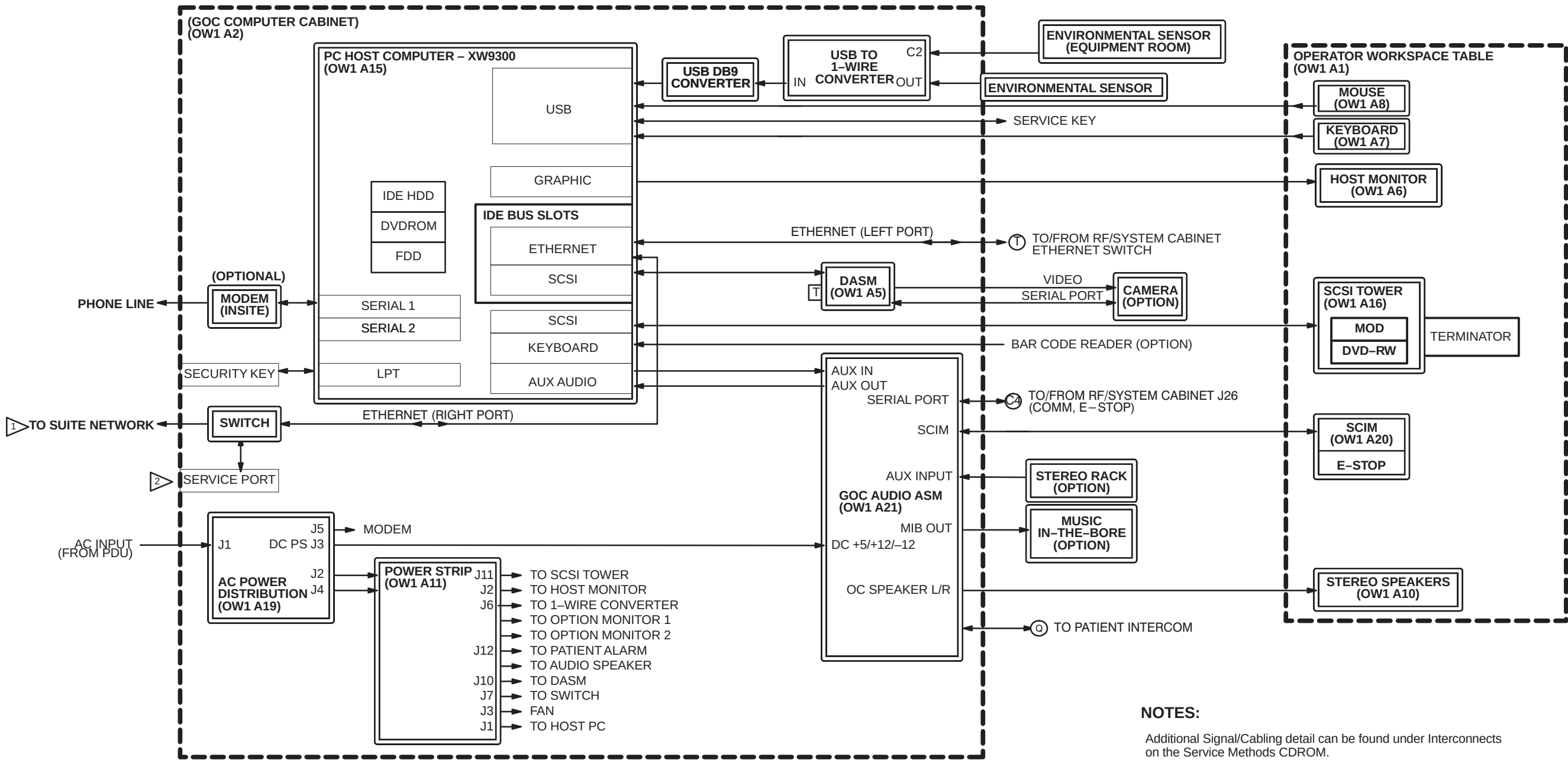
2 network connection available at front of goc assembly for the service laptop

Refer to the following functional block diagram for additional information:

NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF

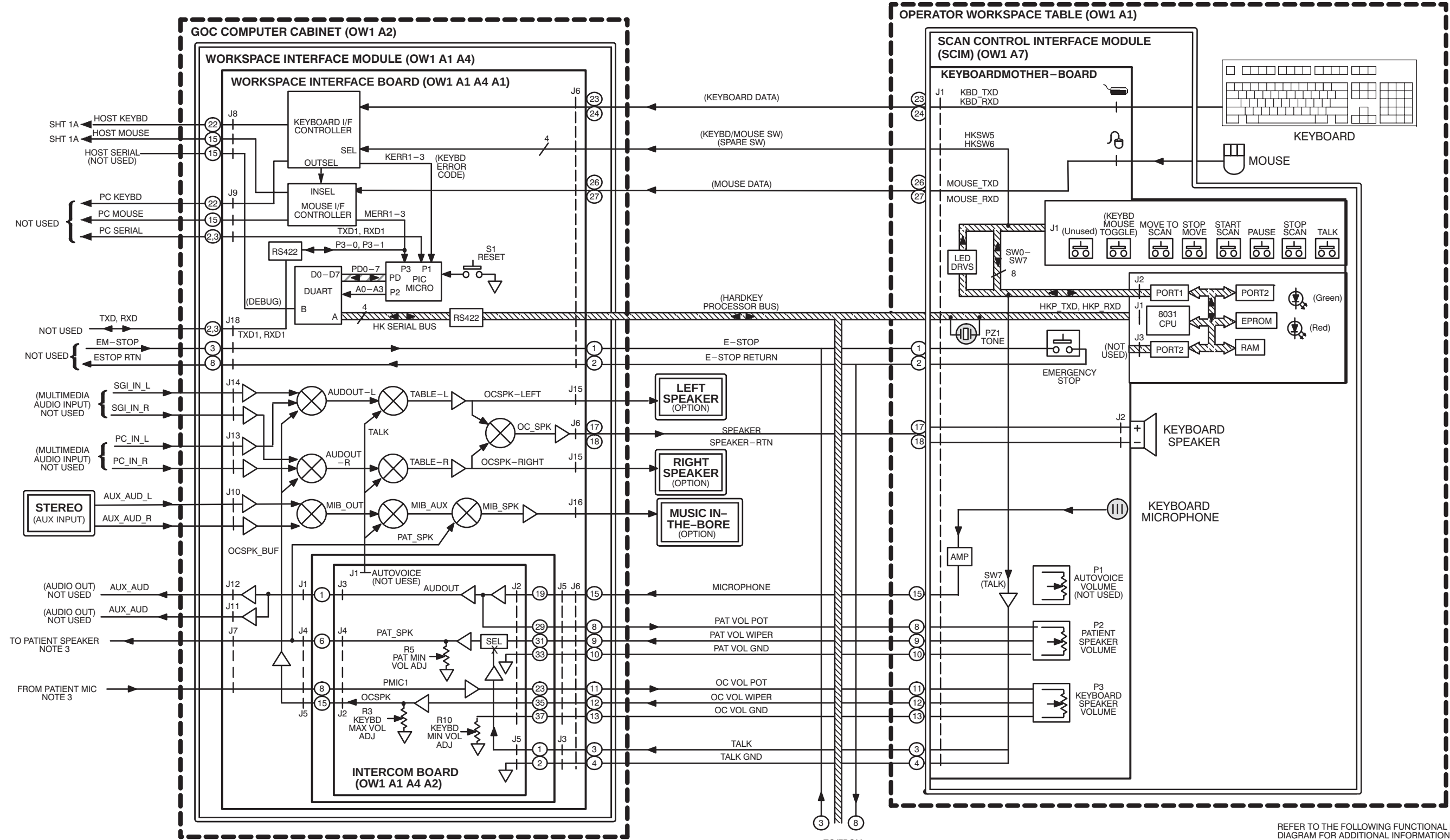


G3 & 3T94 EXCITE HD (12.x) OVERALL SYSTEM
Forward Production or Pre-EXCITE Upgrades
(OPERATOR WORKSPACE CABINET w/8200 & GOC AA and TABLE)
SHEET 1C OF 5



G3 & 3T94 HDx (14.x) & HDxt (15.x) OVERALL SYSTEM
Forward Production, EXCITE, or Pre-EXCITE to HDx Upgrades
(OPERATOR WORKSPACE CABINET w/9300 & GOC AA and TABLE)

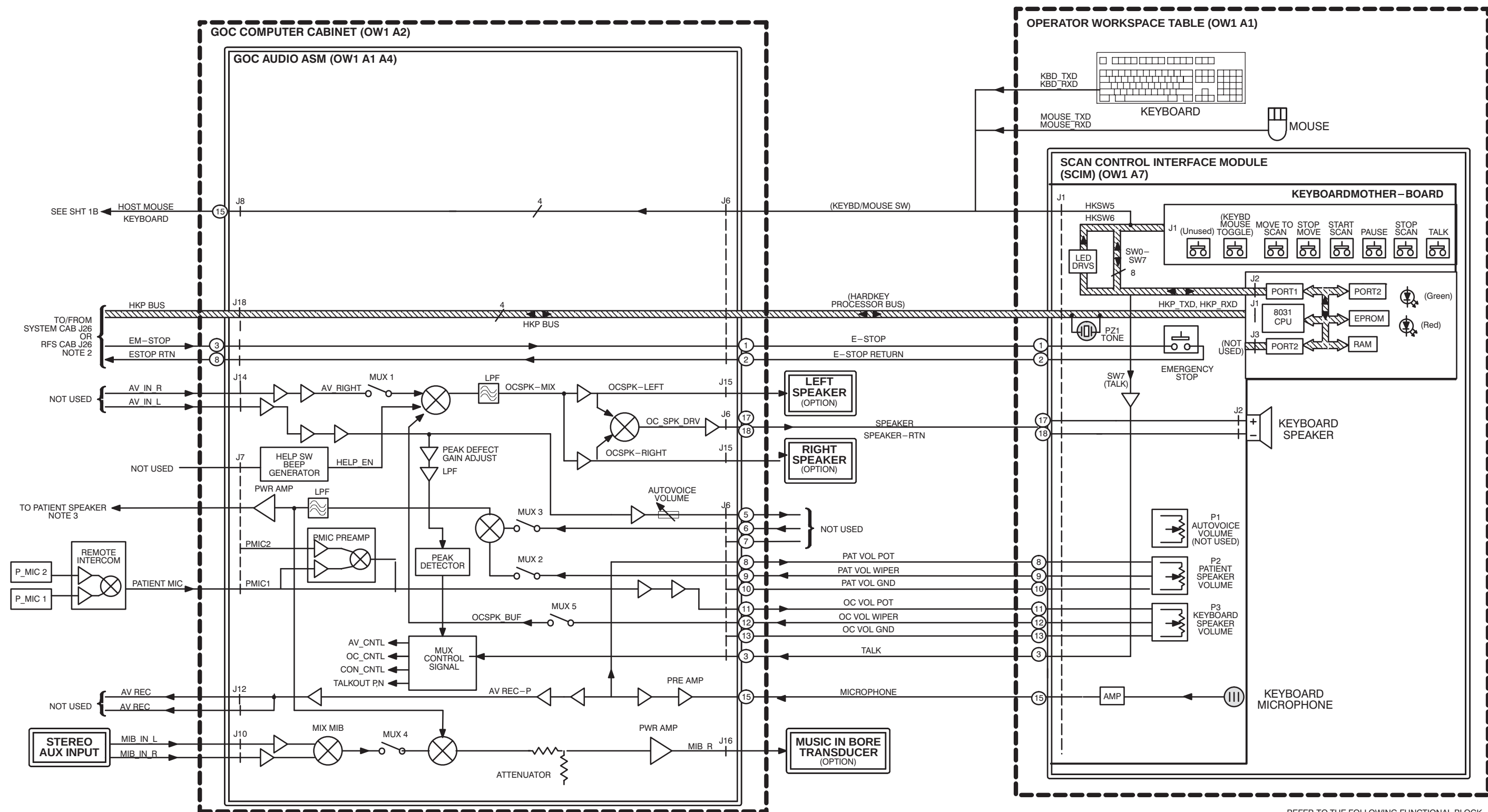
SHEET 1D OF 6



G3 & 3T94 EXCITE HD (12.x) OVERALL SYSTEM
(OPERATOR WORKSPACE CABINET/TABLE with WIM)
SHEET 2A OF 2

REFER TO THE FOLLOWING FUNCTIONAL BLOCK
DIAGRAM FOR ADDITIONAL INFORMATION:

NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF
3) OVERALL SYSTEM

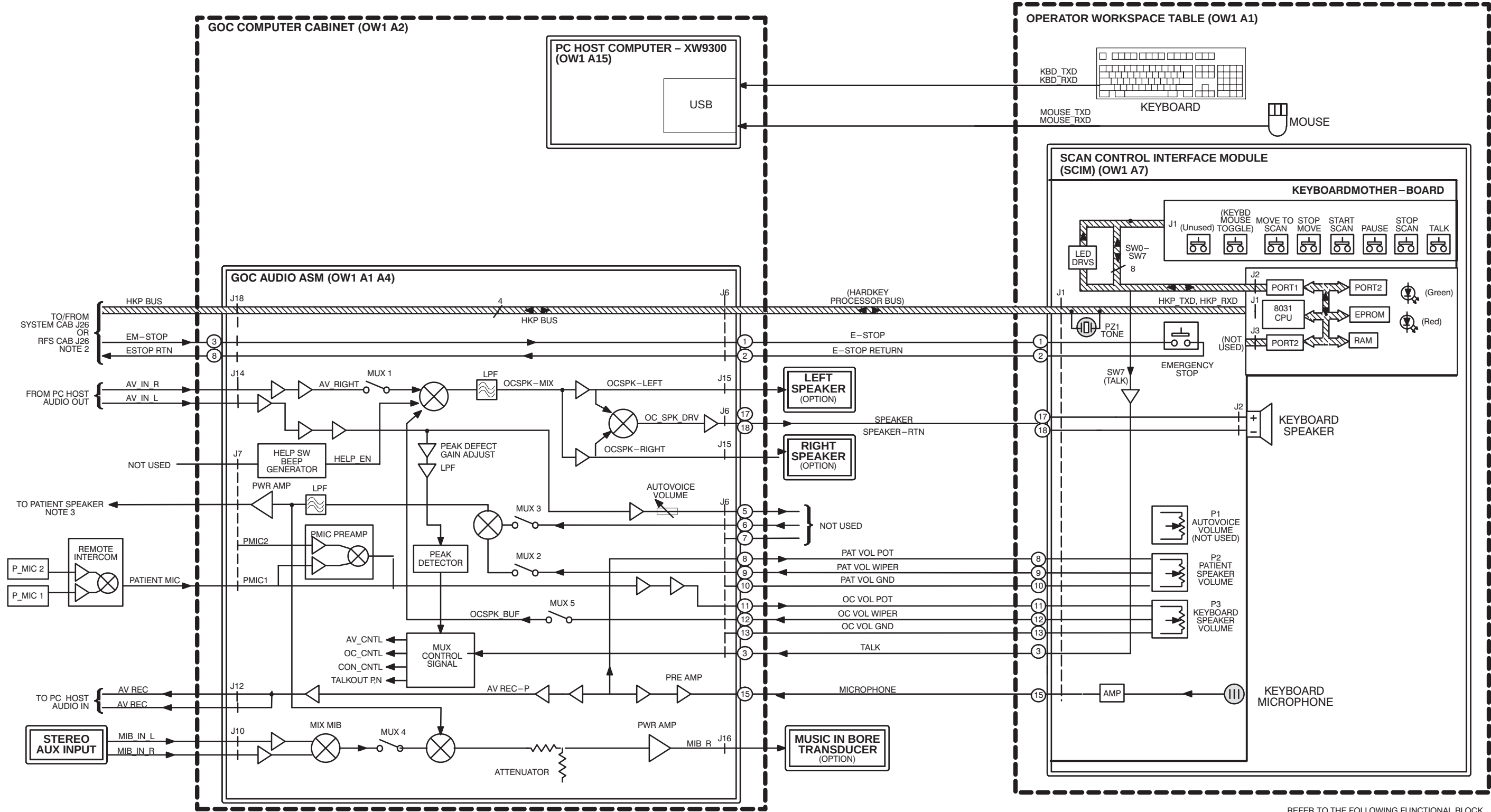


REFER TO THE FOLLOWING FUNCTIONAL BLOCK
DIAGRAM FOR ADDITIONAL INFORMATION:

NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF
3) OVERALL SYSTEM

G3 & 3T94 EXCITE HD (12.x) OVERALL SYSTEM

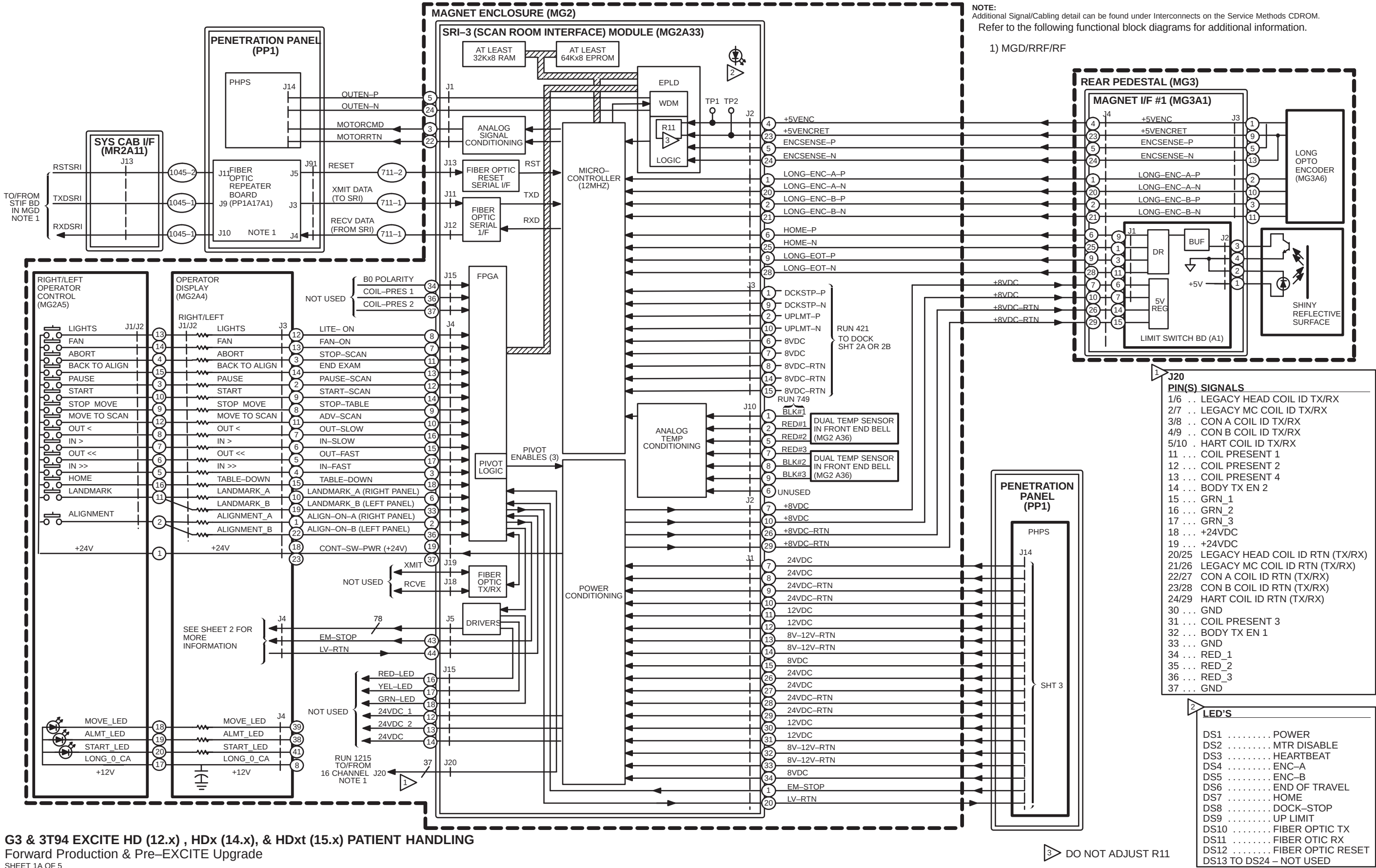
(OPERATOR WORKSPACE CABINET/TABLE with GOC AUDIO ASM)

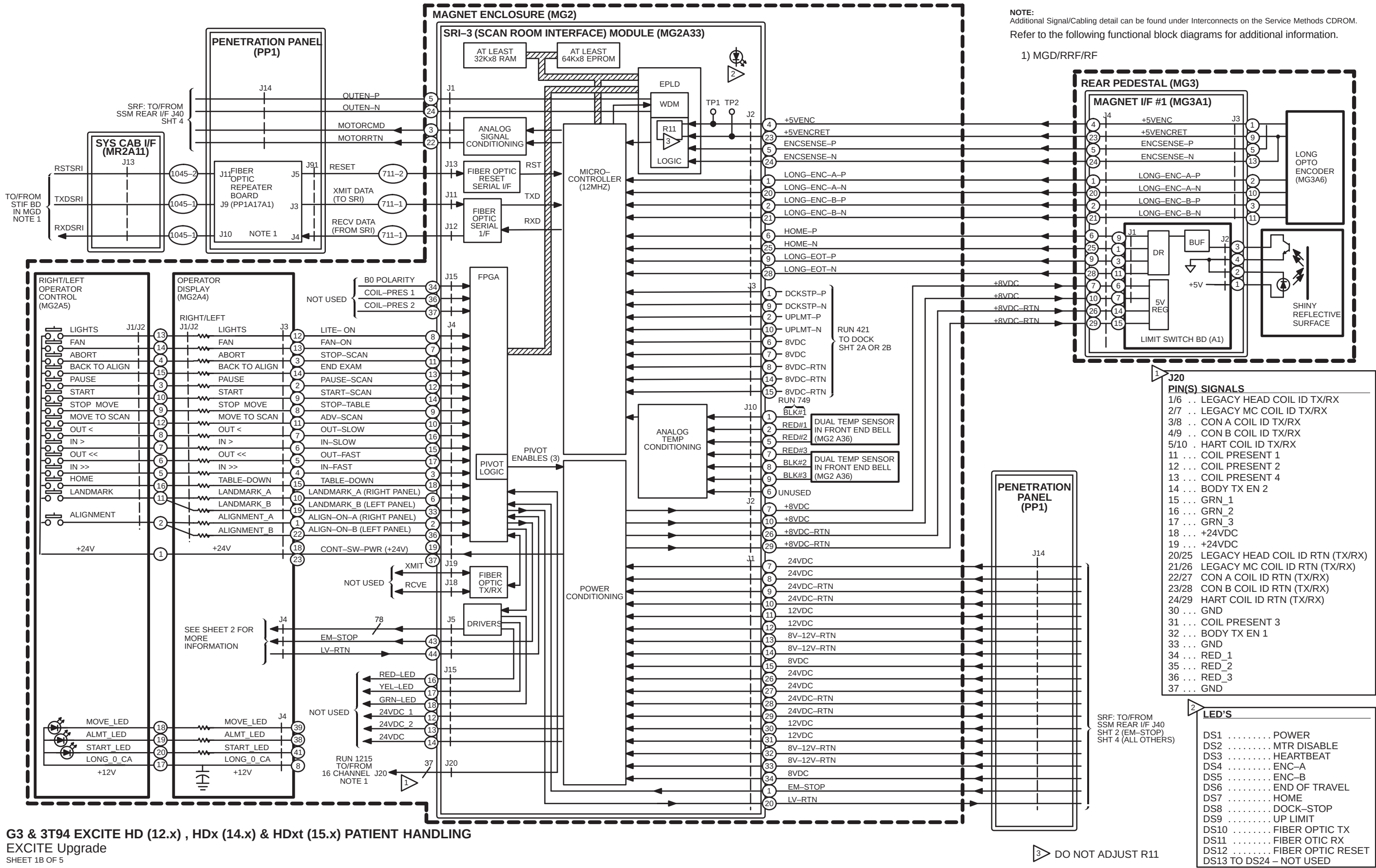


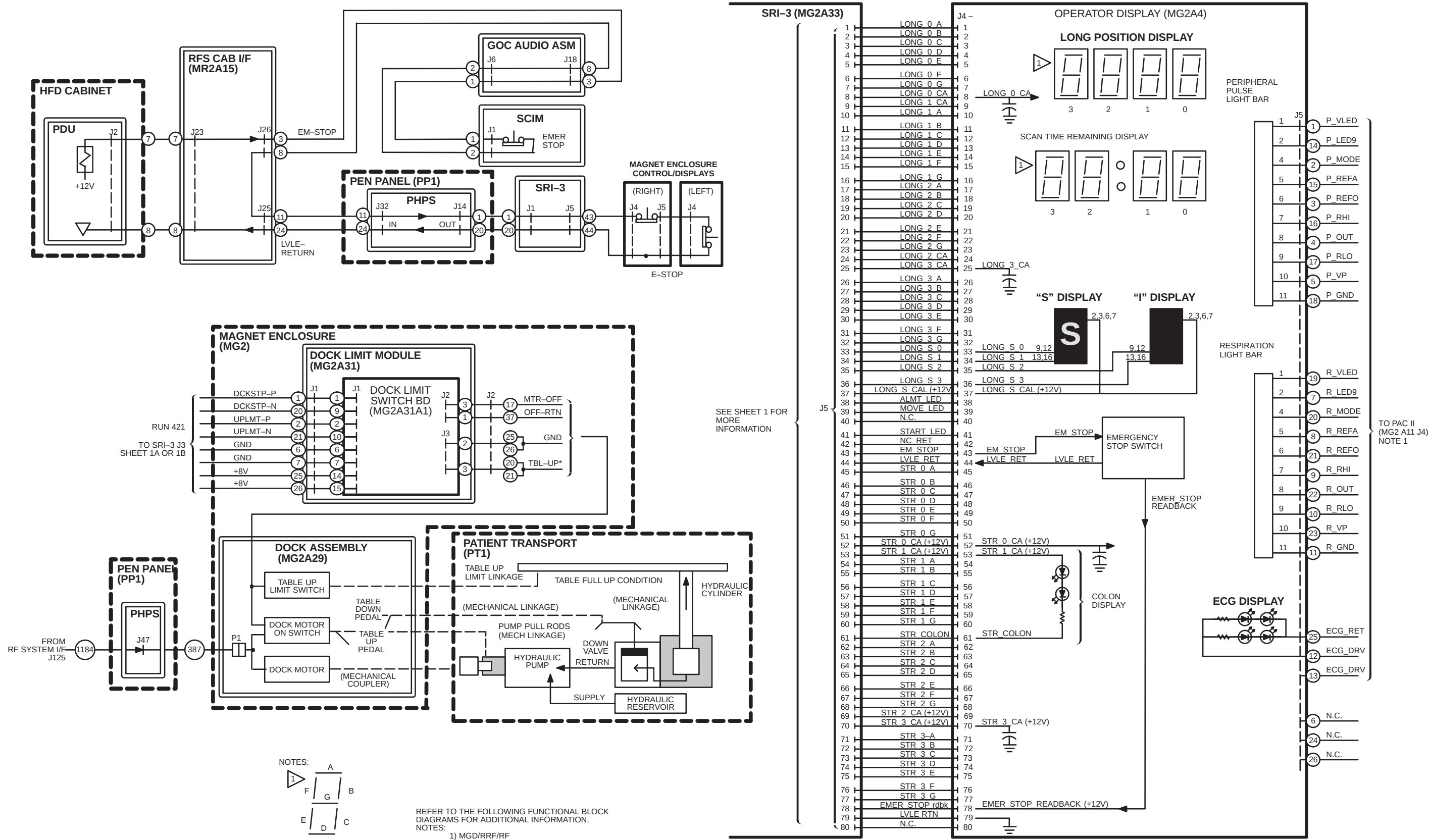
PATIENT HANDLING

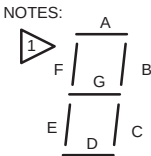
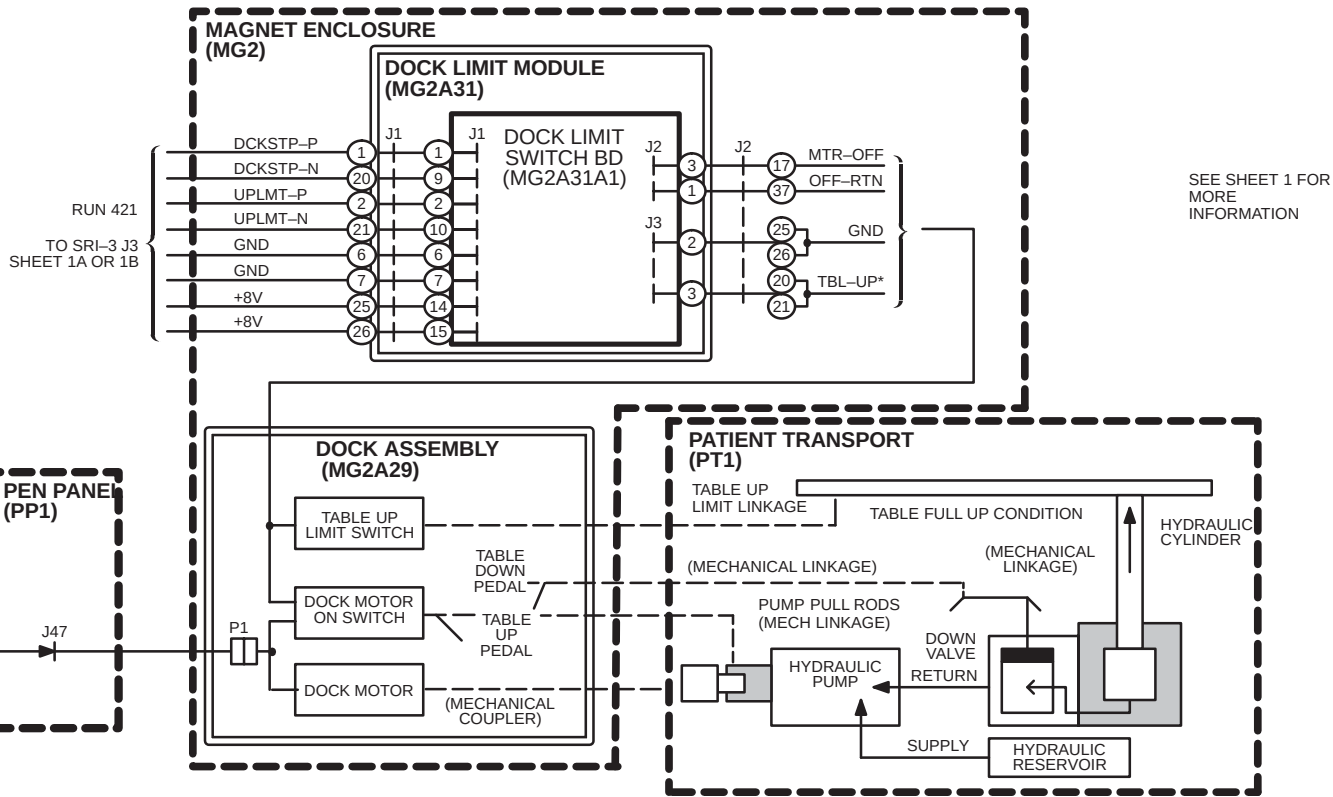
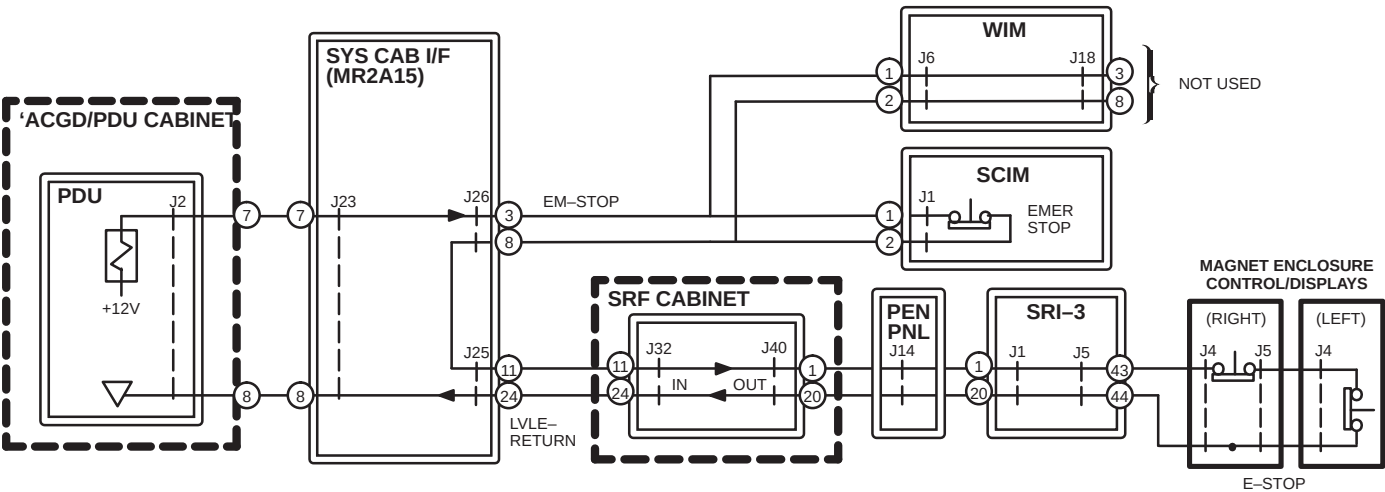
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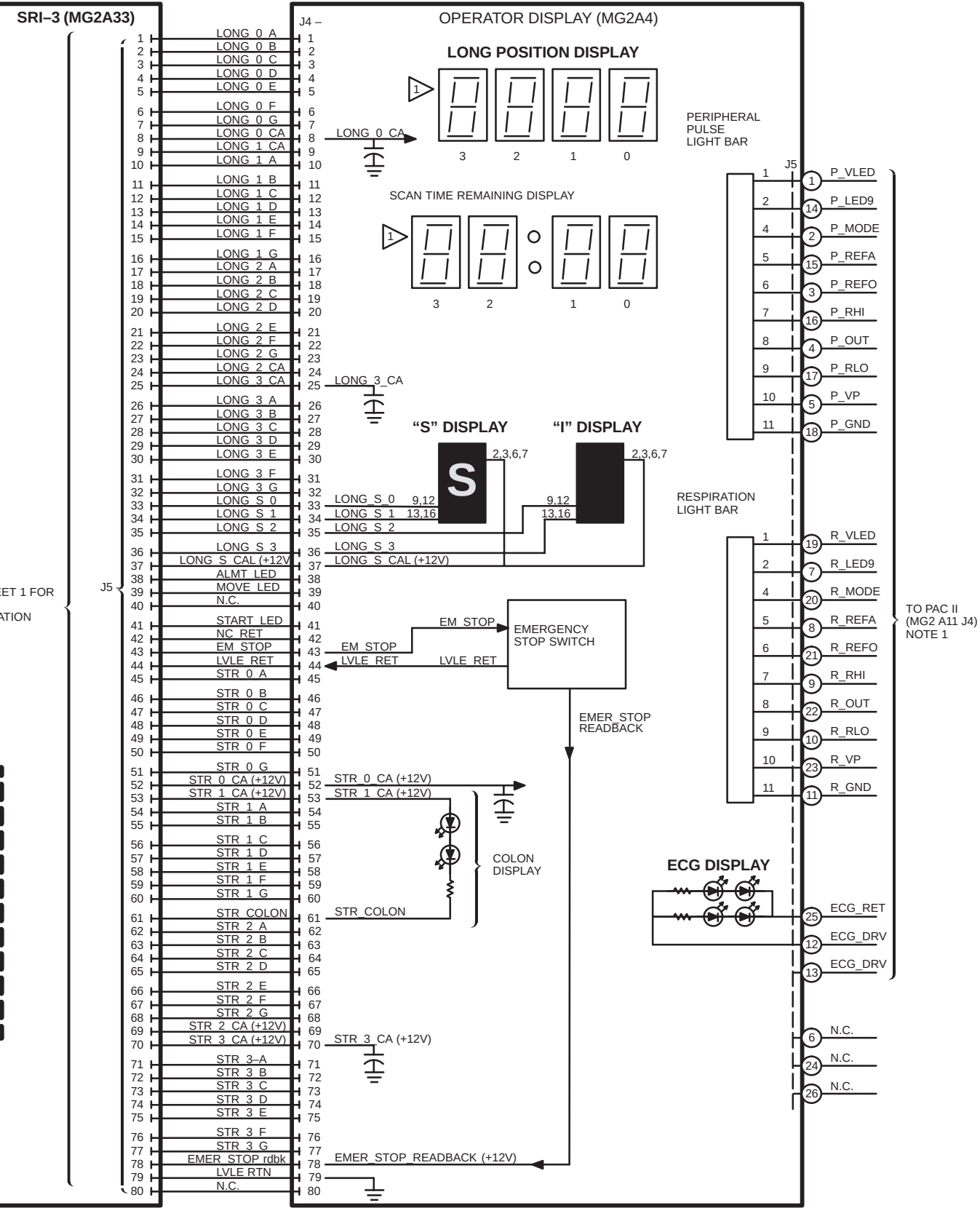


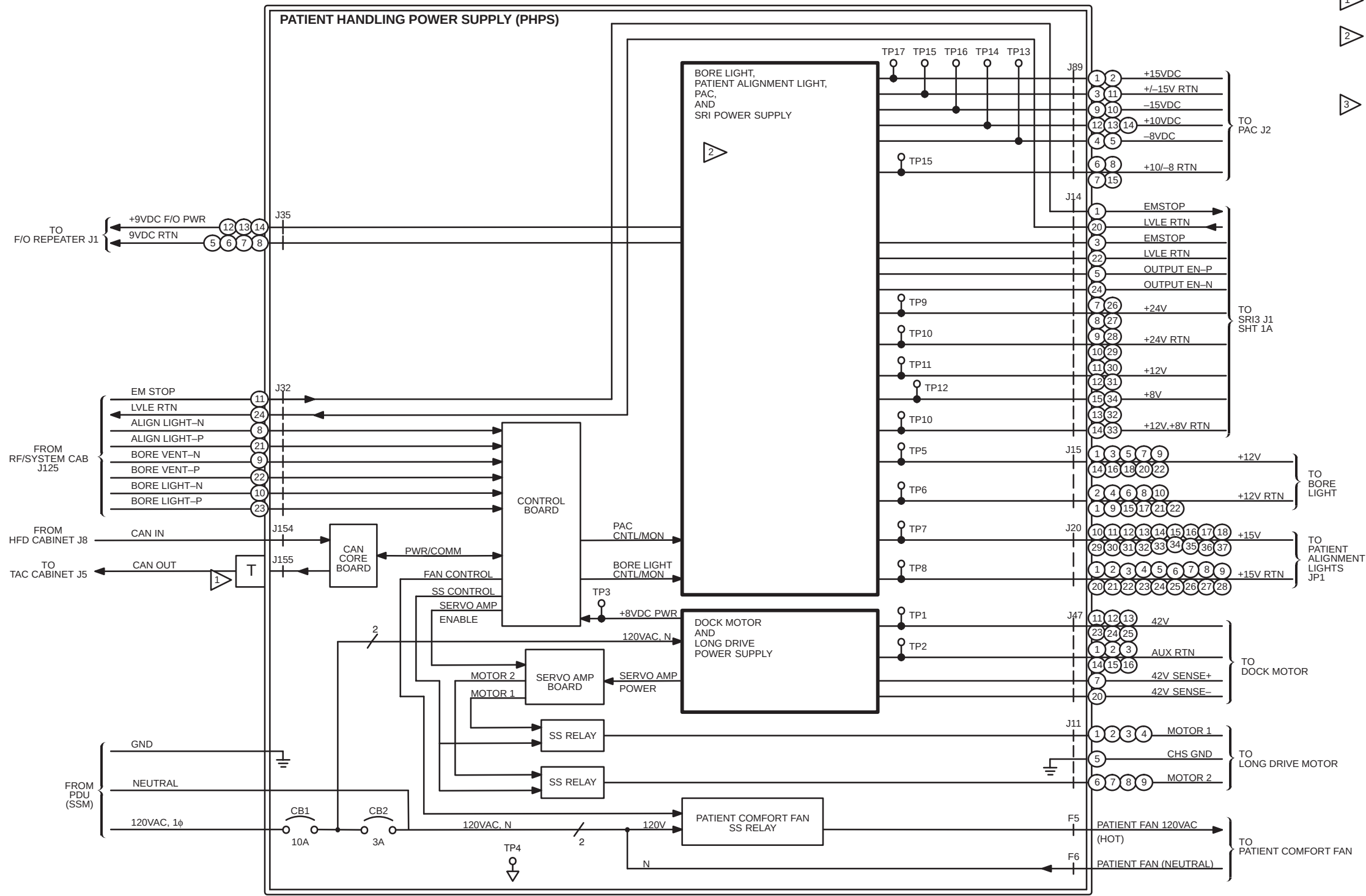


REFER TO THE FOLLOWING FUNCTIONAL BLOCK DIAGRAMS FOR ADDITIONAL INFORMATION.

NOTES:

1) MGD/RRF/RF

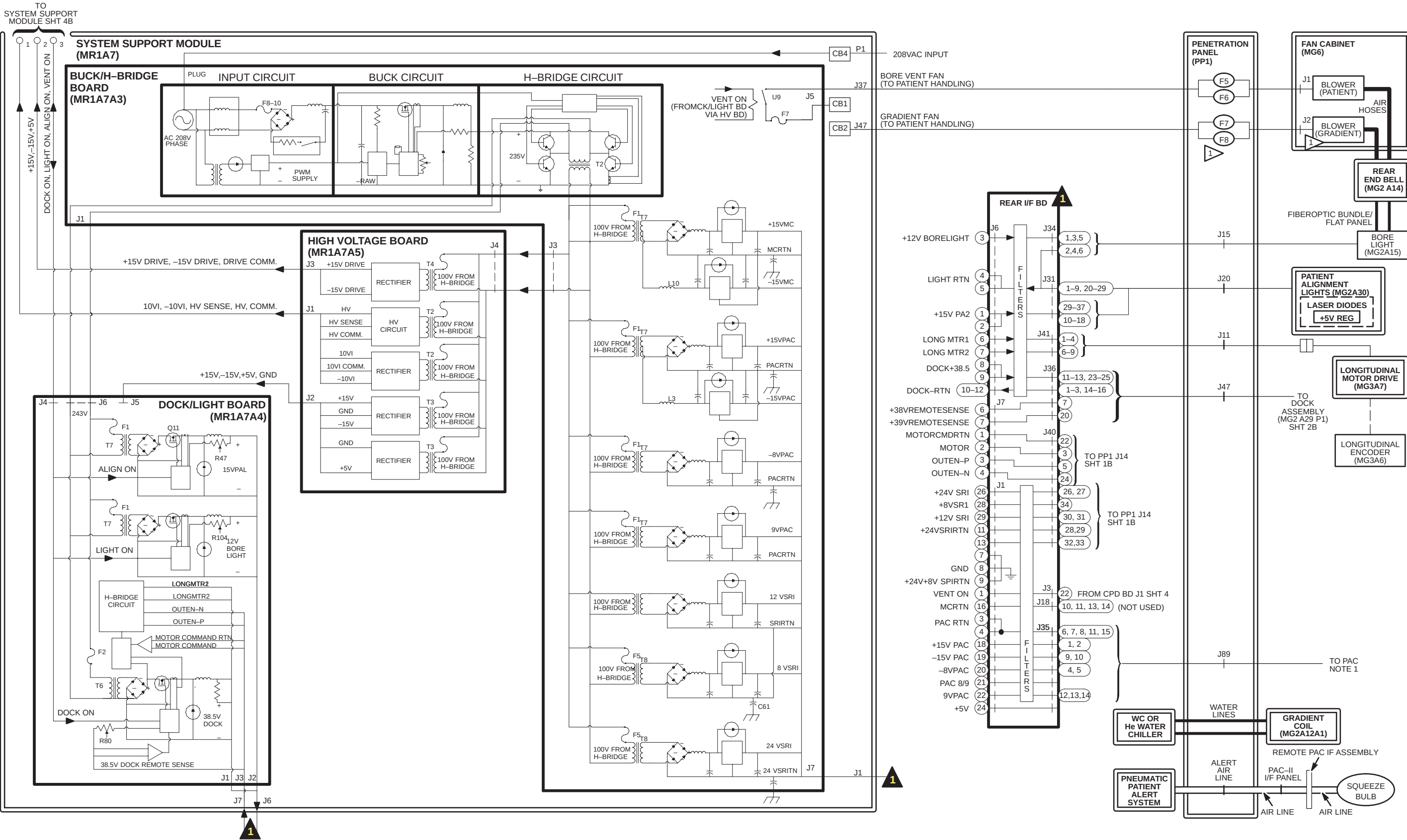




NOTES:

- 1 OPTIONAL: Used only if High Order Shim (HOS) installed. PP1 PHPS J155 Terminator removed when HOS is installed.
- 2 Systems equipped with Laser Patient Alignment lights use one type of power supply while those equipped with Halogen patient Alignment lights must use another. See the Renewal Parts manual.
- 3

TEST POINT	FUNCTION
TP1	TABLE VOLTS (42VDC)
TP2	TABLE COM
TP3	CONTROL BD +8V
TP4	+8V COM
TP5	BORE +12V
TP6	BORE COM
TP7	PAL +15V
TP8	PAL COM
TP9	SRI +24V
TP10	SRI COM
TP11	SRI +12V
TP12	SRI +8V
TP13	PAC -8V
TP14	PAC +10V
TP15	PAC COM
TP16	PAC -15V
TP17	PAC +15V

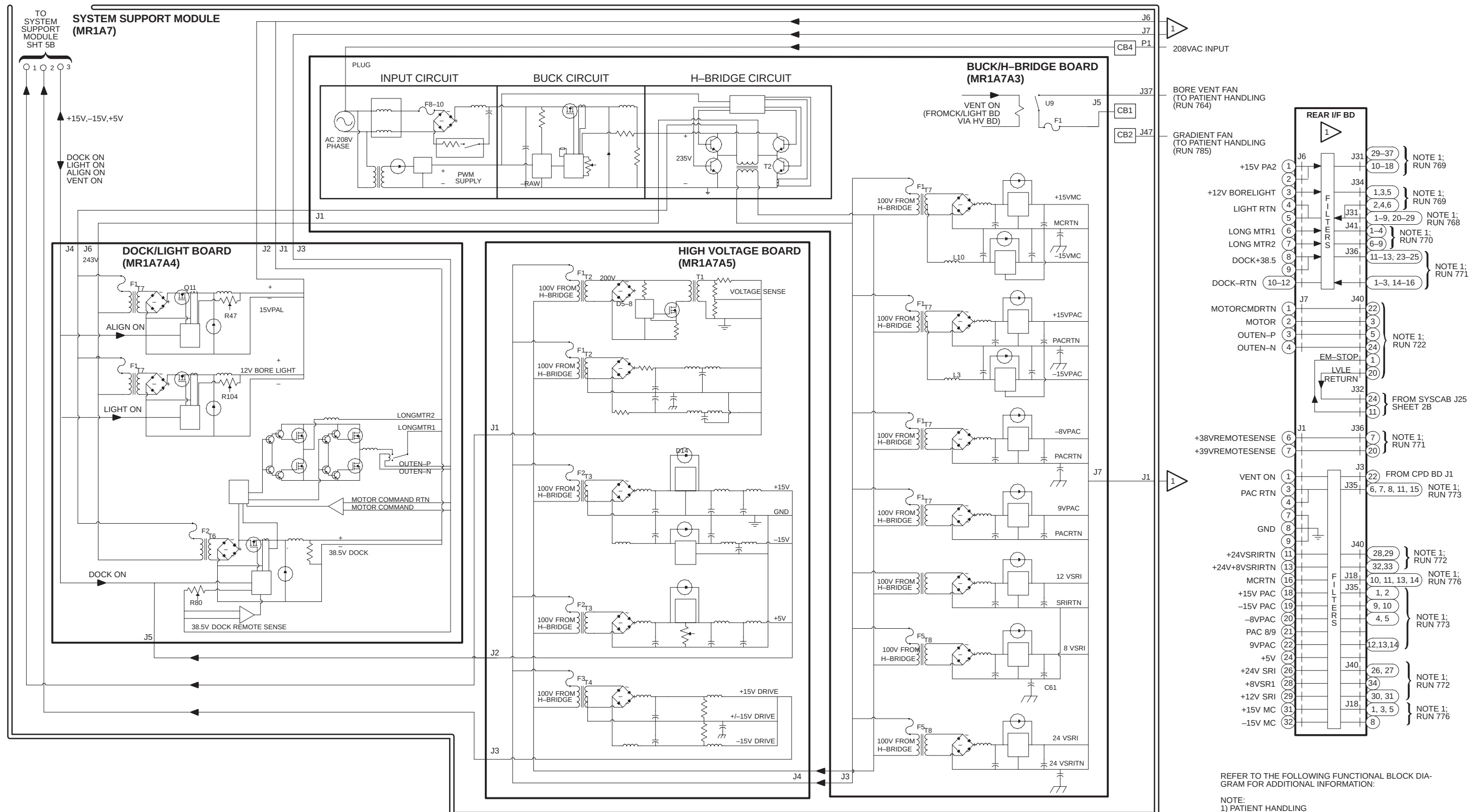


G3 EXCITE HD (12.x), HDx (14.x) & HDxt (15.x) PATIENT HANDLING – SSM
EXCITE Upgrade
SHEET 4A OF 5

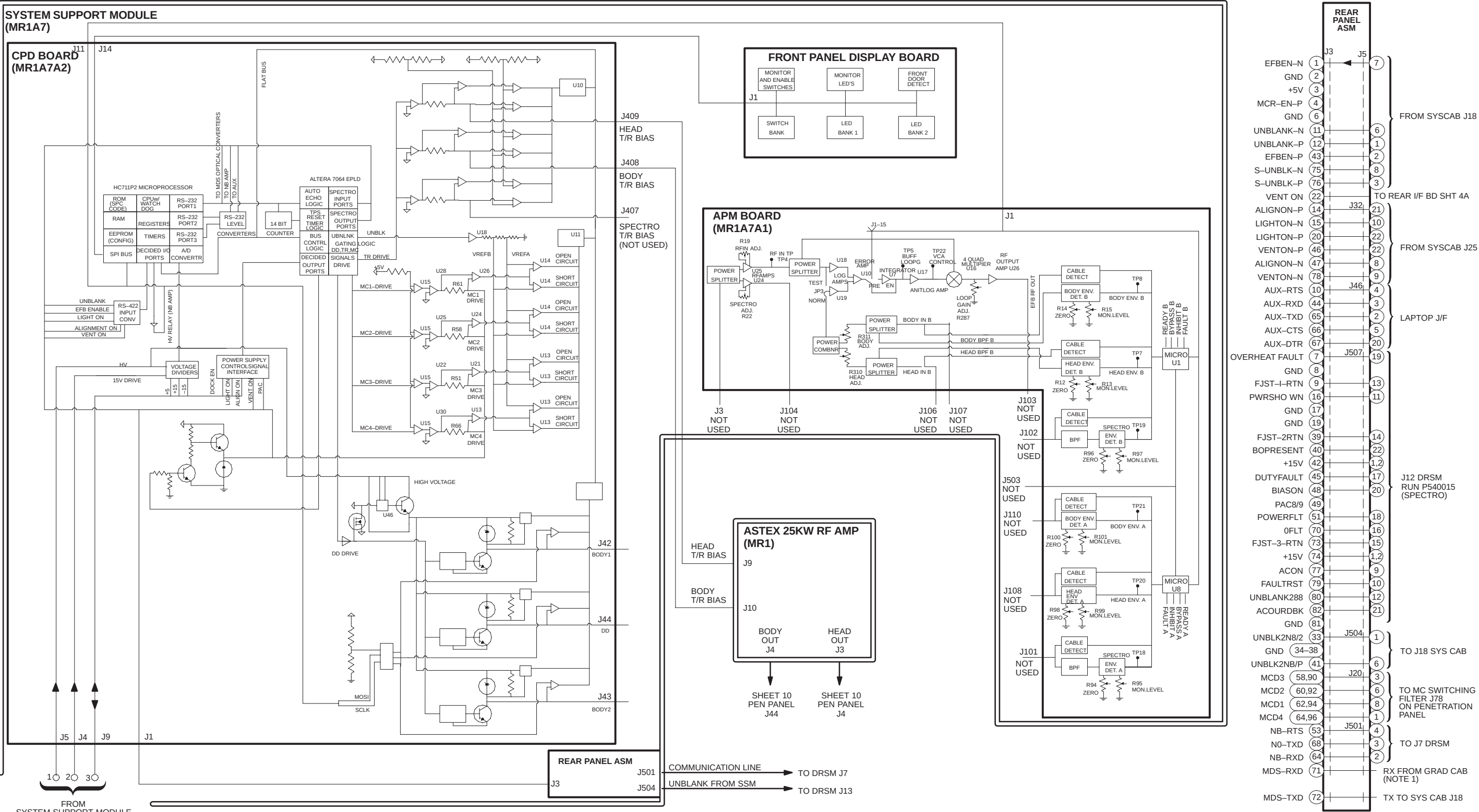
NOTE:
1 F7/F8 AND GRADIENT BLOWER PRESENT
ON SYSTEMS WITH BRM BODY COIL.

REFER TO THE FOLLOWING FUNCTIONAL BLOCK
DIAGRAM FOR ADDITIONAL INFORMATION:
NOTE:
1) MGD/RRF/RF





3T94 EXCITE HD (12.x), HDx (14.x) & HDxt (15.x) PATIENT HANDLING – SSM
EXCITE Upgrade
SHEET 5A OF 5



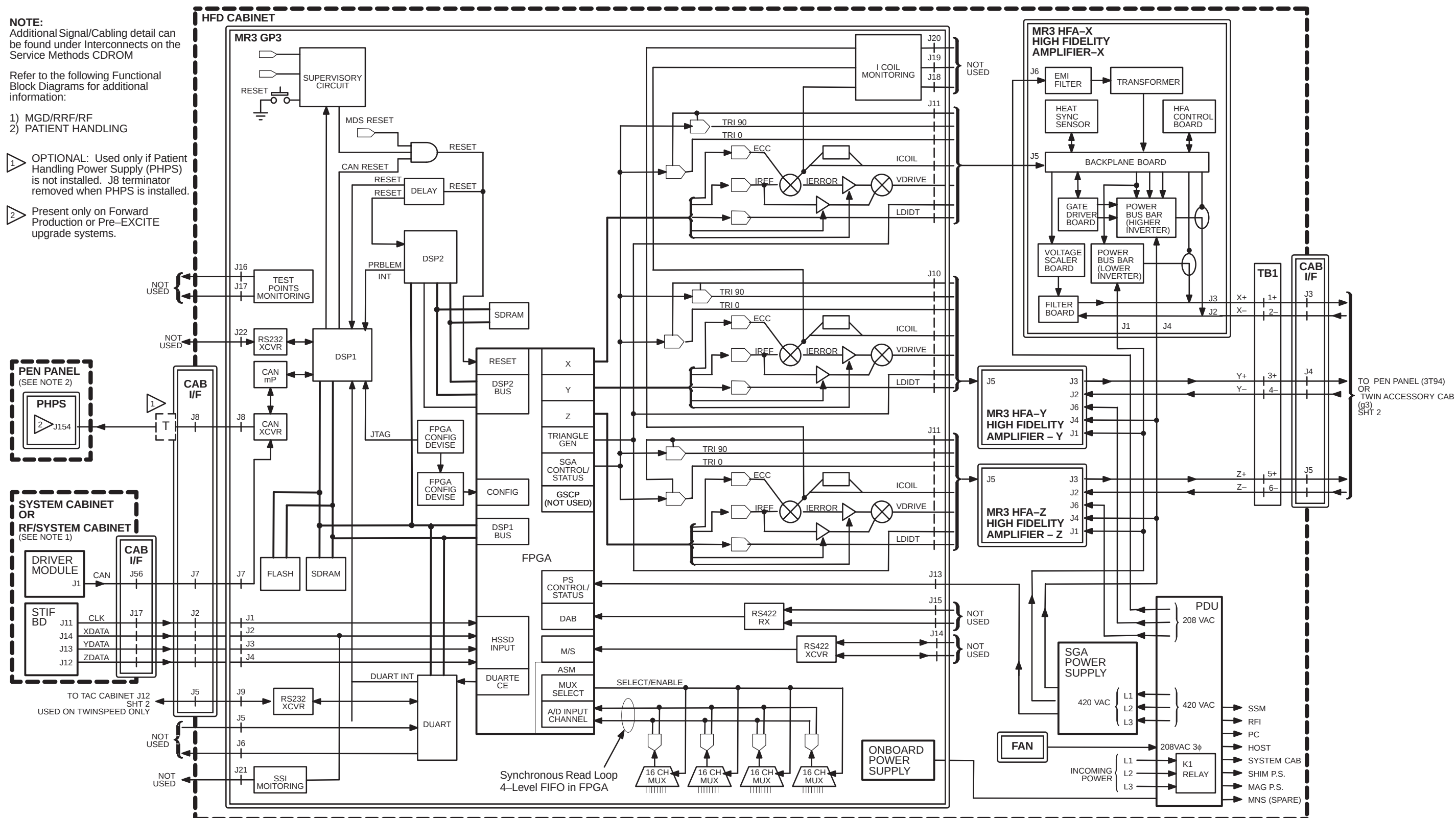
Refer to the following Functional Block Diagrams for additional information:

Refer to the following Functional Block Diagrams for additional information:

- 1) MGD/RRF/RF
- 2) PATIENT HANDLING

1 **OPTIONAL:** Used only if Patient Handling Power Supply (PHPS) is not installed. J8 terminator removed when PHPS is installed.

2 Present only on Forward Production or Pre-EXCITE upgrade systems.



G3 & 3T94 EXCITE HD (12.x) HDx (14.x) & HDxT (15.x) GRADIENT – HFD

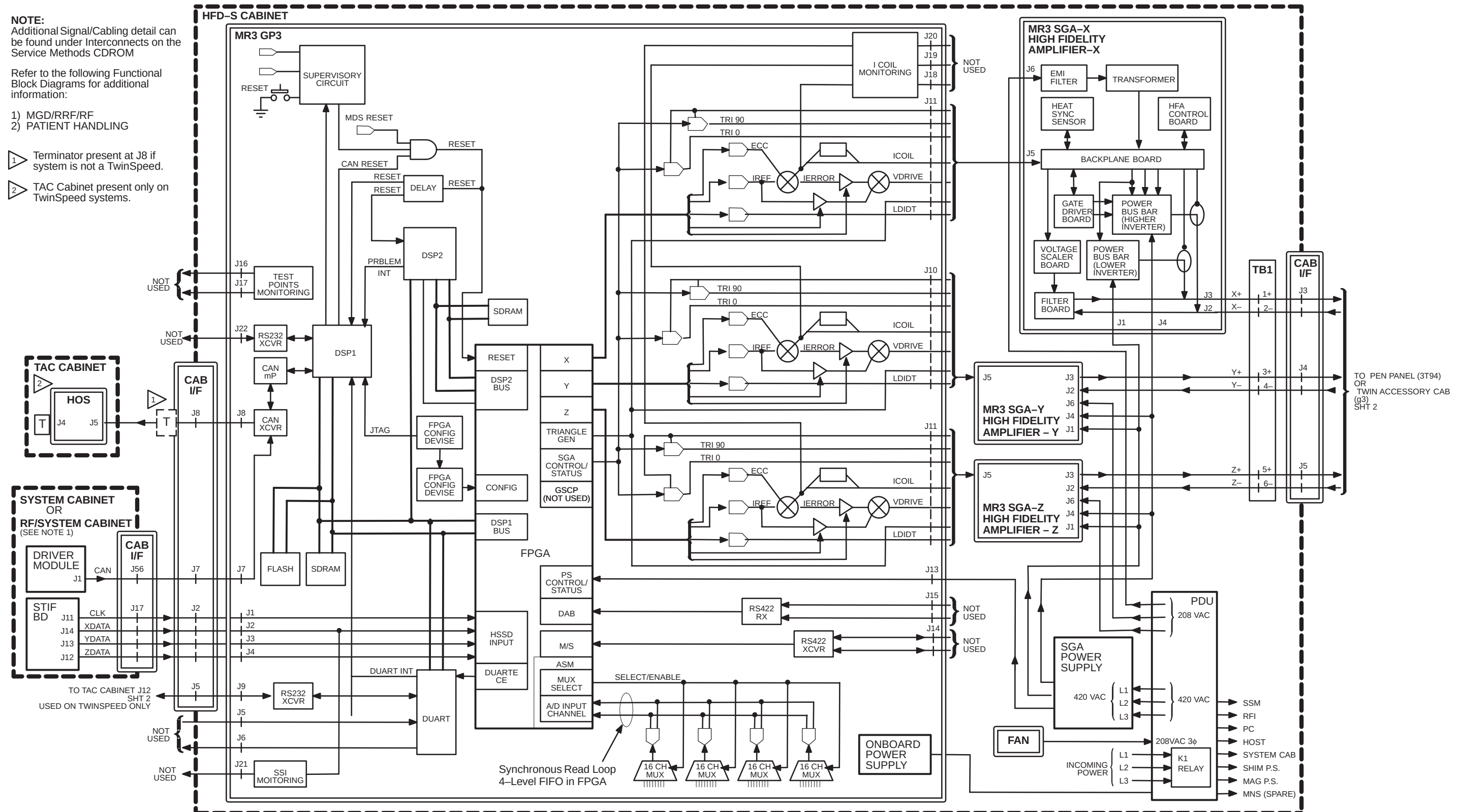
SHEET 1A OF 2

NOTE:
Additional Signal/Cabling detail can
be found under Interconnects on the
Service Methods CDROM

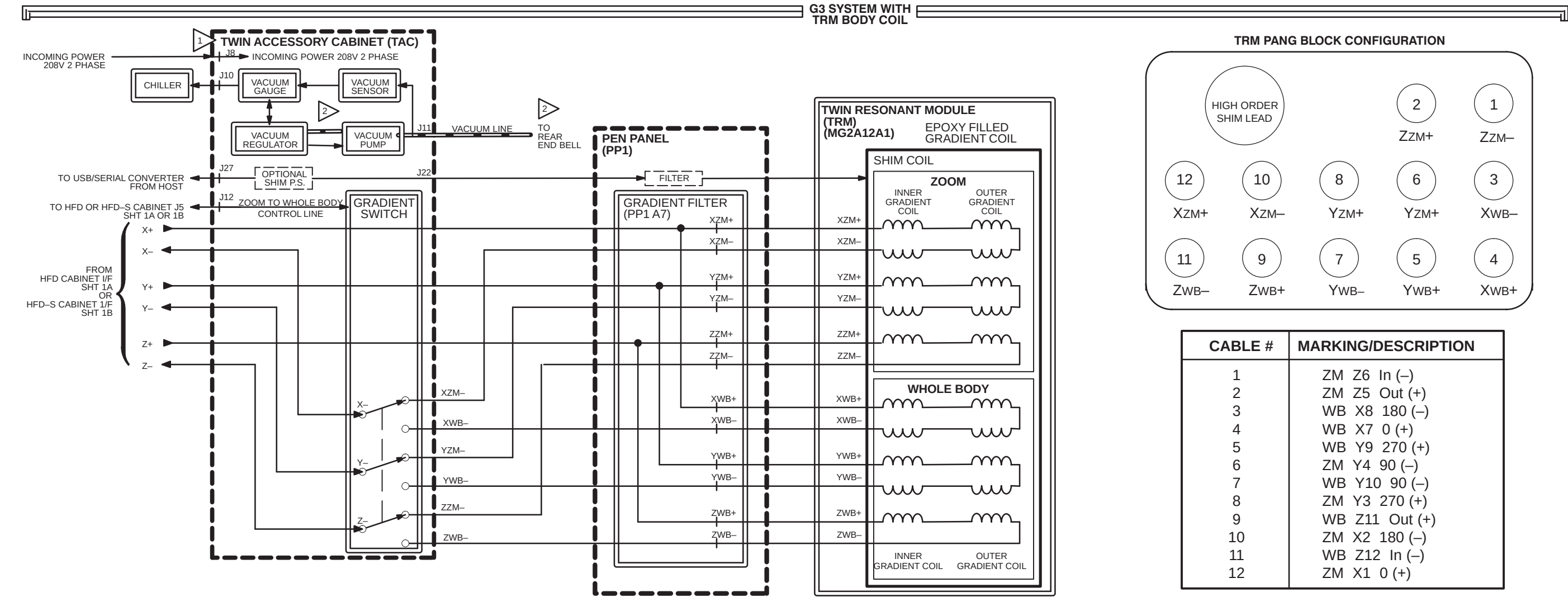
Refer to the following Functional Block Diagrams for additional information:

- 1) MGD/RRF/RF
- 2) PATIENT HANDLING

- 1 Terminator present at J8 if system is not a TwinSpeed.
- 2 TAC Cabinet present only on TwinSpeed systems.



G3 & 3T94 EXCITE HD (12.x), HDx (14.x) & HDxT (15.x) GRADIENT – HFD-S (Upgraded ACGD)



MGD/RRF/RF

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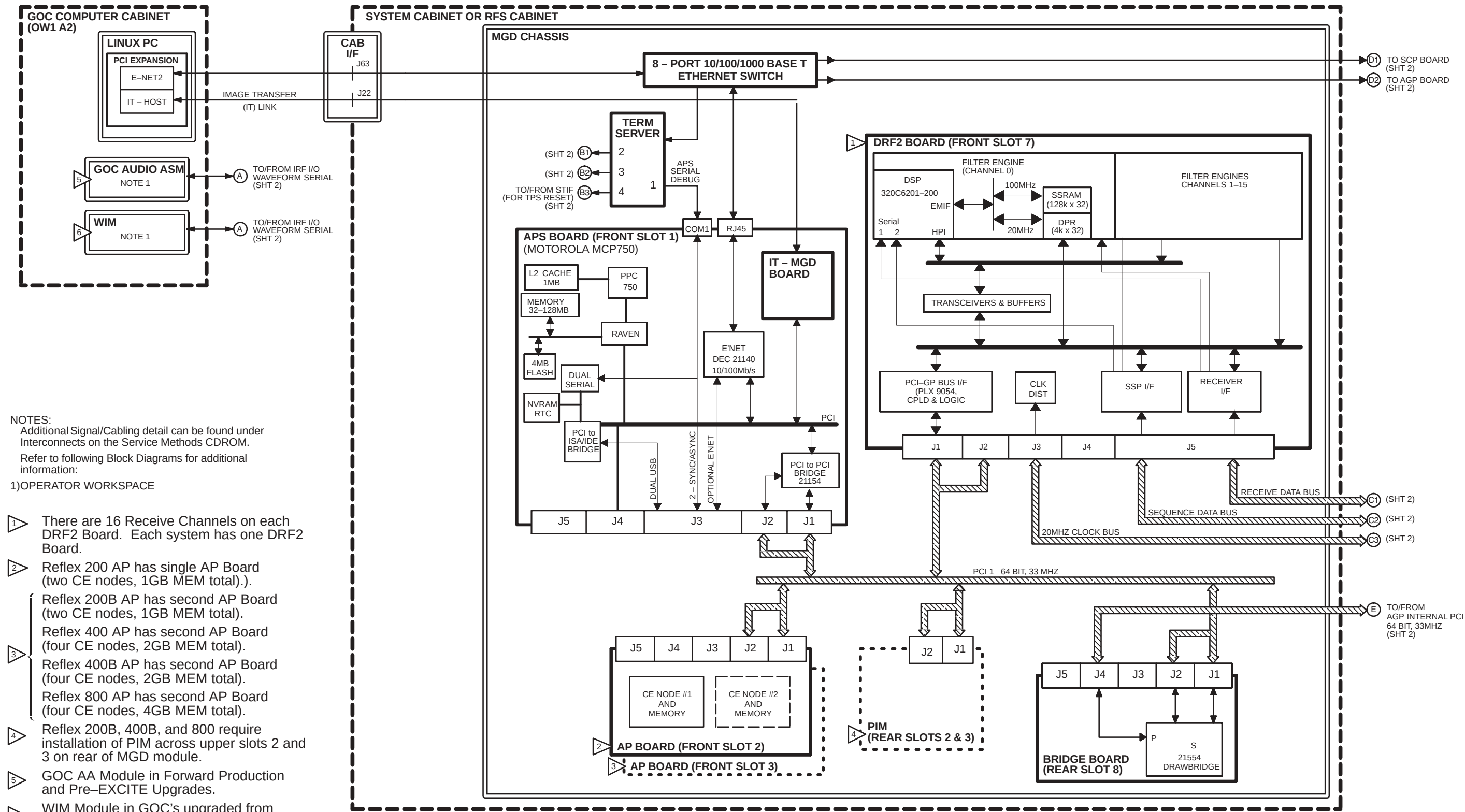
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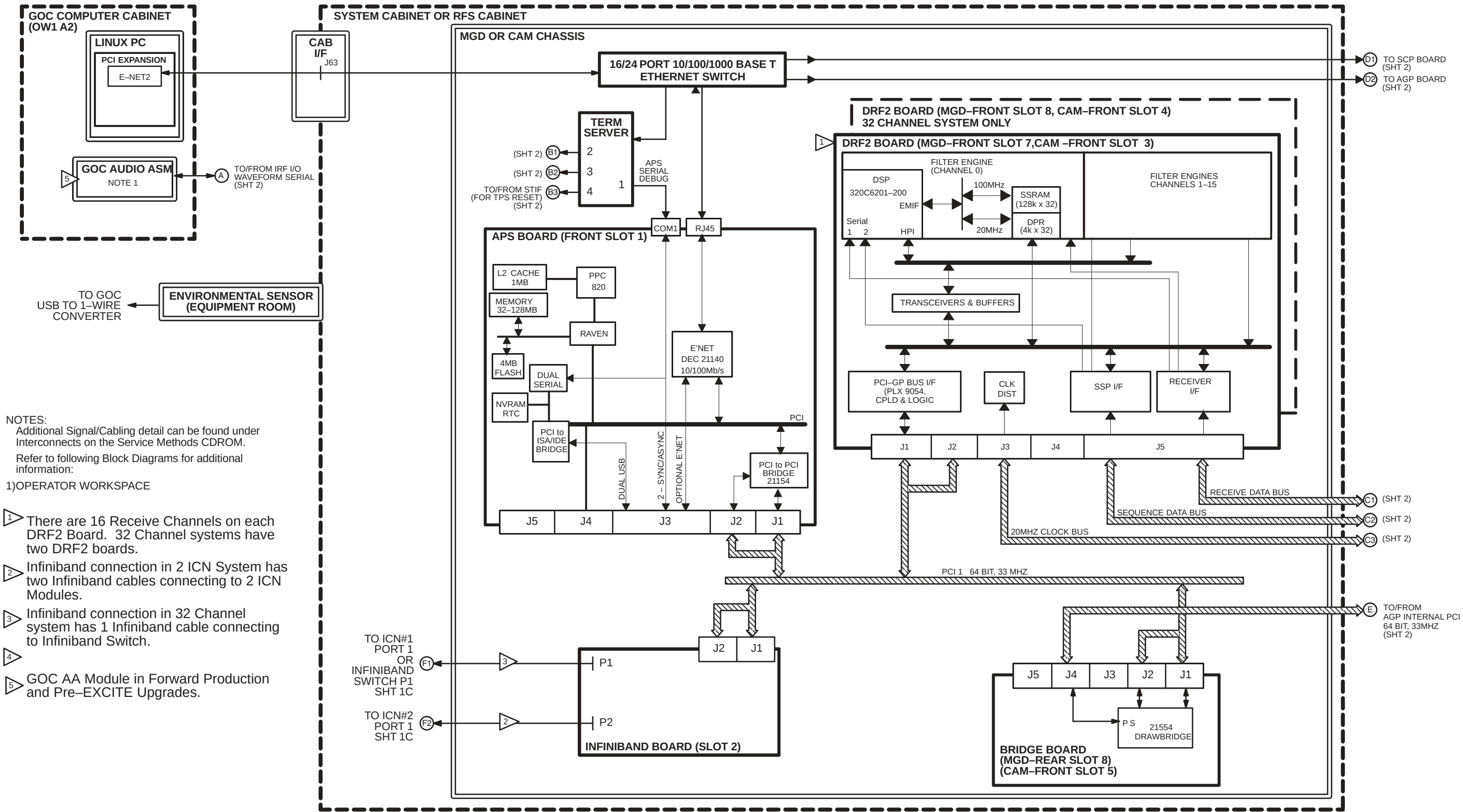
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NOTE:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

SECTION 1 – MGD/RRF/RF SUBSYSTEM (3.0T)

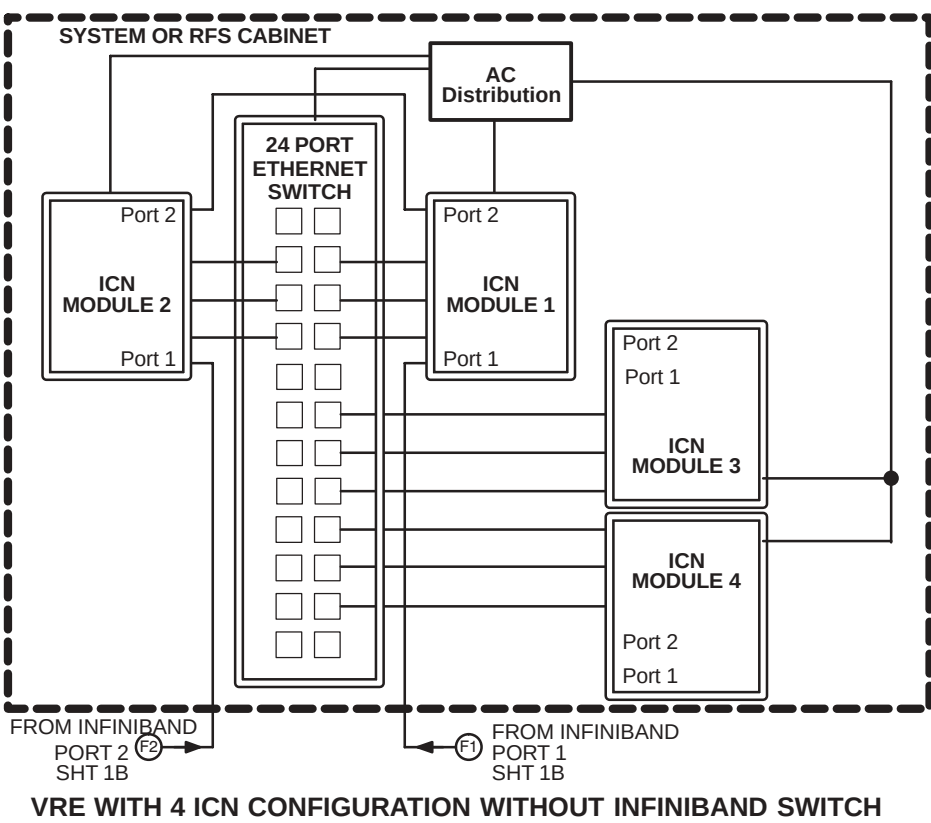
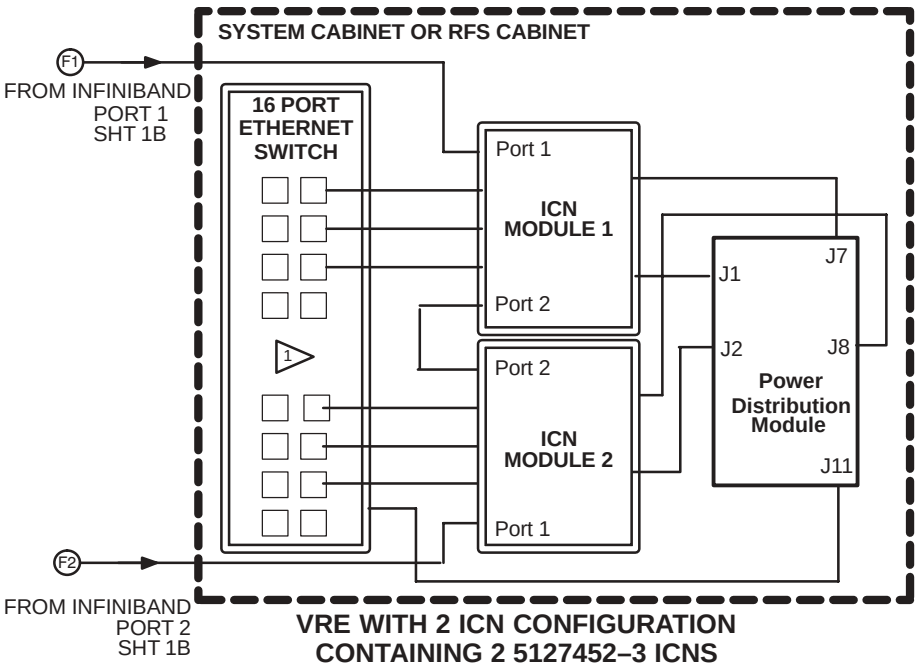
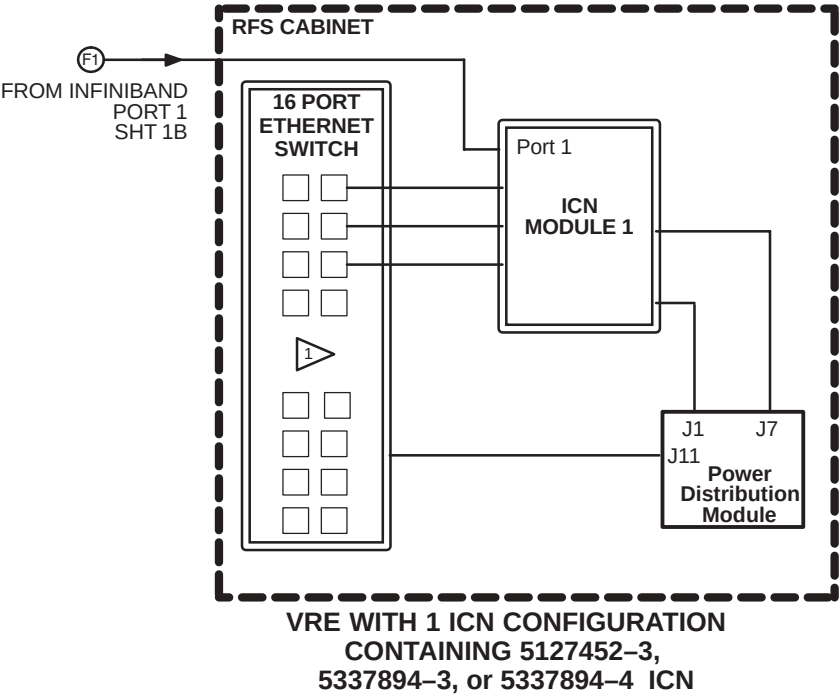
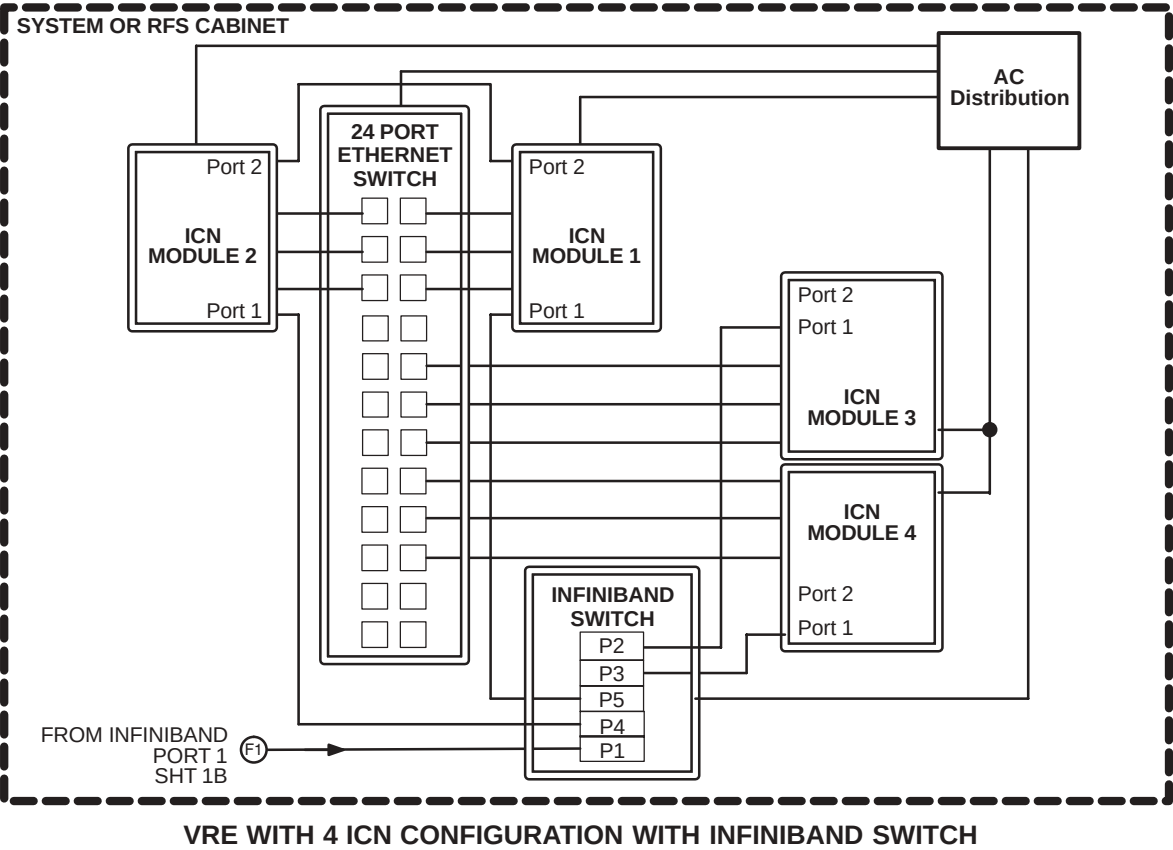
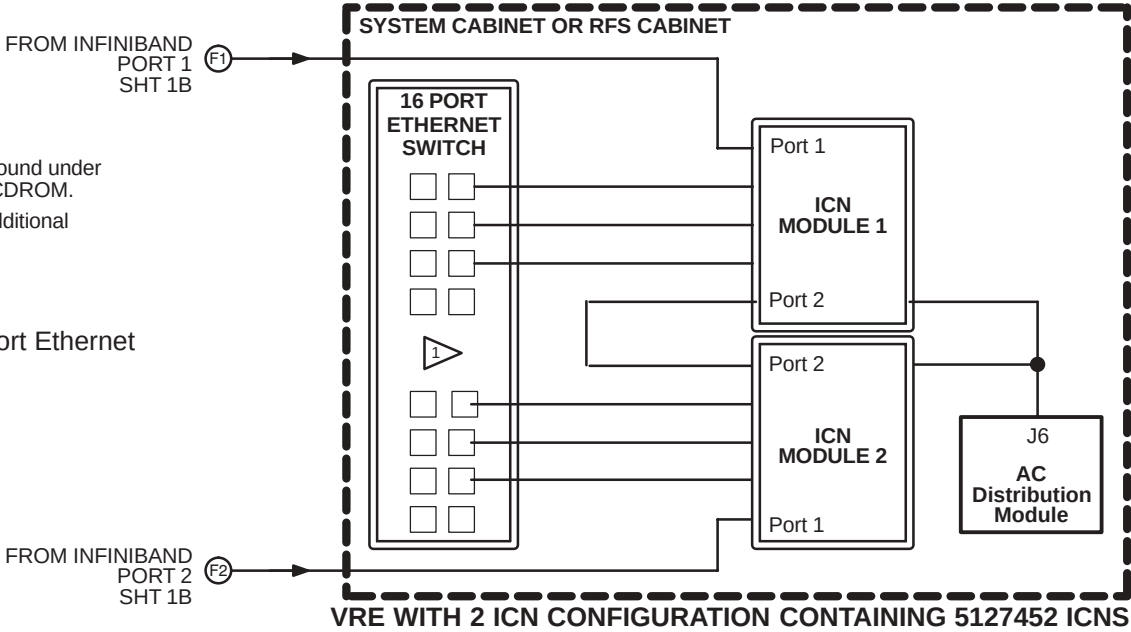


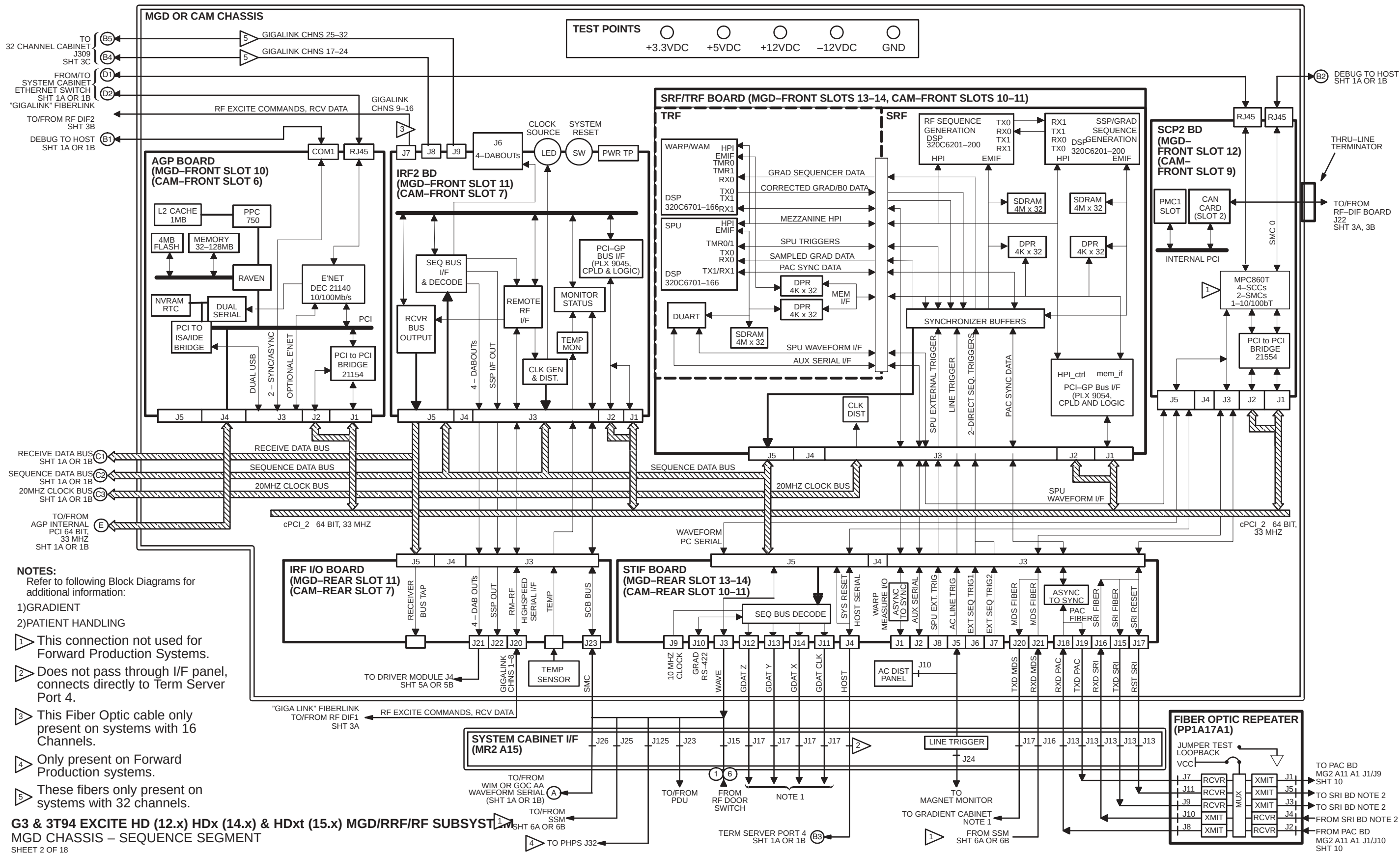


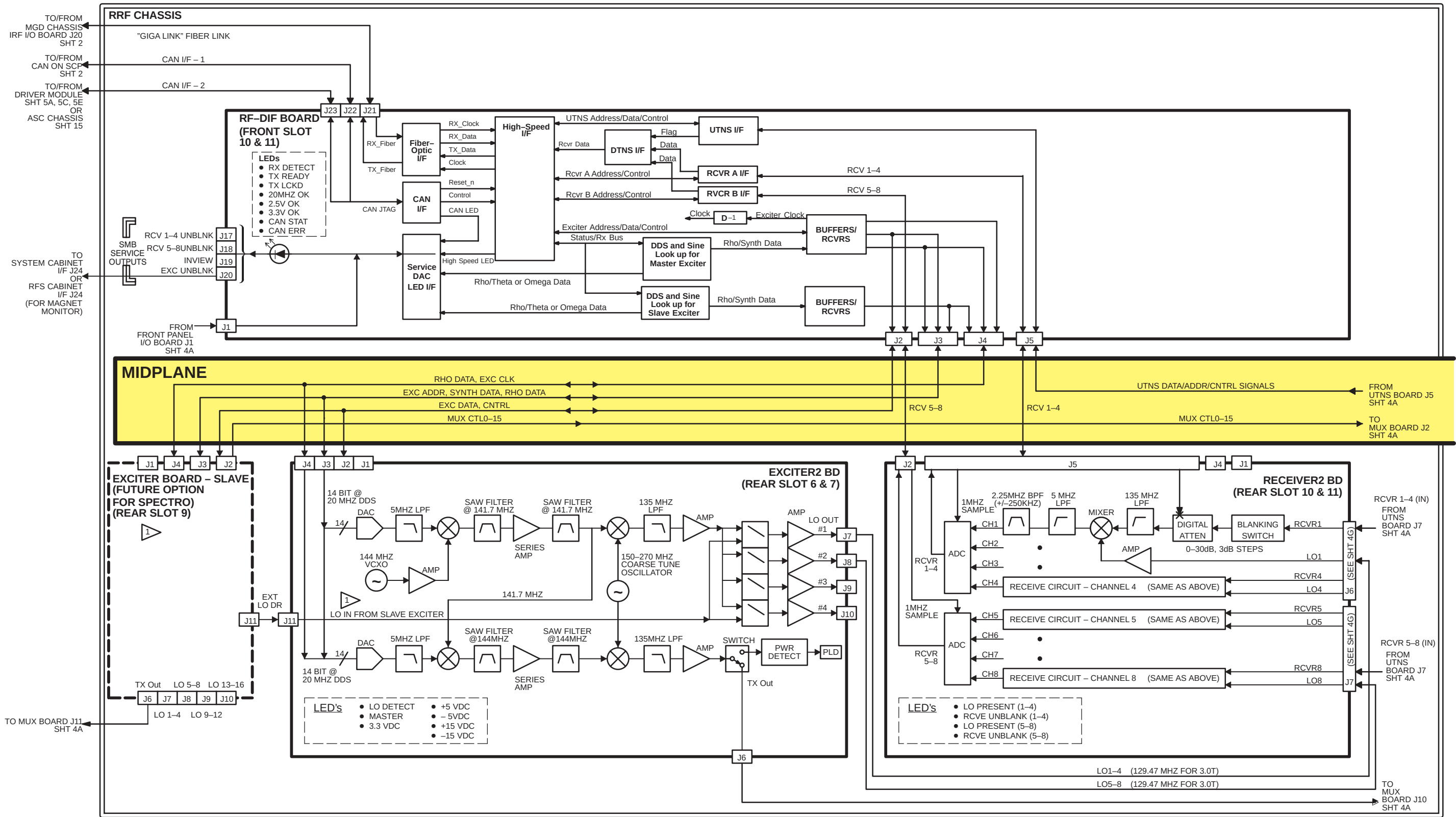
NOTES:
Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.
Refer to following Block Diagrams for additional information:

1) OPERATOR WORKSPACE

Other connections to 16/24 port Ethernet Switch on sheet 1B.



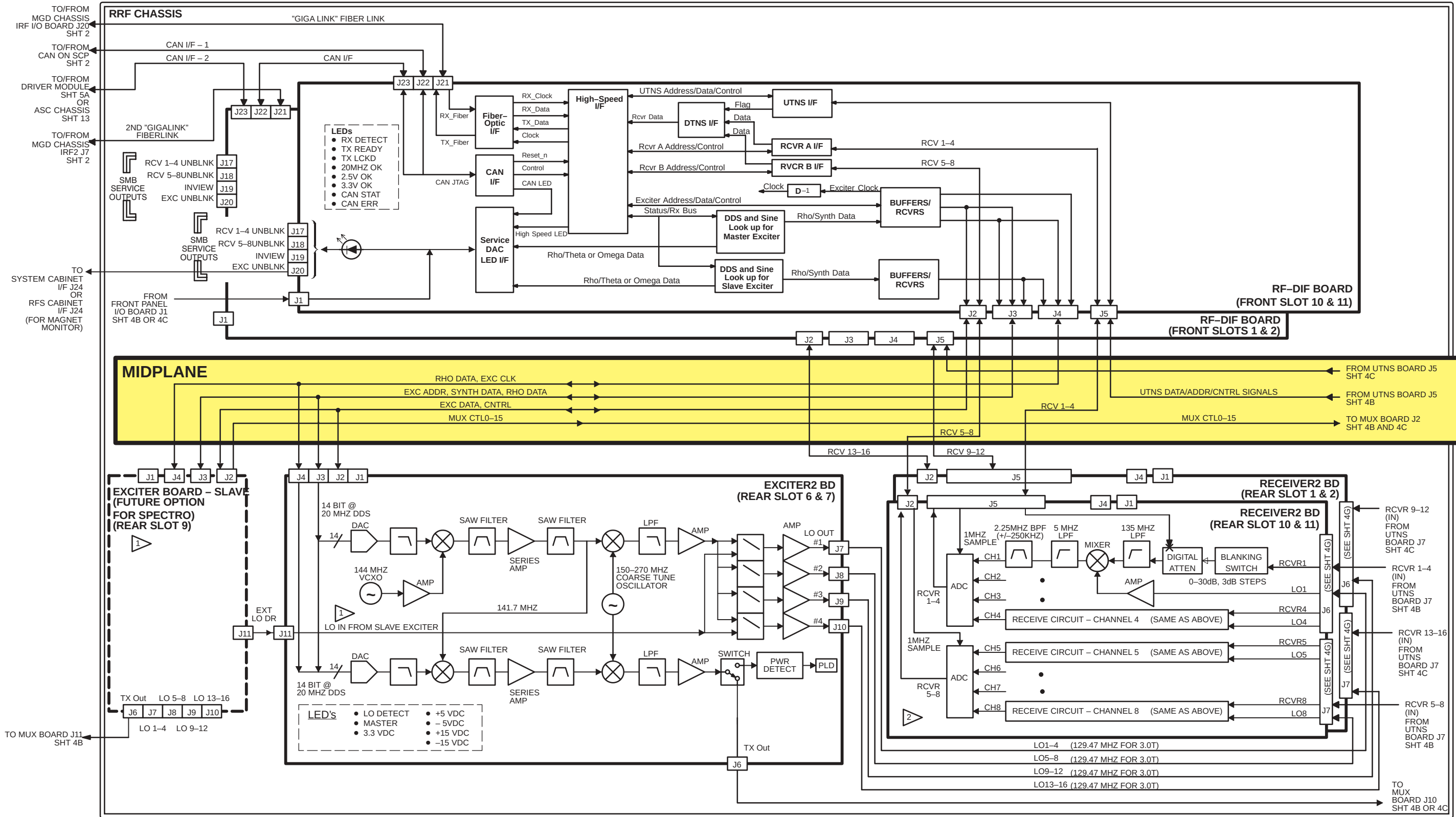




G3 & 3T94 EXCITE HD (12.x), HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
(8 Channel Systems)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 1 OF 2A (8 Channel)
SHEET 3A OF 18

NOTE:

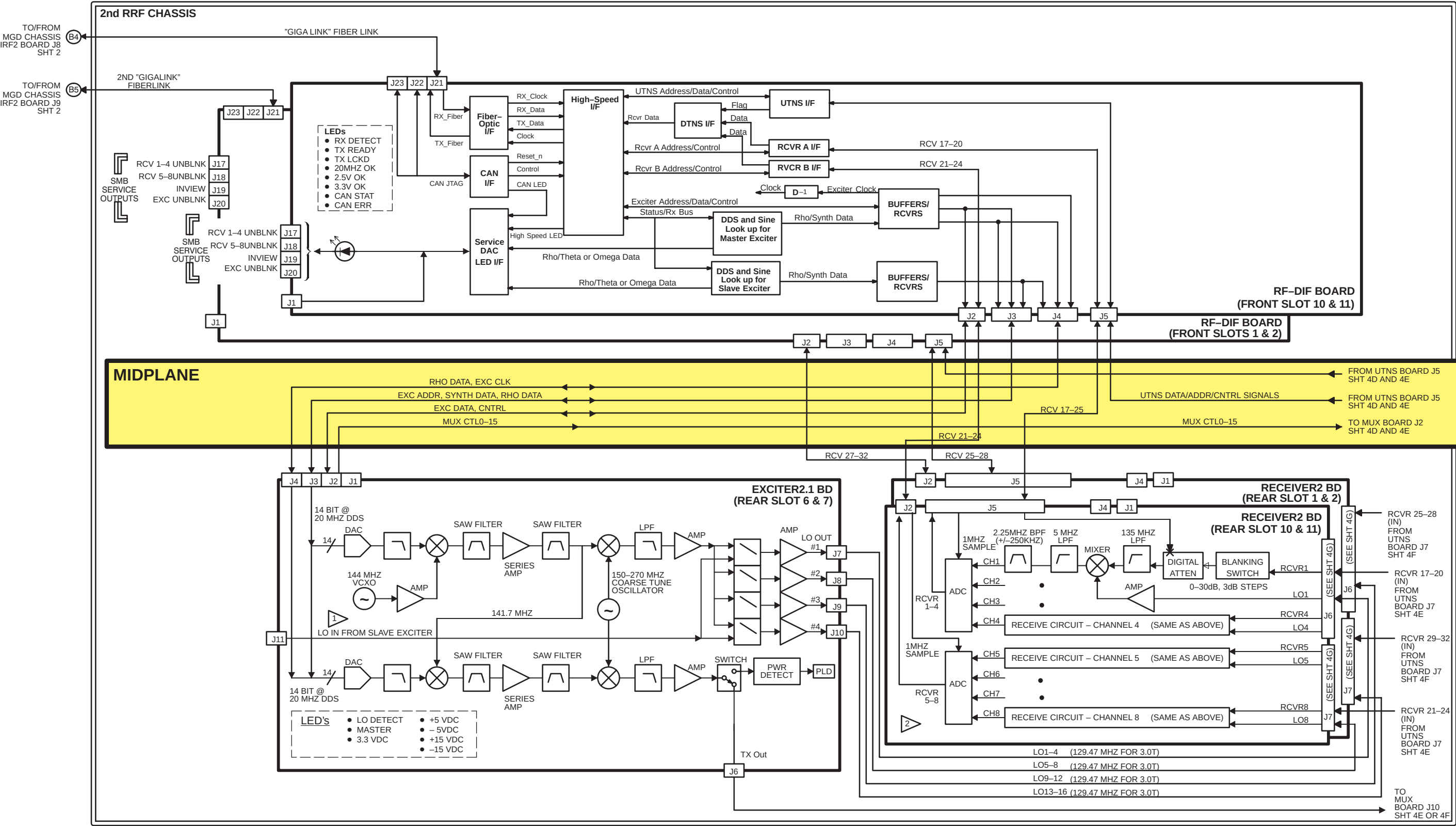
➤ Slave Exciter is a future option for Spectro.



G3 & 3T94 EXCITE HD (12.x) , HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
(16 Channel – Forward Production & Upgrades)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 1 OF 2B (16 Channel)
SHEET 3B OF 18

NOTE:
1 Slave Exciter is a future option for Spectro.

2 LED's

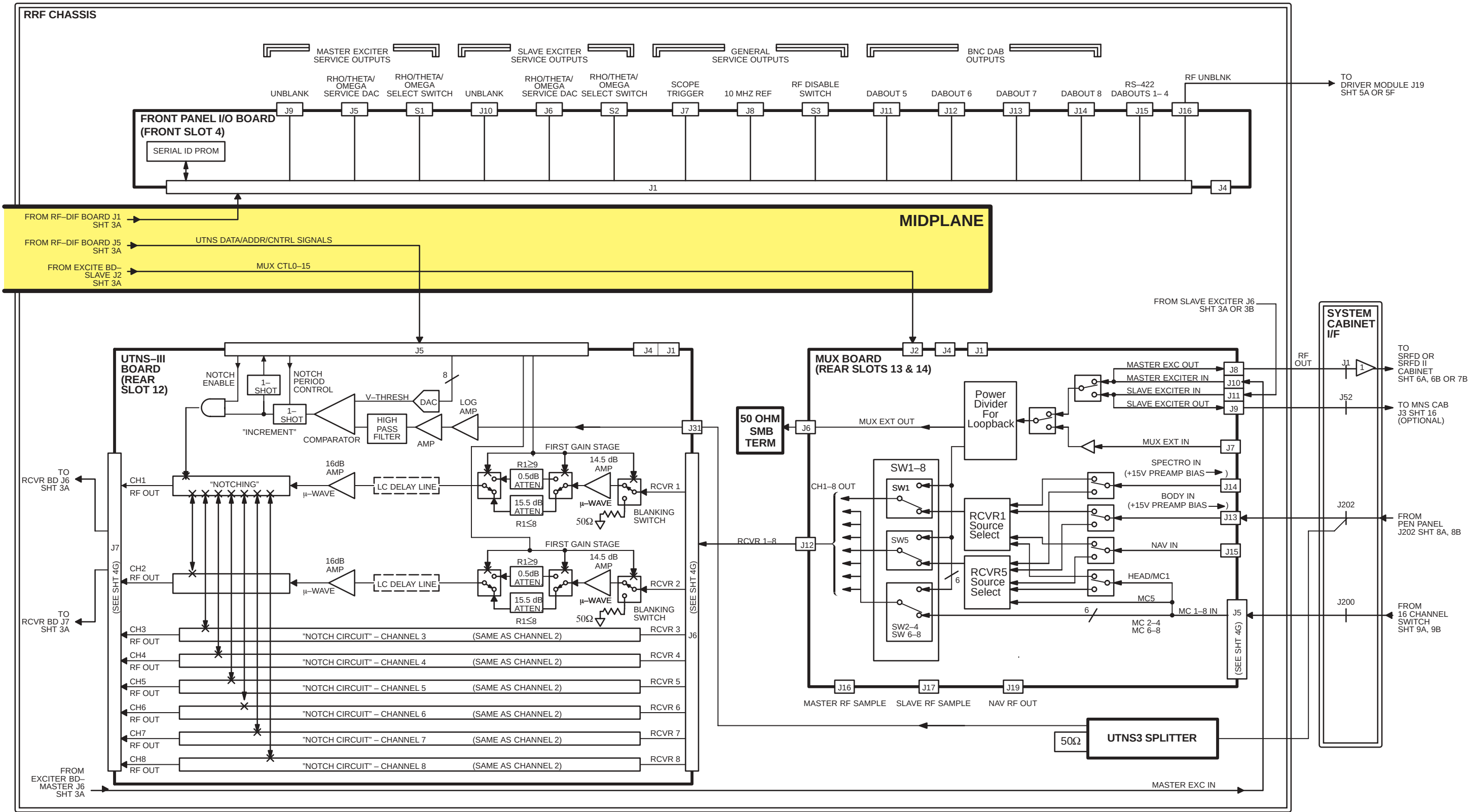


G3 HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
(32 Channel – Forward Production & Upgrades)
2nd RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 1 OF 4C (32 Channel)
SHEET 3C OF 18

NOTE:

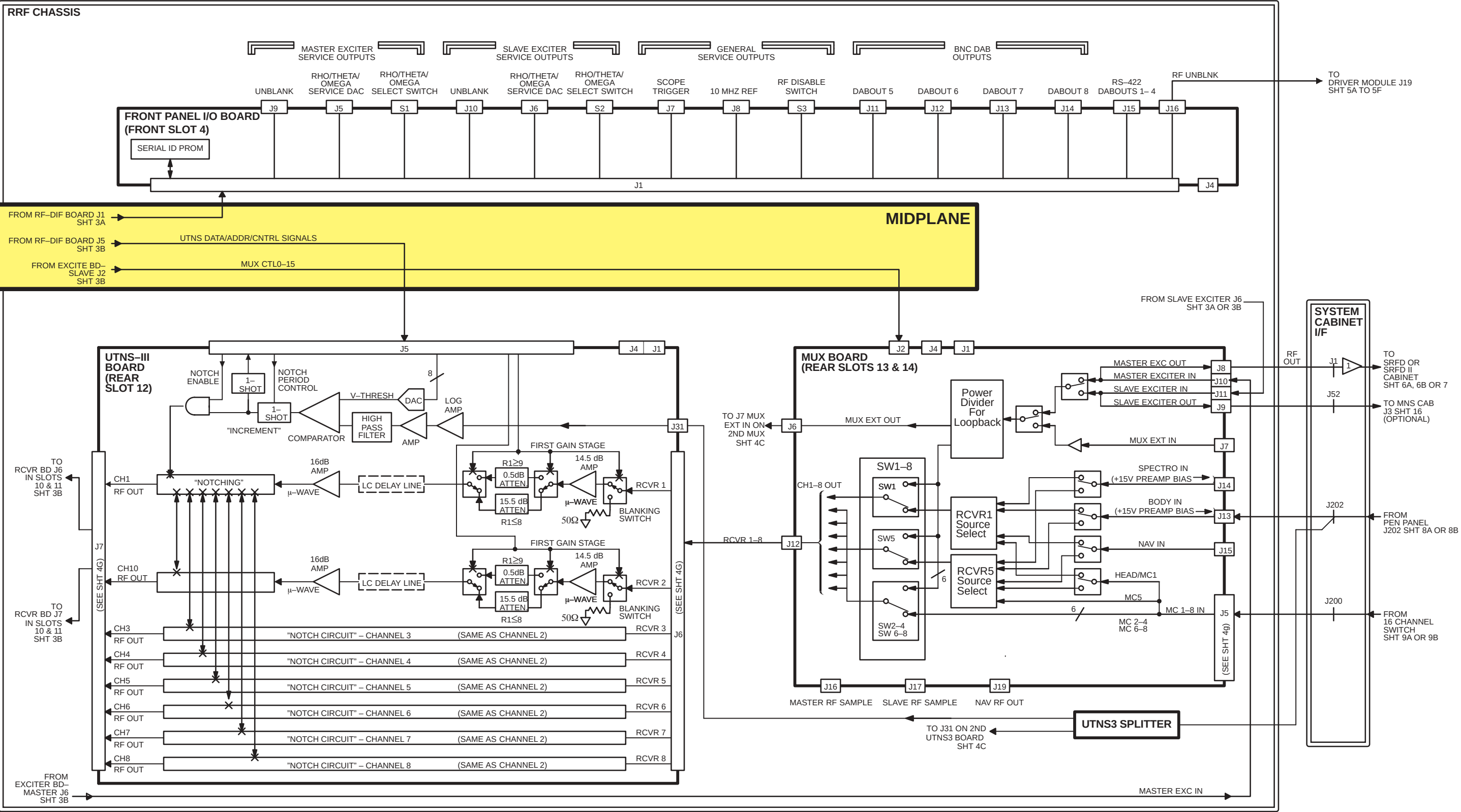
LEDs

- LO PRESENT (1-4)
- RCVE UNBLANK (1-4)
- LO PRESENT (5-8)
- RCVE UNBLANK (5-8)



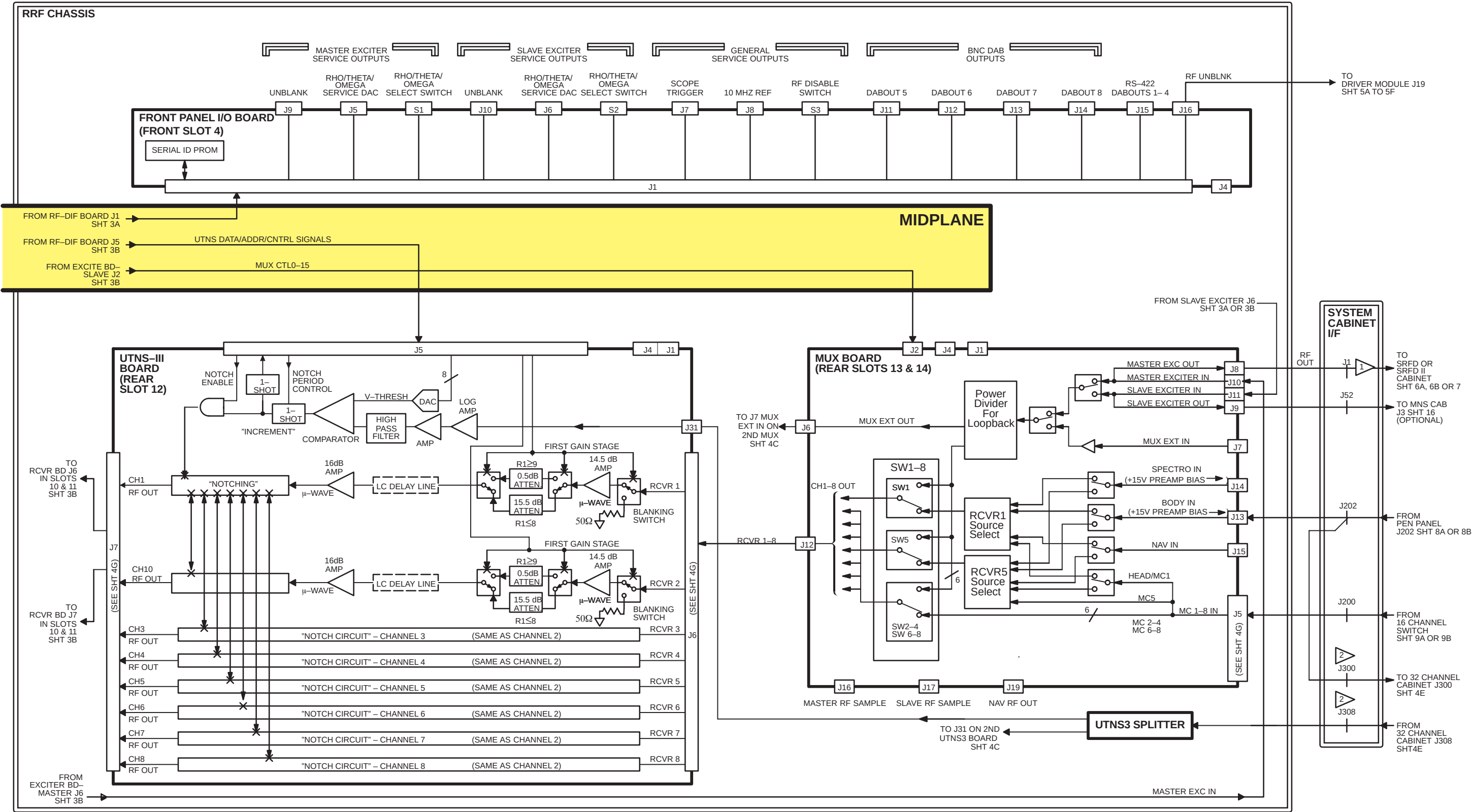
G3 & 3T94 EXCITE HD (12.x), HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
(8 Channel Systems)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 2 OF 2A (8 channel)
SHEET 4A OF 18

NOTE:
1 J1 not present in RF System Cabinet.



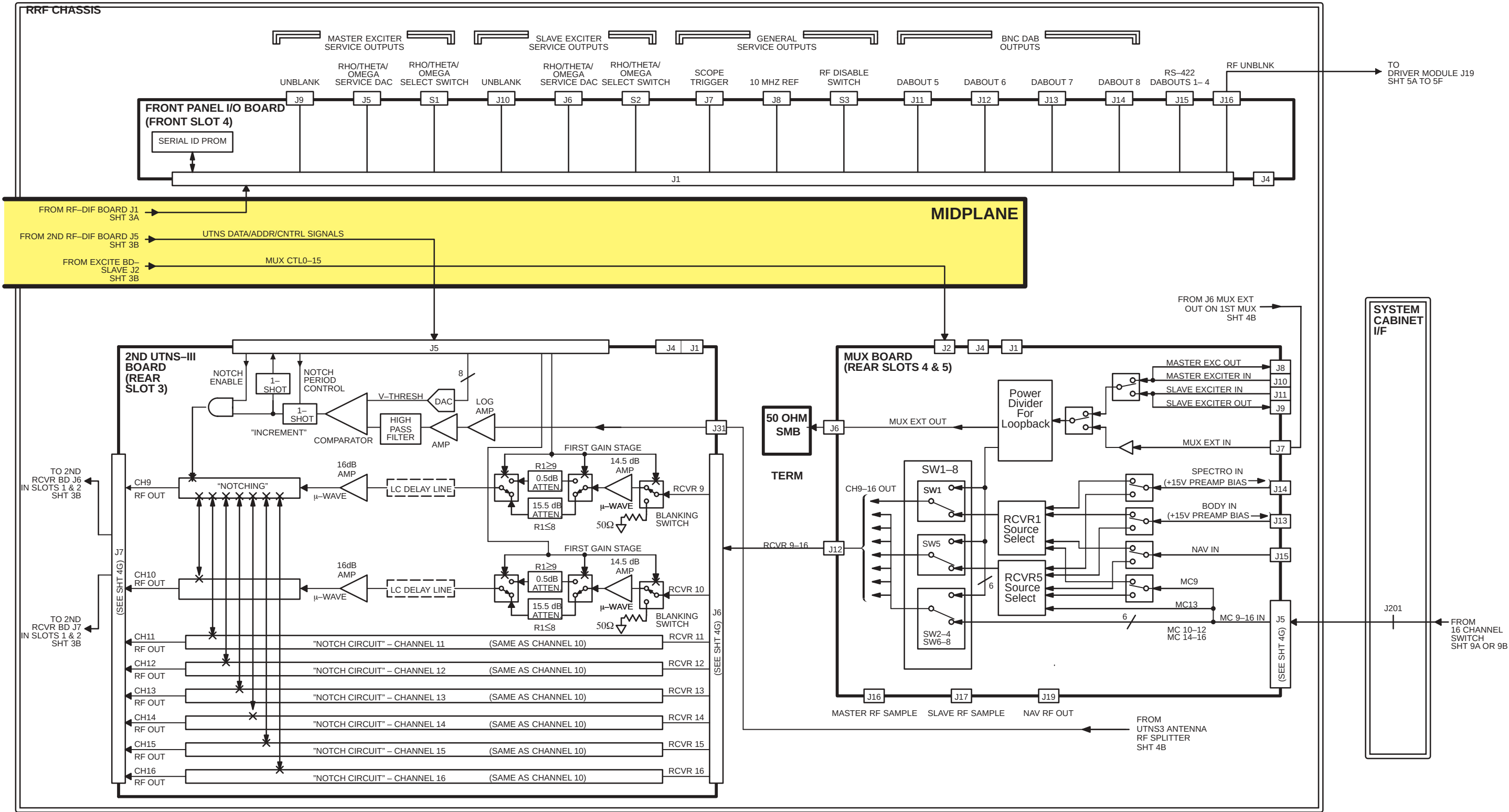
G3 & 3T94 EXCITE HD (12.x), HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
(Channels 1-8 of 16 Channel - Forward Production & Upgrades)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 2a OF 2B (16 Channel)
SHEET 4B OF 18

NOTE:
1 J1 not present in RF System Cabinet.

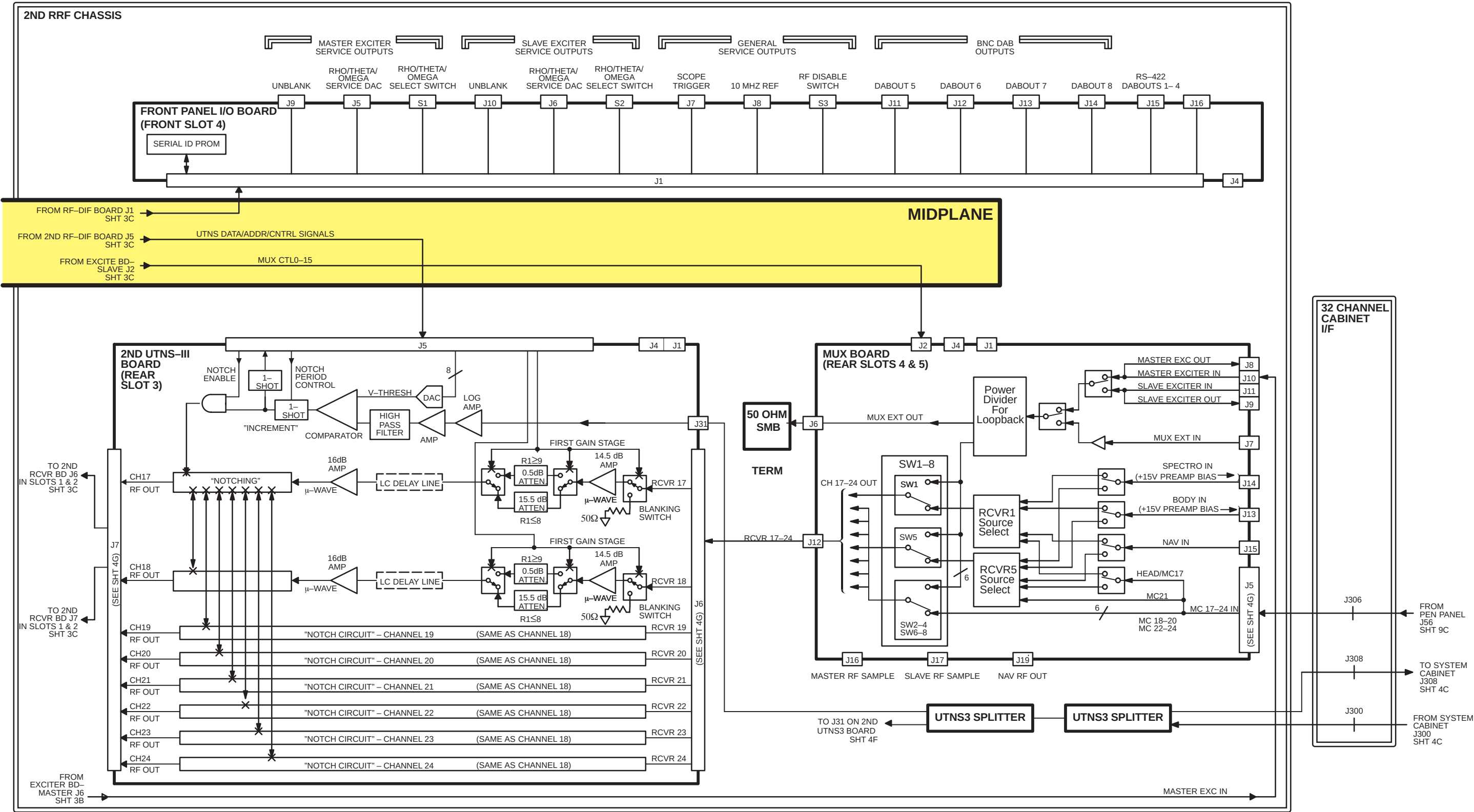


G3 & 3T94 EXCITE HD (12.x), HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
(Channels 1–8 of 32 Channel – Forward Production & Upgrades)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 2a OF 4C (32 Channel)
SHEET 4C OF 18

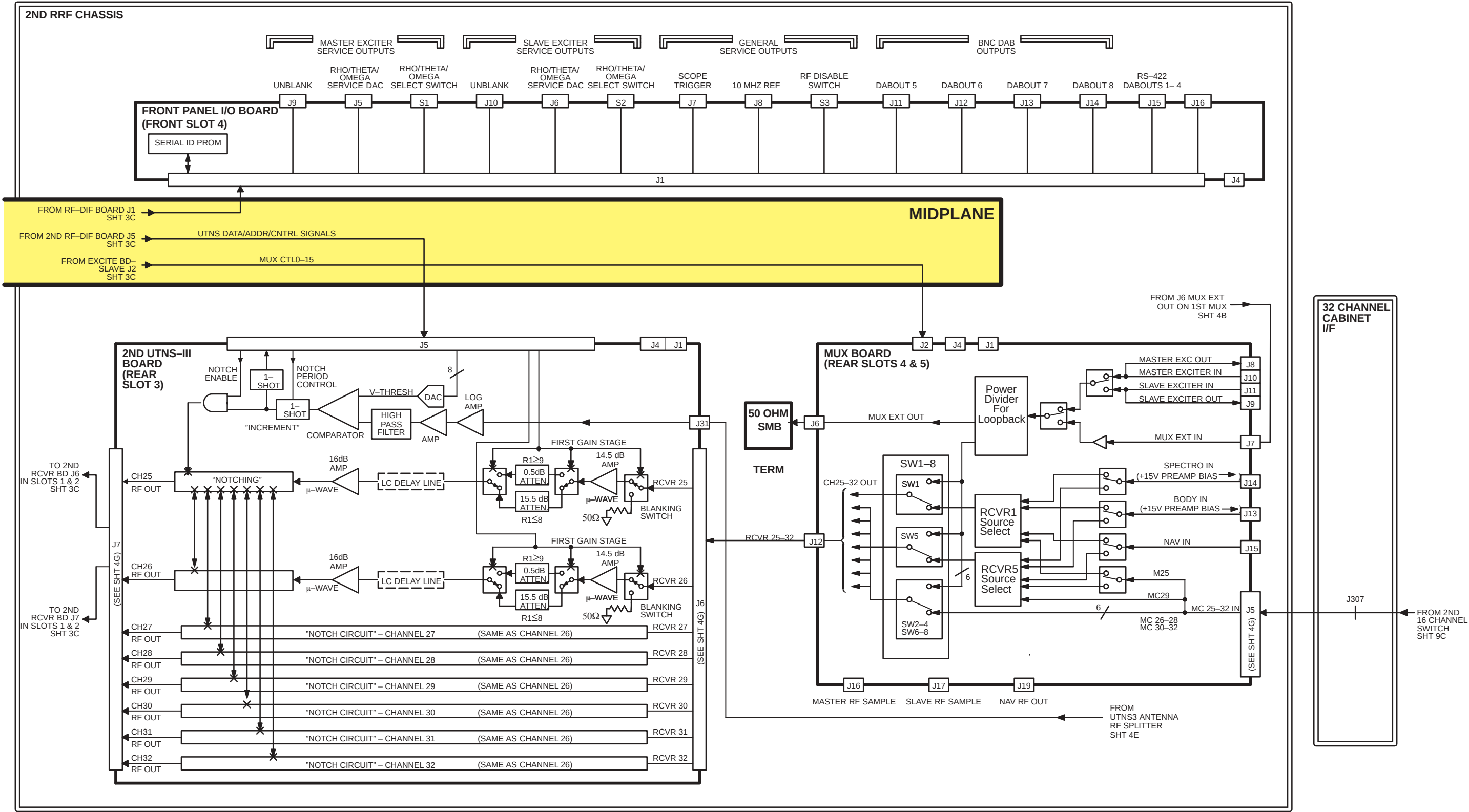
NOTE:
1 J1 not present in RF System Cabinet.
2 32 Channel systems only.



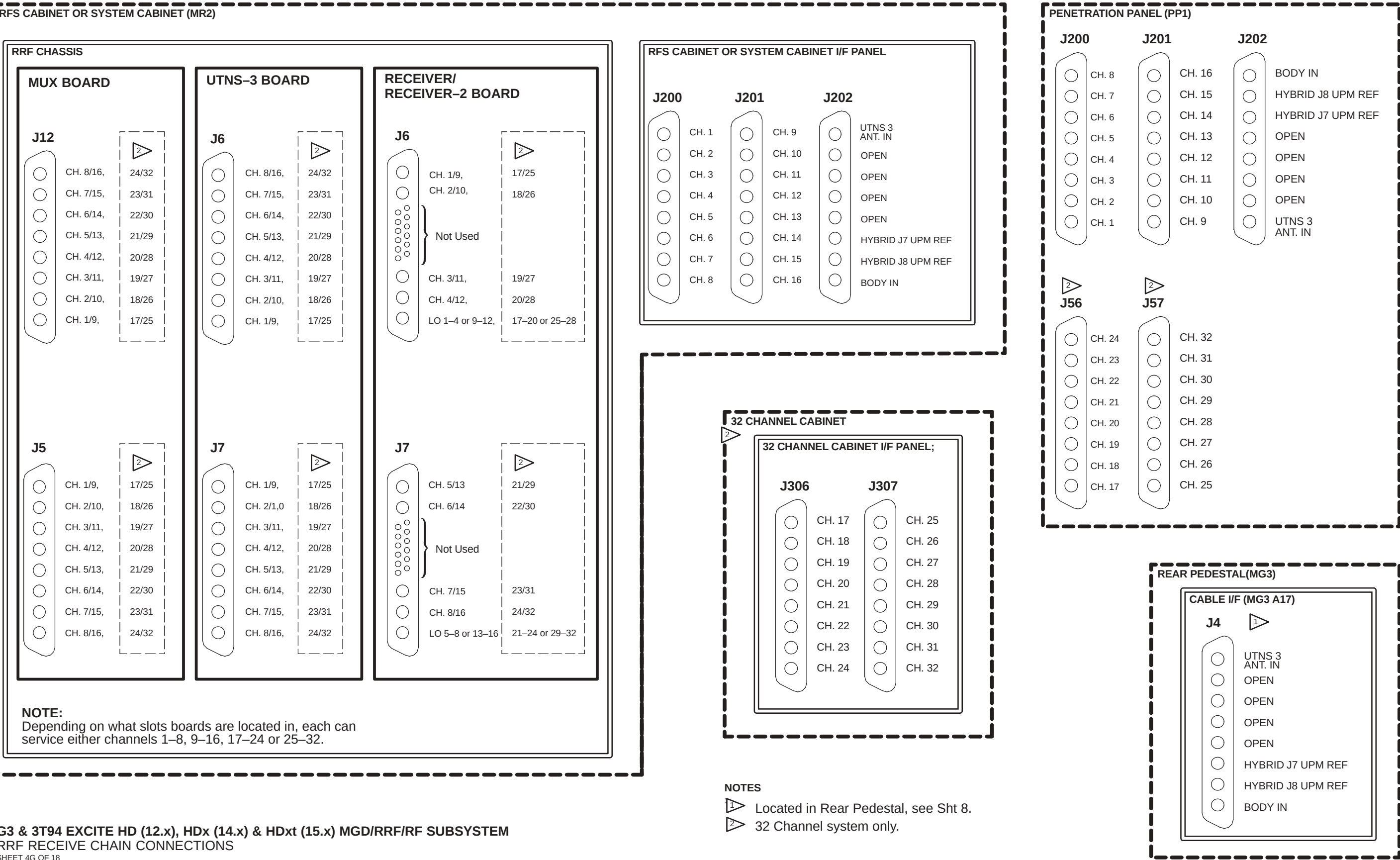
G3 & 3T94 EXCITE HD (12.x) , HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
(Channels 9-16 of 16 or 32 Channel – Forward Production & Upgrades)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 2b OF 2B (16 Channel)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 2b OF 4C (32 Channel)
SHEET 4D OF 18



G3 HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
(Channels 17-24 of 32 Channel – Forward Production & Upgrades)
2nd RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 2c OF 4C (32 Channel)
SHEET 4E OF 18



G3 HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
(Channels 25-32 of 32 Channel - Forward Production & Upgrades)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 2d OF 4C (32 Channel)
SHEET 4F OF 18



NOTES

1

 Located in Rear Pedestal, see Sht 8.

2

 32 Channel system only.

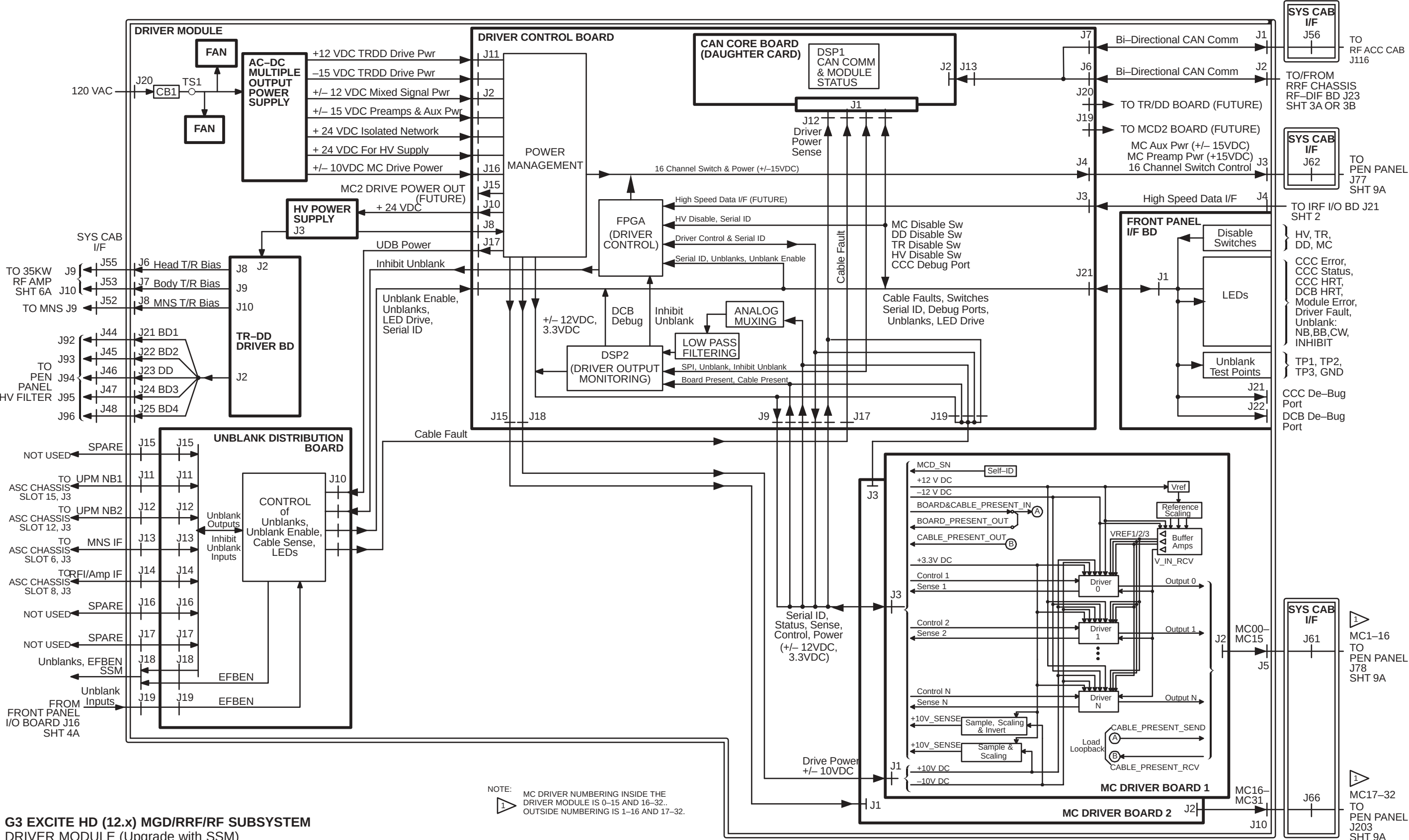
G3 & 3T94 EXCITE HD (12.x), HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM

RRF RECEIVE CHAIN CONNECTIONS

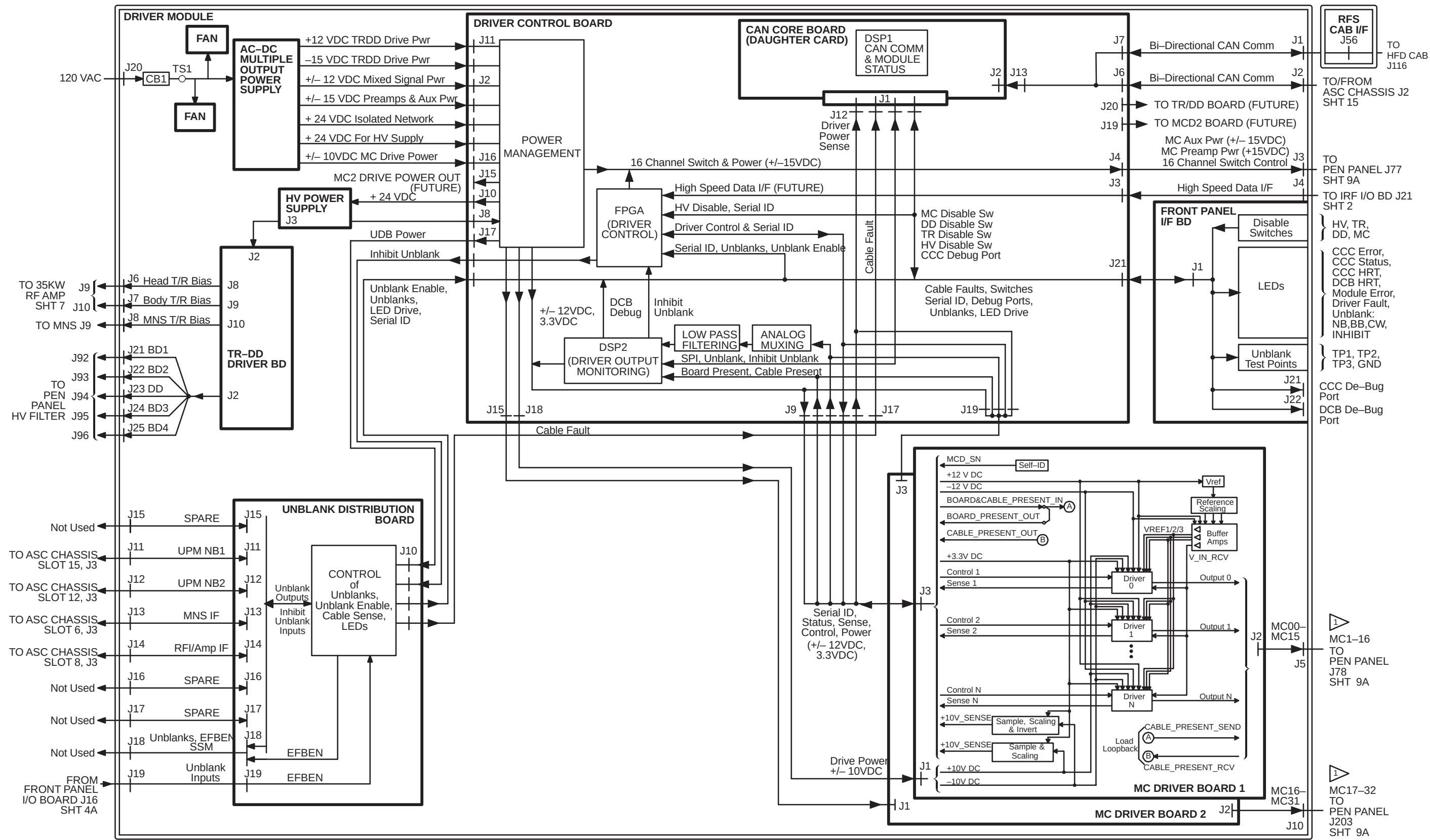
SHEET 4G OF 18

MGD/RRF/RF

1-14

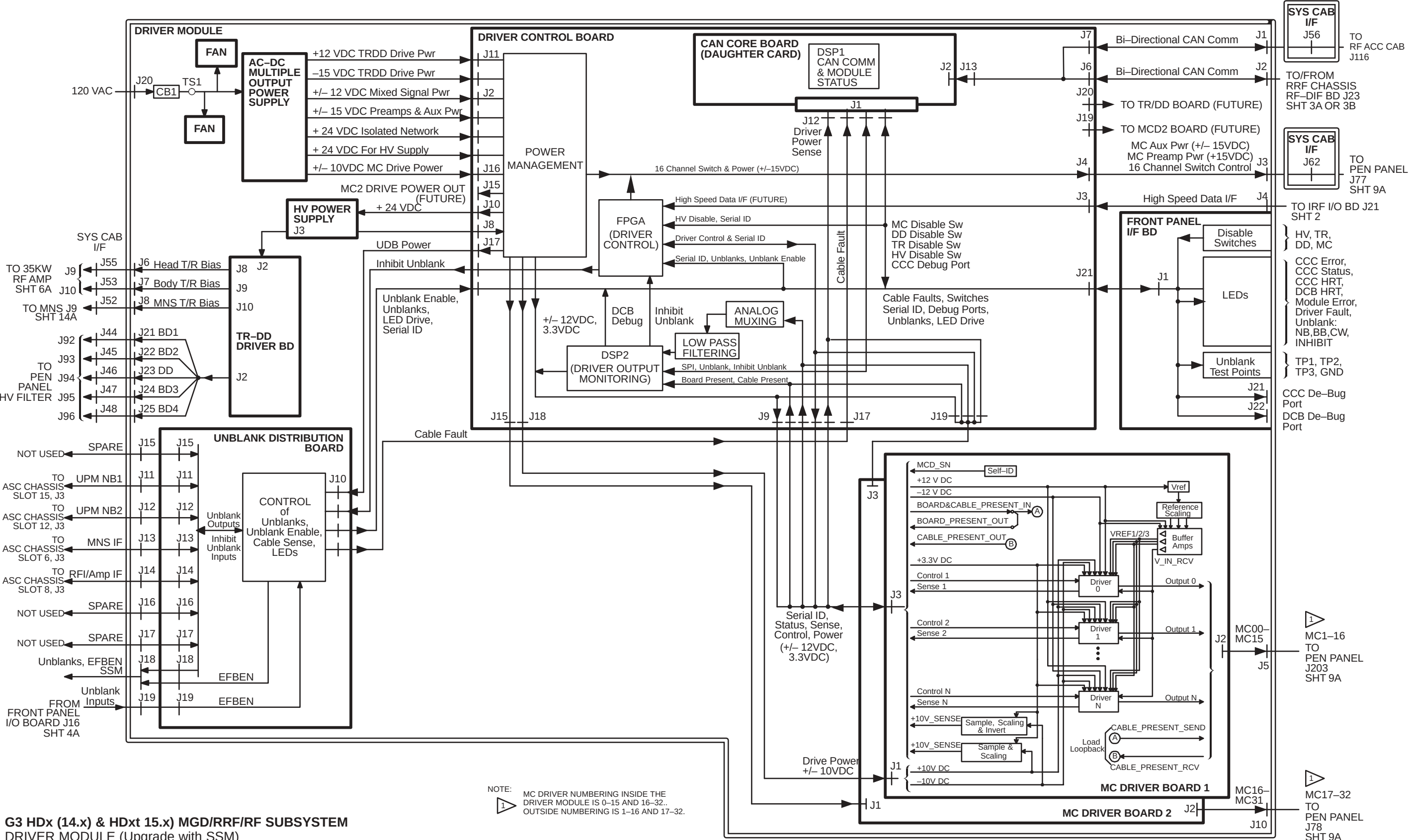


G3 EXCITE HD (12.x) MGD/RRF/RF SUBSYSTEM
DRIVER MODULE (Upgrade with SSM)
SHEET 5A OF 18



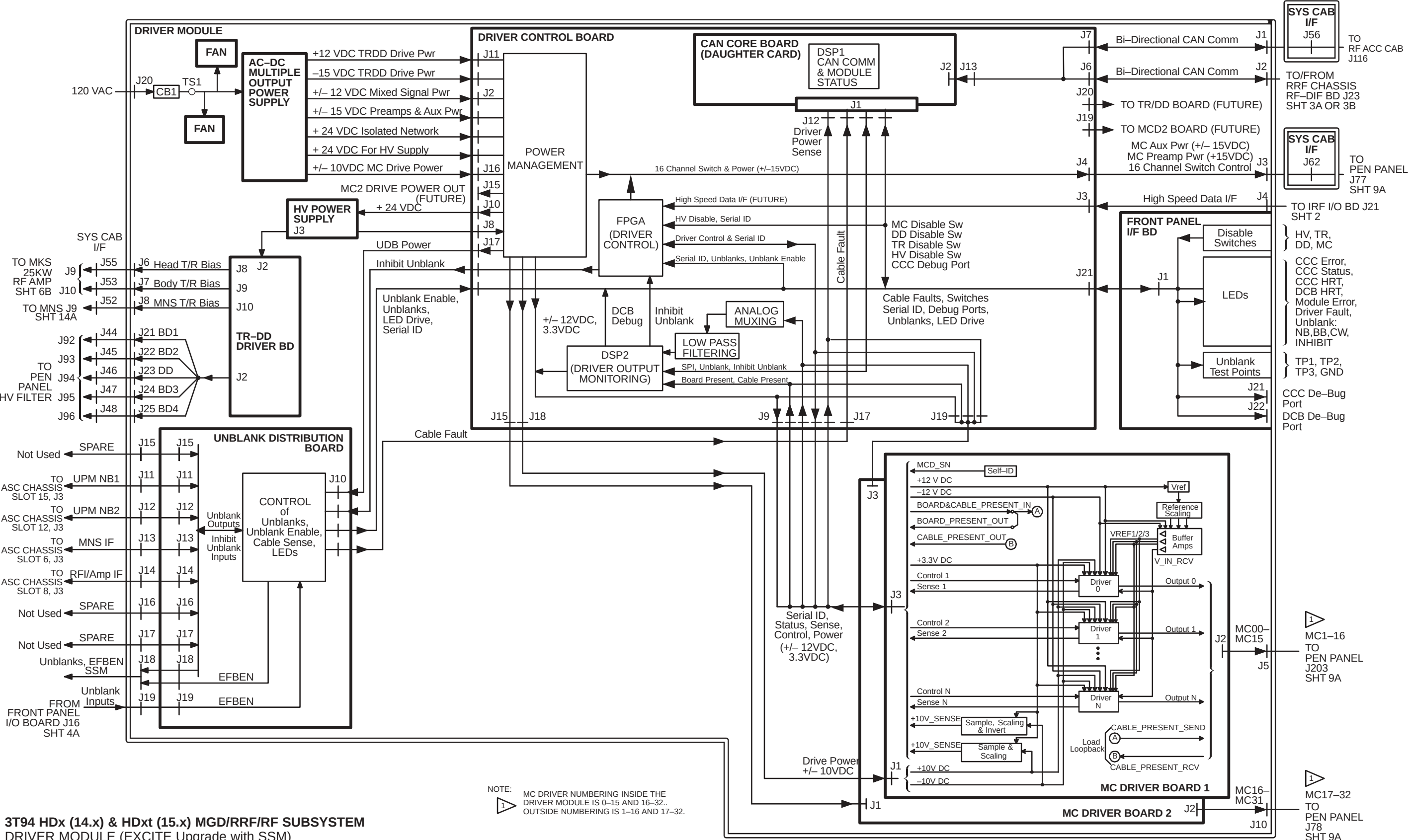
G3 EXCITE HD (12.x) MGD/RRF/RF SUBSYSTEM
DRIVER MODULE (Forward Production, RFS Cabinet, no SSM)
SHEET 5B OF 18

NOTE:
1 MC DRIVER NUMBERING INSIDE THE DRIVER MODULE IS 0-15 AND 16-32. OUTSIDE NUMBERING IS 1-16 AND 17-32.

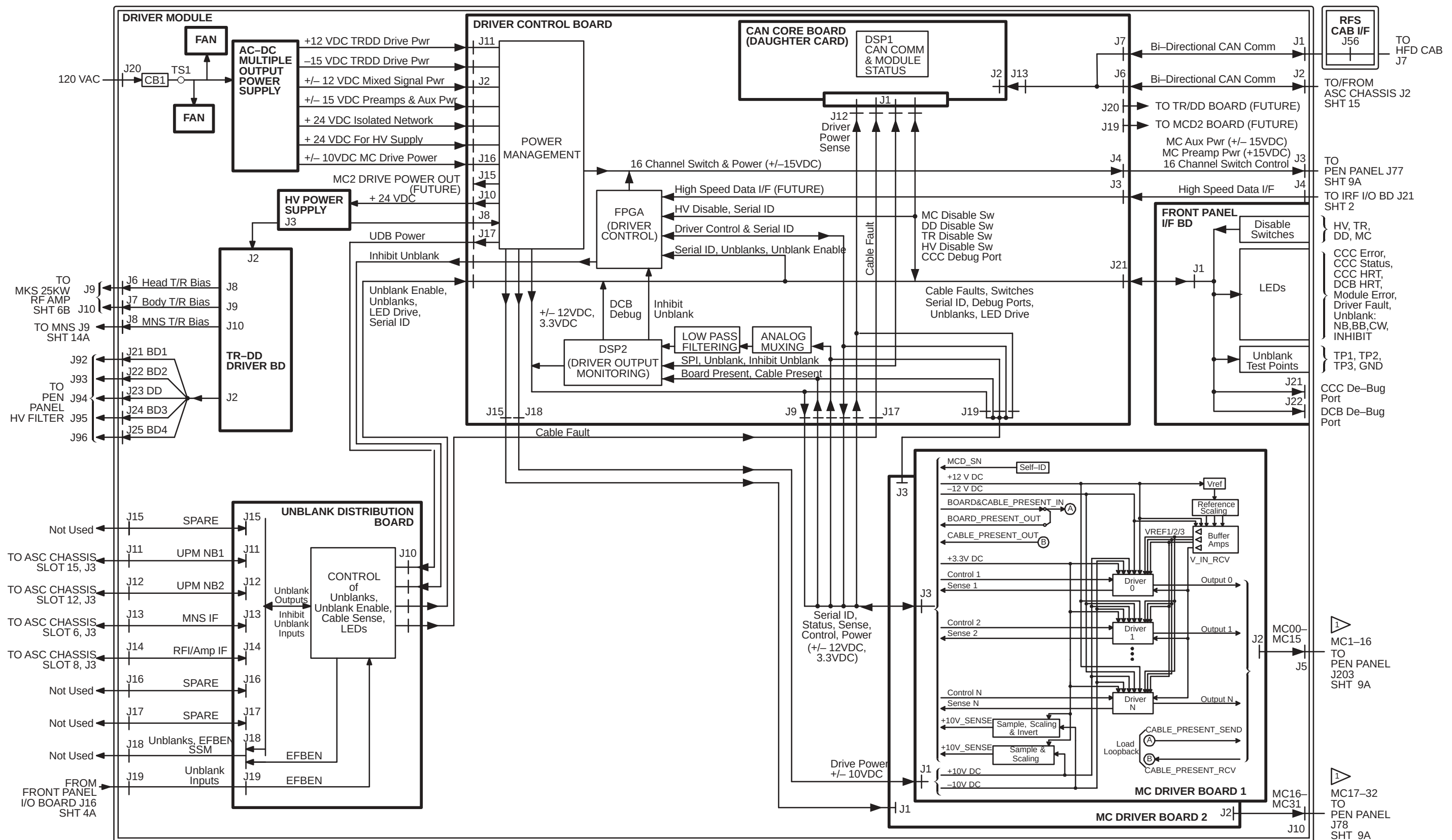


G3 HDx (14.x) & HDxt 15.x MGD/RRF/RF SUBSYSTEM
DRIVER MODULE (Upgrade with SSM)
SHEET 5C OF 18



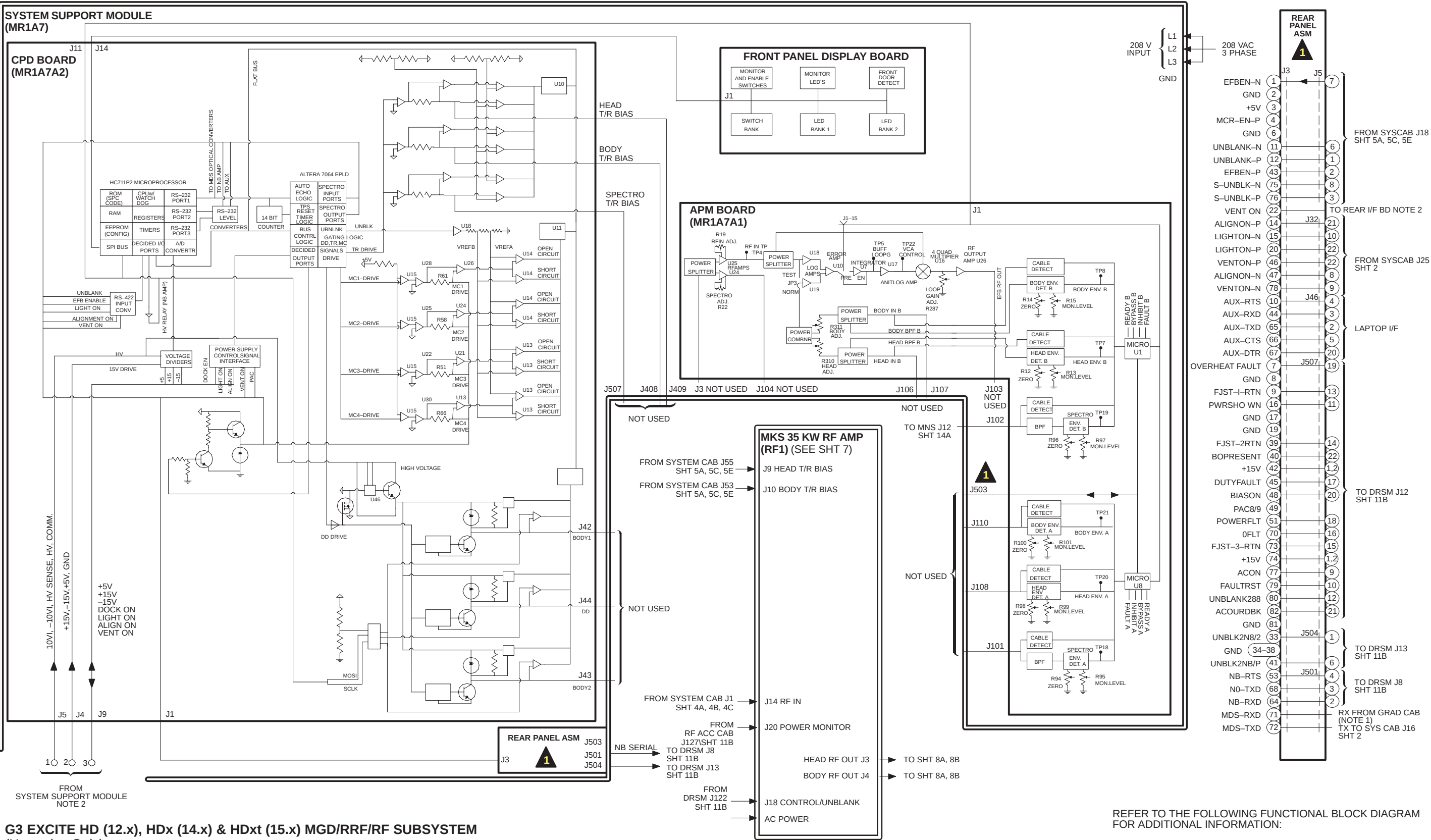


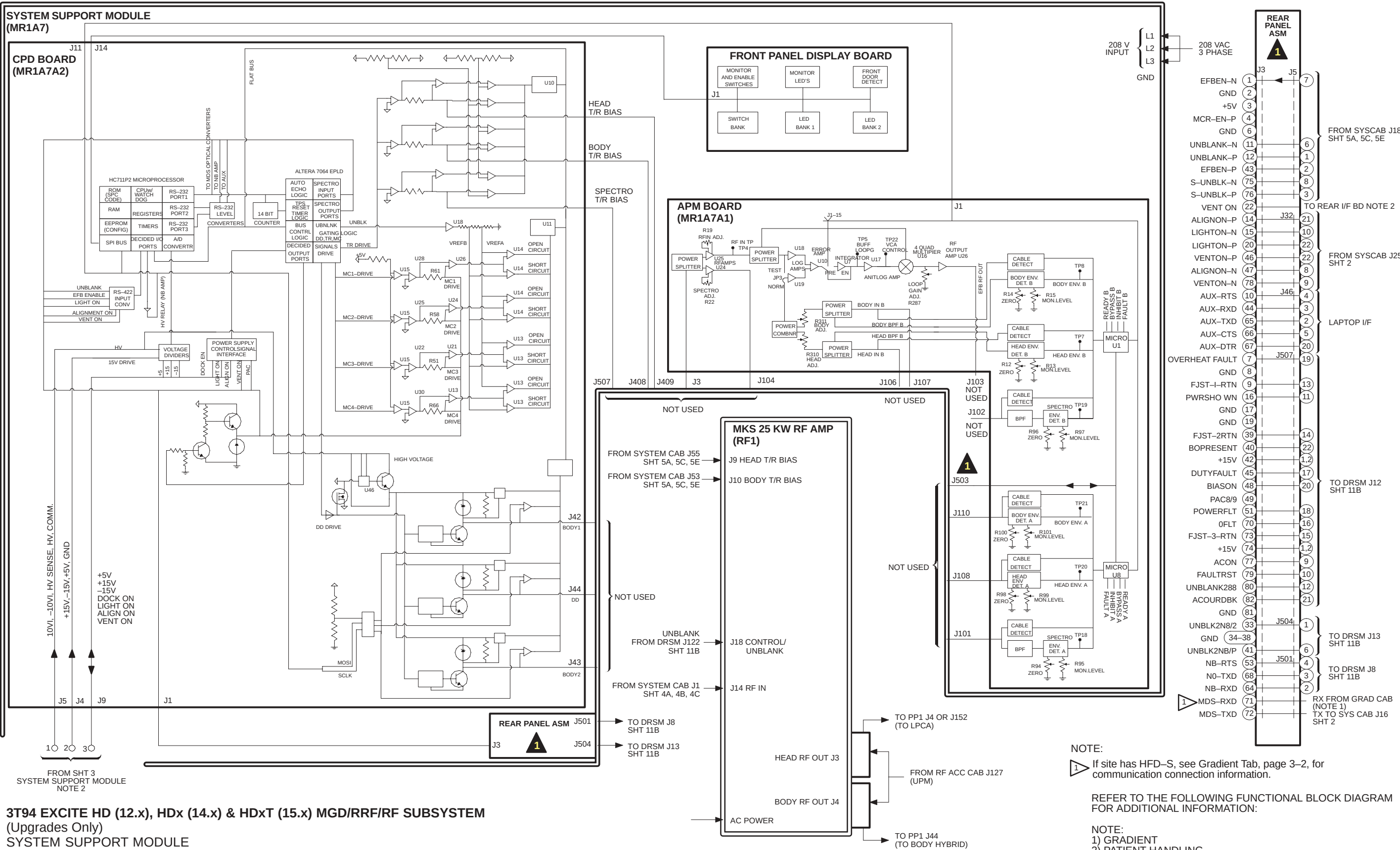
3T94 HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
DRIVER MODULE (EXCITE Upgrade with SSM)
SHEET 5E OF 18

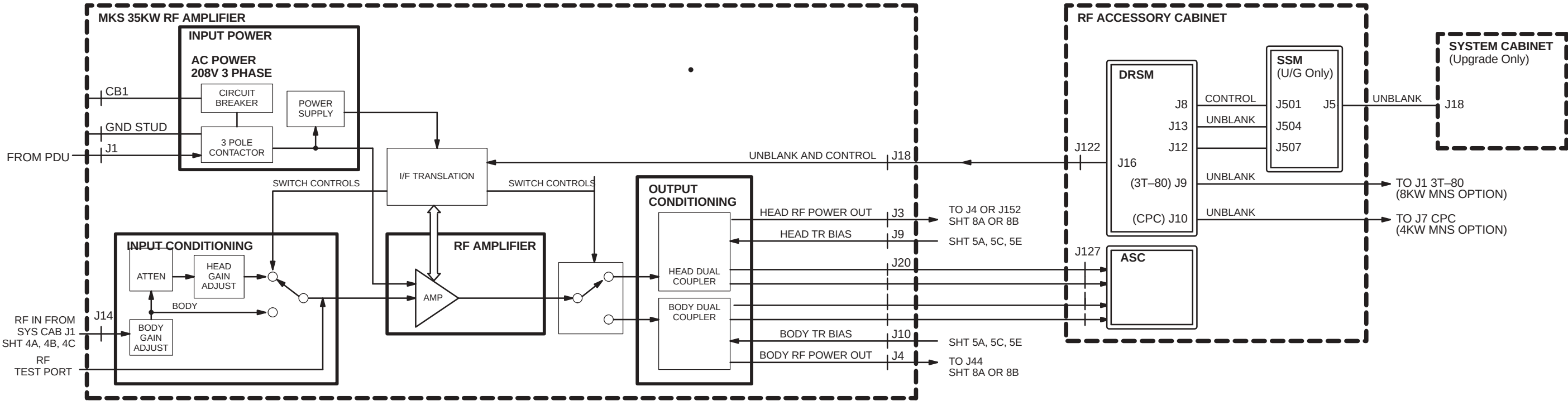


3T94 HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
DRIVER MODULE (Pre-EXCITE UPGRADE, RFS Cabinet, no SSM)
 SHEET 5F OF 18

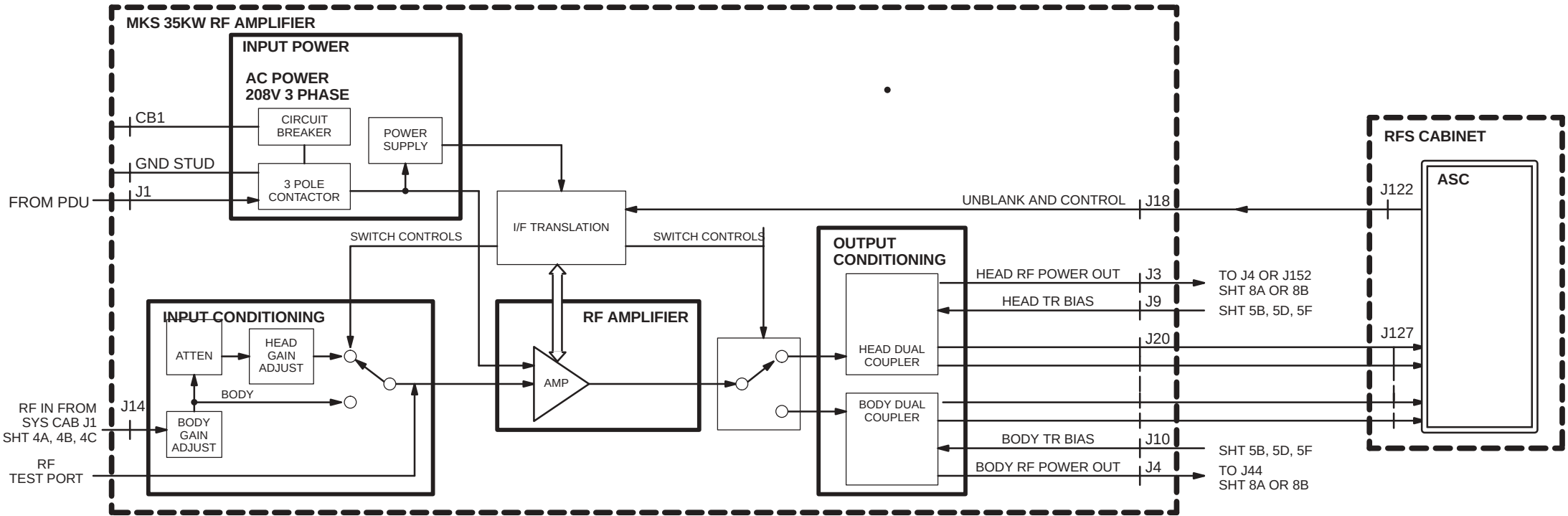
NOTE:  MC DRIVER NUMBERING INSIDE THE DRIVER MODULE IS 0-15 AND 16-32.. OUTSIDE NUMBERING IS 1-16 AND 17-32.

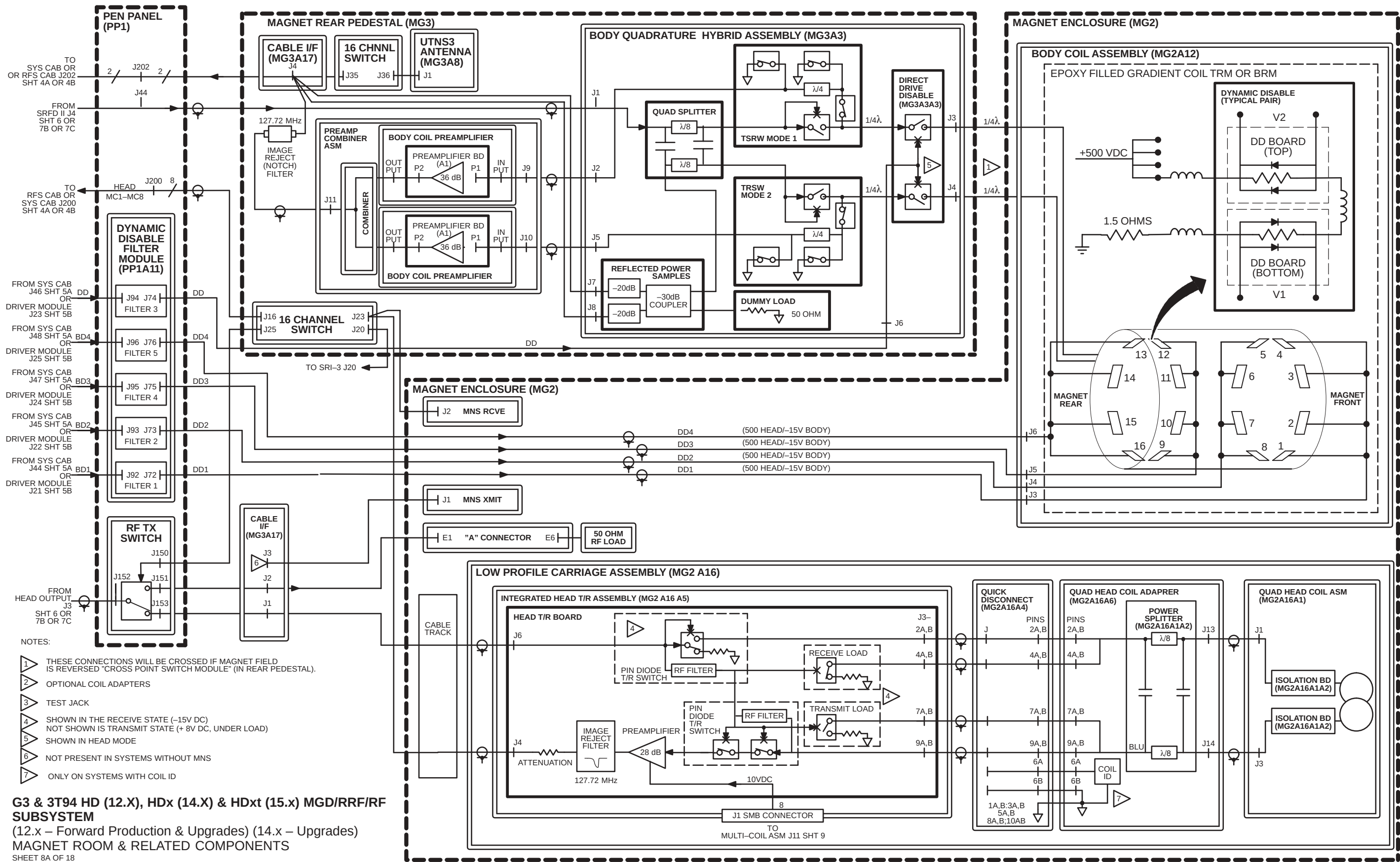


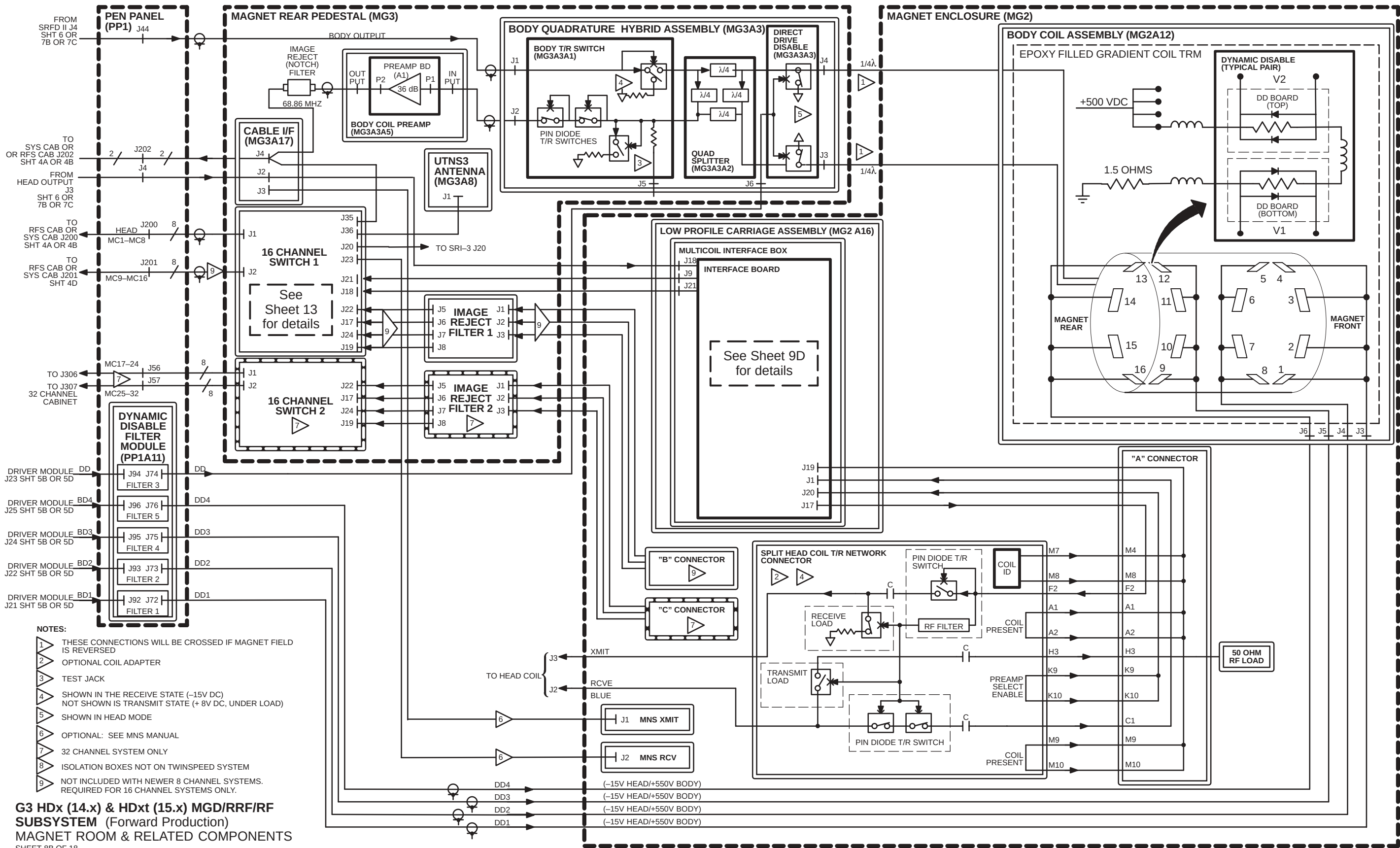


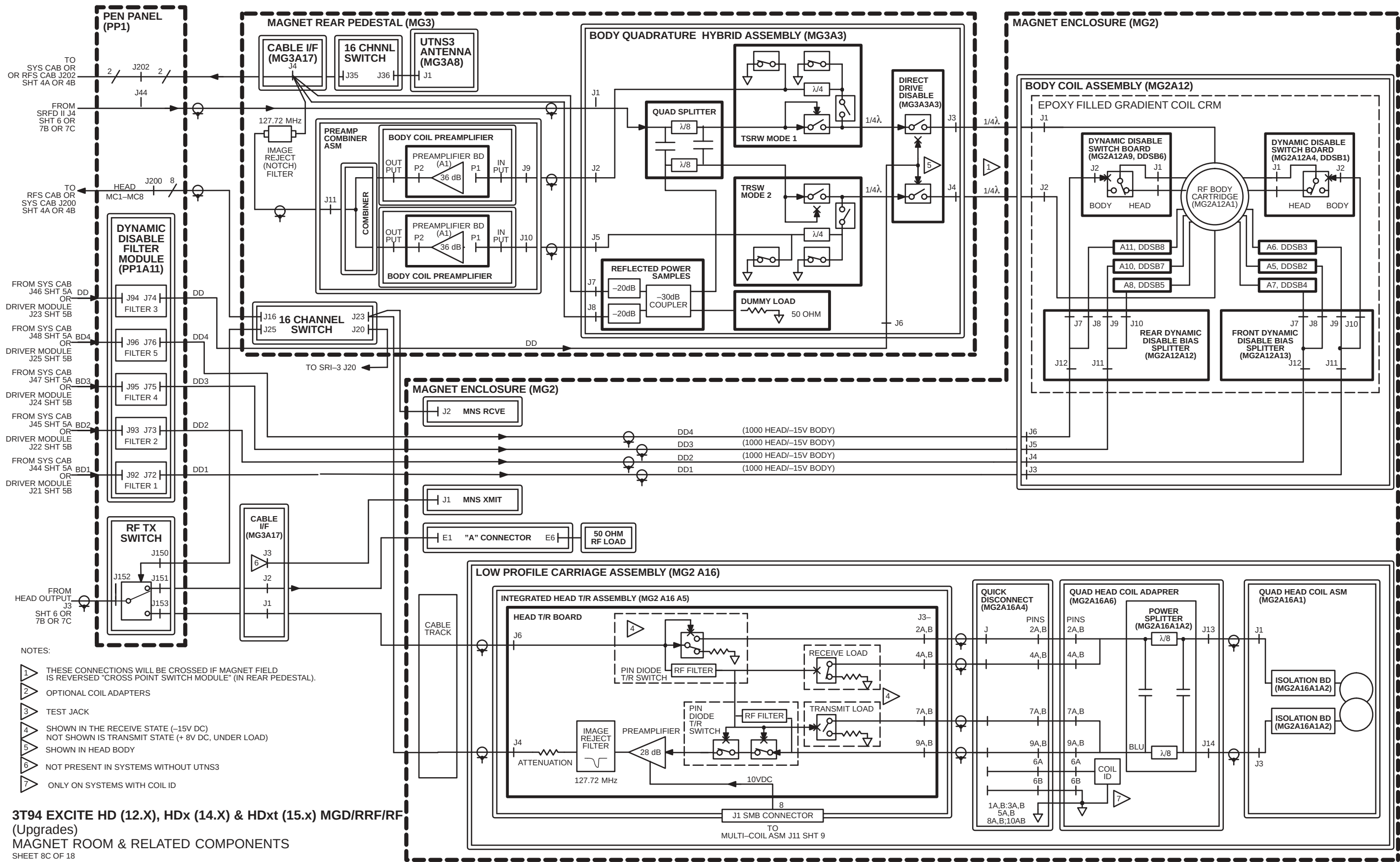


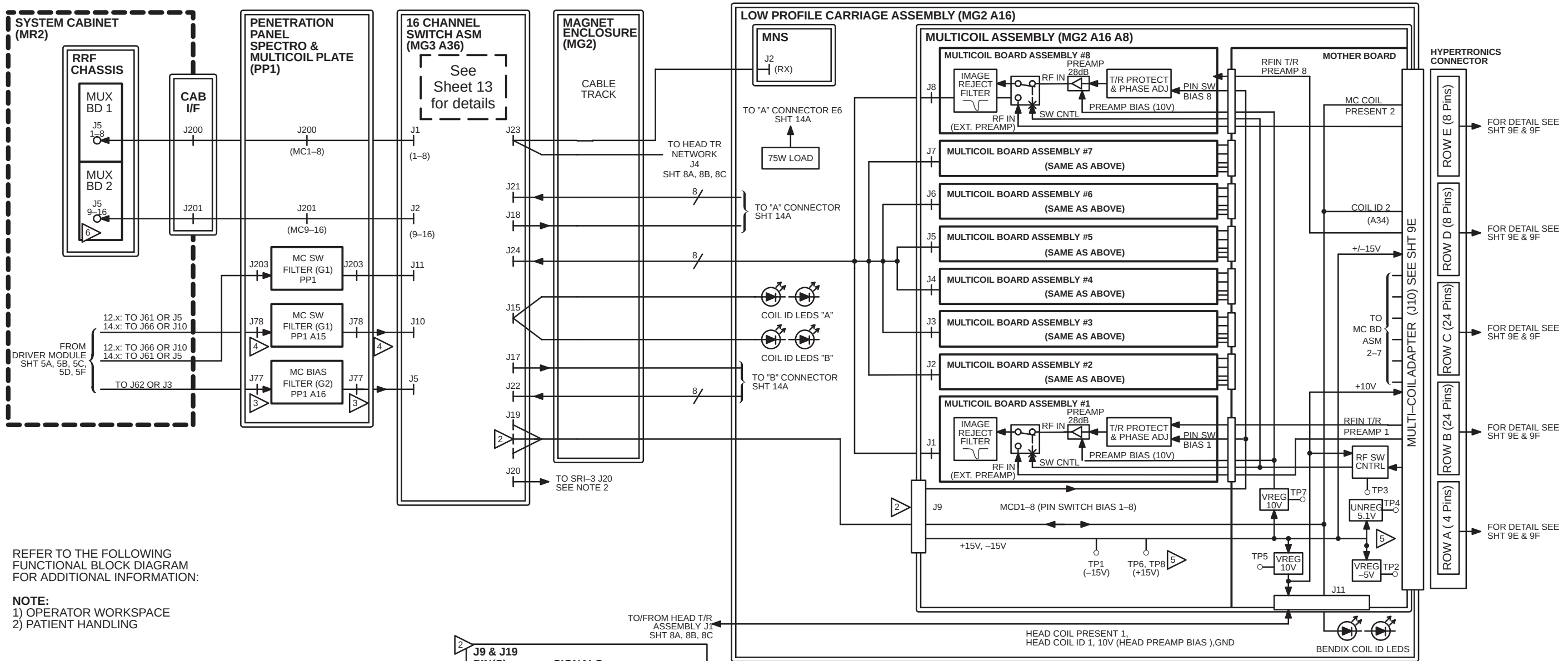
OR











G3 & 3T94 EXCITE HD (12.x), HDx (14.x) & HDxt (15.x) MGD/RRF/RF
(12.x – Forward Production & Upgrades)
(14.x – Upgrades)
MULTI-COIL MODULES
SHEET 9A OF 18

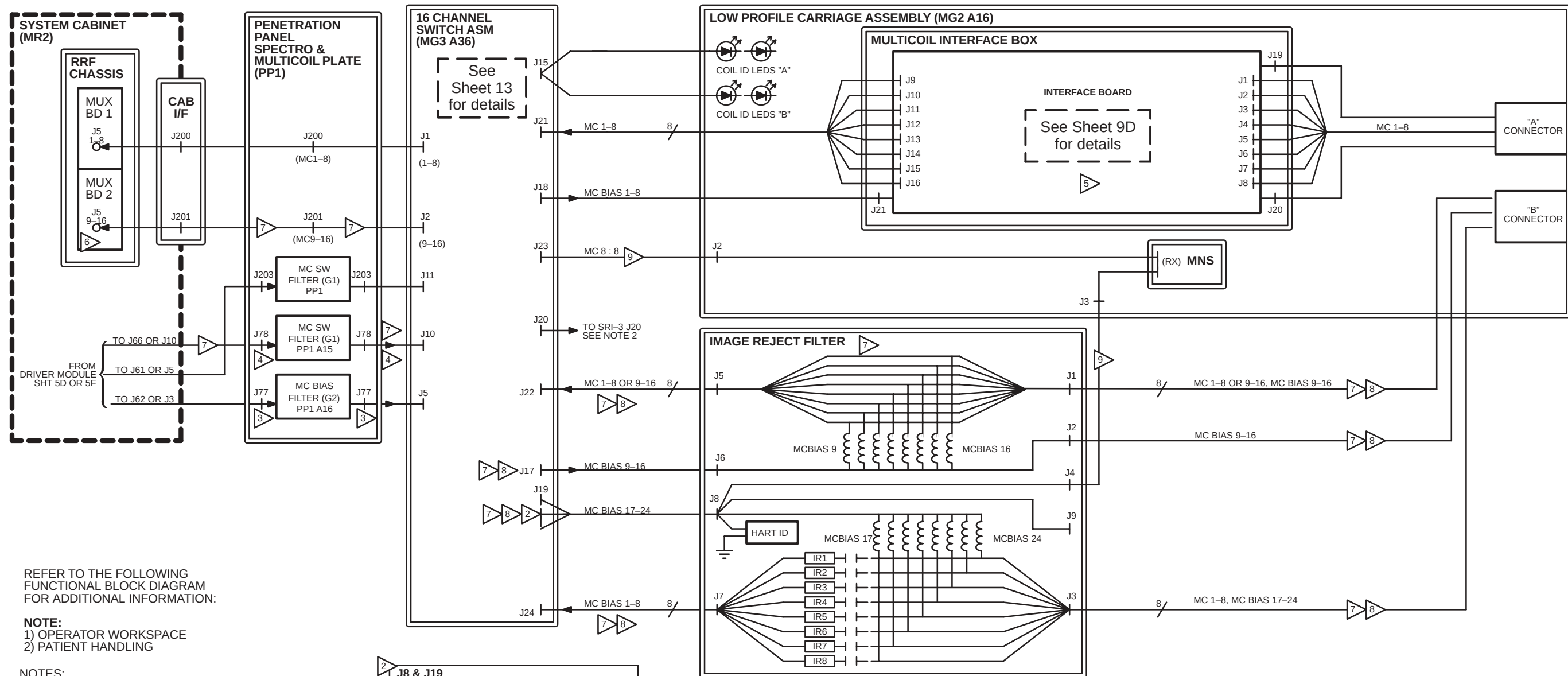
2	J9 & J19 PIN(S) SIGNALS
1,14	 -15V
2,15	 +15V
3, 4, 23-25	 GND
5	 MCD8
6	 MCD4
7	 MCD7
8	 MCD3
9	 MCD6
10	 MCD2
11	 MCD5
12	 MCD1
13	 CTL_PREAMP (NOTE 1)
16	 RED_LED
17	 YEL_LED
18	 GRN_LED
19	 COIL PRESENT #1
20	 COIL PRESENT #2
21	 COIL ID #1
22	 COIL ID #2
NOTE 1: PIN 13 DEFAULT LOW. LOW: PREAMP POWER ON. HIGH: PREAMP POWER OFF.	

3	J62 OR J3 & J77 PIN SIGNAL NAME
1	 XPT_DATA_POS
2	 XPT_STROBE_POS
3	 XPT_OE_N_POS
4	 AGND
5	 AGND
6	 +15V (NOTE 1)
7	 +15V (NOTE 1)
8	 +15V (NOTE 1)
9	 XPT_DATA_NEG
10	 XPT_STROBE_NEG
11	 XPT_OE_N_NEG
12	 AGND
13	 AGND
14	 -15V (MC COILS)
15	 -15V (MC COILS)
NOTE 1: ANALOG POWER FOR PREAMPS (MC & HEAD), CROSSPOINT & MC COILS.	

4	J10 & J78 PIN(S) SIGNAL
1-8	 MCD1-8 (NOTE 1)
9	 MC CABLE PRESENT_O IN (NOTE 2)
10,11	 NO CONNECT
12-19	 MCD9-16 (NOTE 3)
20-27	 GND
29	 MC CABLE PRESENT_1 IN (NOTE 2)
30-37	 GND
NOTE 1: MULTICOIL BIAS (+3V,+5V,+7V,-5V). NOTE 2: THESE TWO PINS CONNECTED TOGETHER ON J10 FOR DIAGNOSTIC LOOPBACK. NOTE 3: NOT USED/CONNECTED AT J10 INSIDE CROSSPOINT SWITCH ASSEMBLY.	

5	LOCATED BEHIND MULTICOIL TEST BD ASSY	POINT	FUNCTION
7	 TP1		-15V
6	 TP2		-5V
6	 TP3		RF SW CNTRL SIG
2, 3	 TP4		5.1V
1, 2	 TP5		+10V_B
1, 2	 TP6		+15V
7, 8	 TP7		+10V_A
7, 8	 TP8		+15V

6 Only 16 Channel Systems have a second MUX board. J201 not used in 8 and 4 Channel systems.



REFER TO THE FOLLOWING
FUNCTIONAL BLOCK DIAGRAM
FOR ADDITIONAL INFORMATION:

NOTE:

- 1) OPERATOR WORKSPACE
- 2) PATIENT HANDLING

NOTES:

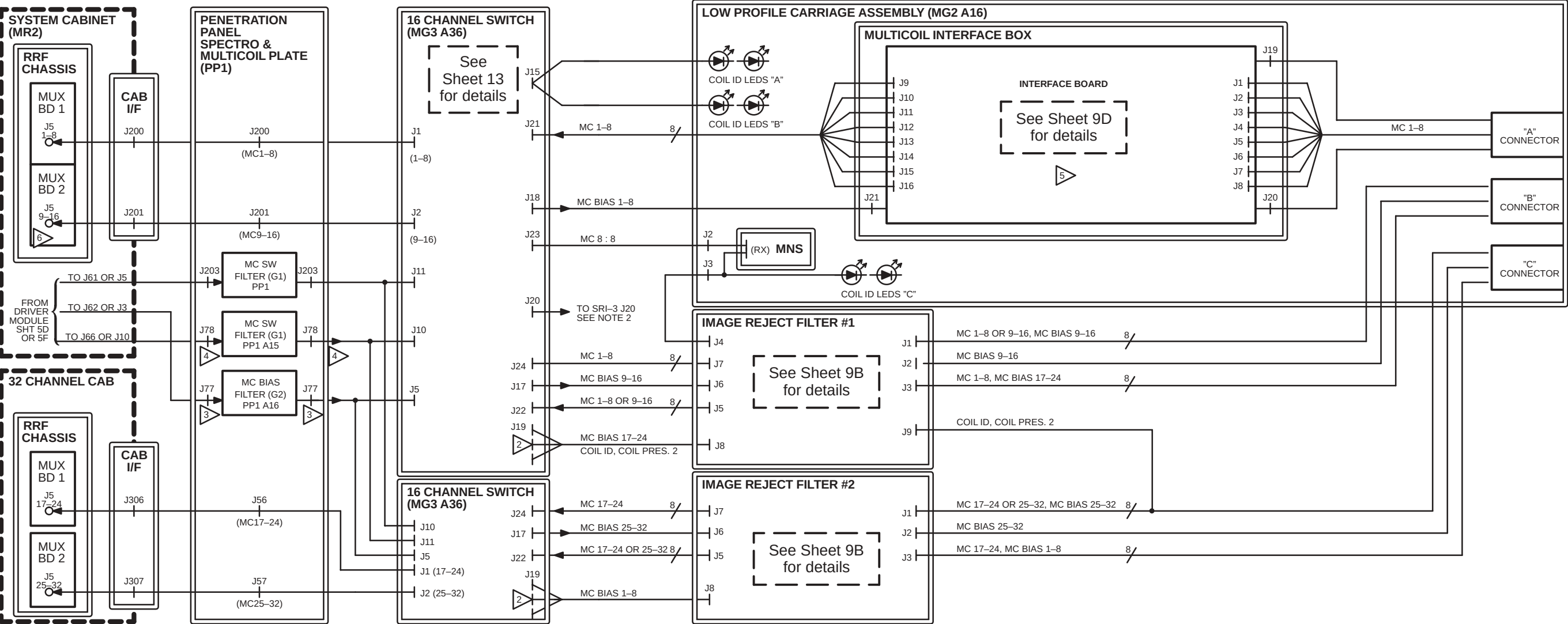
2	J8 & J19
	PIN(S) SIGNALS
1,14	-15V
2,15	+15V
3, 4, 23-25	... GND
5	 MCD8
6	 MCD4
7	 MCD7
8	 MCD3
9	 MCD6
10	 MCD2
11	 MCD5
12	 MCD1
13	 CTL_PREAMP (NOTE 1)
16	 RED_LED
17	 YEL_LED
18	 GRN_LED
19	 COIL PRESENT #1
20	 COIL PRESENT #2
21	 COIL ID #1
22	 COIL ID #2

MC PREAMP BIAS FILTER PINOUTS	
3	J62 OR J3 & J77
PIN	SIGNAL NAME
1	XPT_DATA_POS
2	XPT_STROBE_POS
3	XPT_OE_N_POS
4	AGND
5	AGND
6	+15V (NOTE 1)
7	+15V (NOTE 1)
8	+15V (NOTE 1)
9	XPT_DATA_NEG
10	XPT_STROBE_NEG
11	XPT_OE_N_NEG
12	AGND
13	AGND
14	-15V (MC COILS)
15	-15V (MC COILS)

4	J10 OR J66 & J78 PIN(S) SIGNAL
1-8	MCD1-8 (NOTE 1)
9	MC CABLE PRESENT_O IN (NOTE 2)
10,11	NO CONNECT
12-19	MCD9-16 (NOTE 3)
20-27	GND
29	MC CABLE PRESENT_1 IN (NOTE 2)
30-37	GND
NOTE 1:	MULTICOIL BIAS (+3V,+5V,+7V,-5V).
NOTE 2:	THESE TWO PINS CONNECTED TOGETHER ON J10 FOR DIAGNOSTIC LOOPBACK.
NOTE 3:	NOT USED/CONNECTED AT J10 INSTEAD CROSSPOINT SWITCH ASSEMBLY.

TESTPOINT . FUNCTION

- 6 Only 16 or 32 Channel Systems have a second MUX board.
- 7 Required on 16 or 32 Channel systems.
- 8 B Connector and MC Bias lines are not included in newer 8 Channel systems.
- 9 Optional: See MNS Manual



REFER TO THE FOLLOWING
FUNCTIONAL BLOCK DIAGRAM
FOR ADDITIONAL INFORMATION:

NOTE:
1) OPERATOR WORKSPACE
2) PATIENT HANDLING

NOTES:

1

G3 HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
32 CHANNEL MULTI-COIL
SHEET 9C OF 18

J8 & J19	PIN(S) SIGNALS
1,14	 -15V
2,15	 +15V
3, 4, 23-25	... GND
5	 MCD8
6	 MCD4
7	 MCD7
8	 MCD3
9	 MCD6
10	 MCD2
11	 MCD5
12	 MCD1
13	 CTL_PREAMP (NOTE 1)
16	 RED_LED
17	 YEL_LED
18	 GRN_LED
19	 COIL PRESENT #1
20	 COIL PRESENT #2
21	 COIL ID #1
22	 COIL ID #2

NOTE 1: PIN 13 DEFAULT LOW.
LOW: PREAMP POWER ON.
HIGH: PREAMP POWER OFF.

J62 OR J3 & J77	PIN SIGNAL NAME
1	 XPT_DATA_POS
2	 XPT_STROBE_POS
3	 XPT_OE_N_POS
4	 AGND
5	 AGND
6	 +15V (NOTE 1)
7	 +15V (NOTE 1)
8	 +15V (NOTE 1)
9	 XPT_DATA_NEG
10	 XPT_STROBE_NEG
11	 XPT_OE_N_NEG
12	 AGND
13	 AGND
14	 -15V (MC COILS)
15	 -15V (MC COILS)

NOTE 1: ANALOG POWER FOR
PREAMPS (MC & HEAD),
CROSSPOINT & MC COILS.

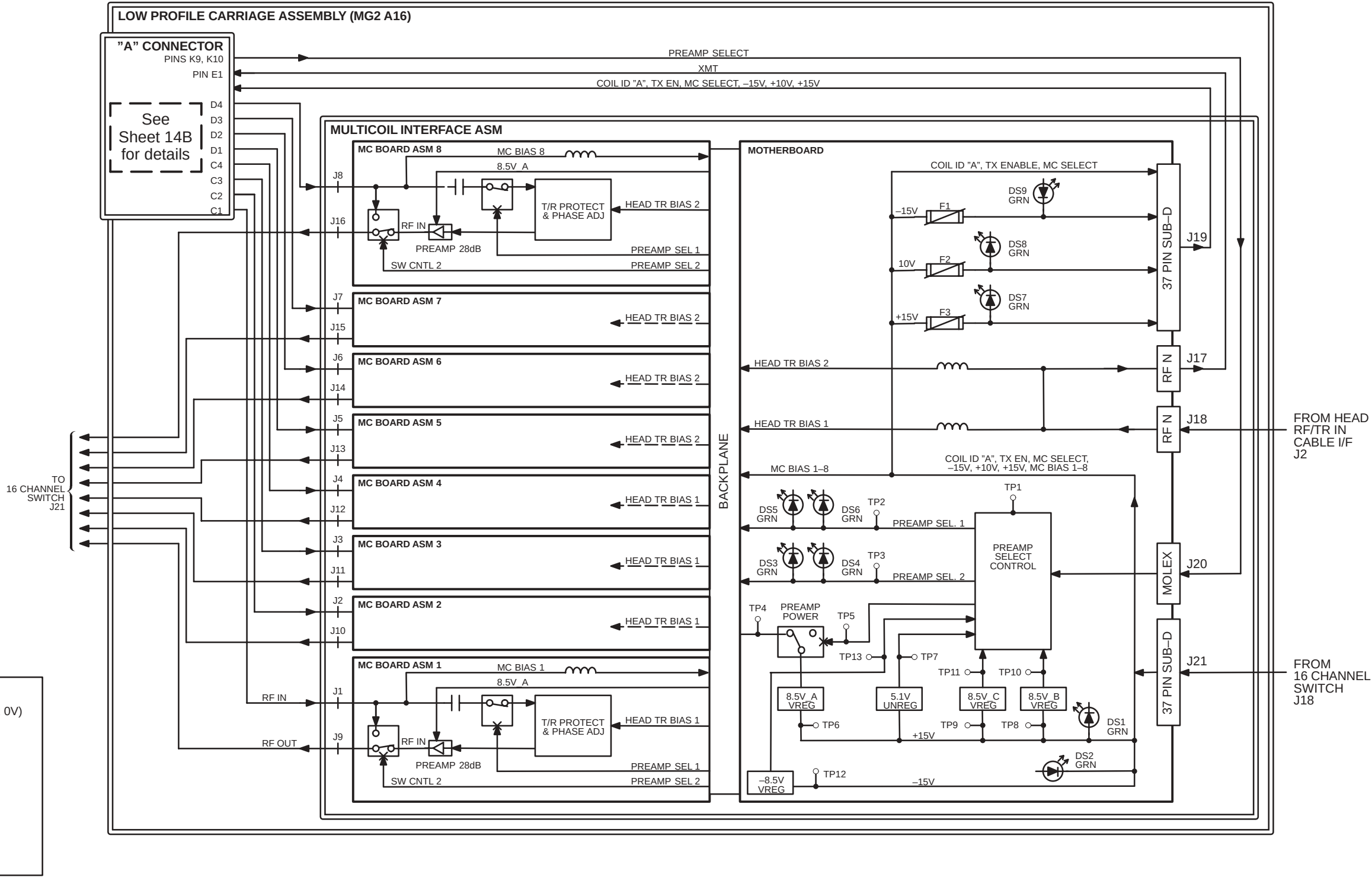
J10 OR J66 & J78	PIN(S) SIGNAL
1-8	 MCD1-8 (NOTE 1)
9	 MC CABLE PRESENT_O IN (NOTE 2)
10,11	 NO CONNECT
12-19	 MCD9-16 (NOTE 3)
20-27	 GND
29	 MC CABLE PRESENT_1 IN (NOTE 2)
30-37	 GND

NOTE 1: MULTICOIL BIAS (+3V,+5V,+7V,-5V).
NOTE 2: THESE TWO PINS CONNECTED TOGETHER
ON J10 FOR DIAGNOSTIC LOOPBACK.
NOTE 3: NOT USED/CONNECTED AT J10 INSIDE
CROSSPOINT SWITCH ASSEMBLY.

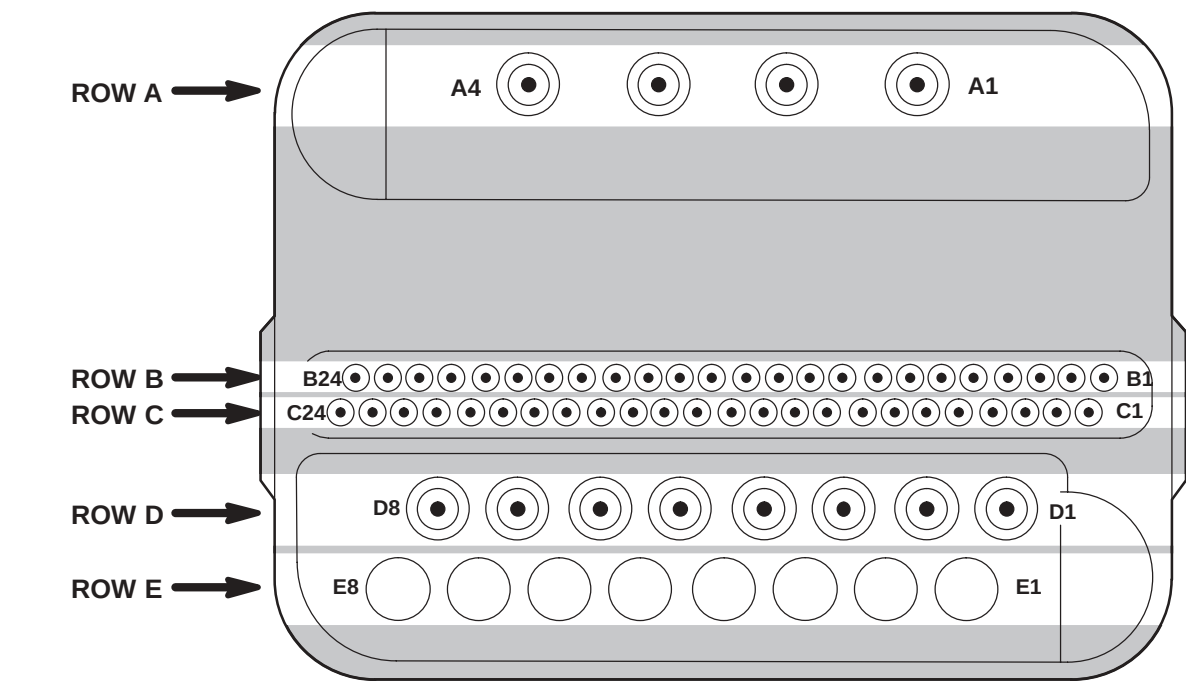
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TESTPOINT	FUNCTION
TP1	SYSTEM PREAMP ENABLE (+15V OR 0V)
TP2	PREAMP SELECT 1
TP3	PREAMP SELECT 2
TP4	8.5V_A
TP5	+10V_B CNTRL
TP6	+15V
TP7	+5.1V
TP8	+15V
TP9	+15V
TP10	8.5V_B
TP11	8.5V_C
TP12	-15V
TP13	-8.5V

6 Only 16 Channel Systems have a
second MUX board. J201 not used
in 8 and 4 Channel systems.



TESTPOINT	FUNCTION
TP1	SYSTEM PREAMP ENABLE (+15V OR 0V)
TP2	PREAMP SELECT 1
TP3	PREAMP SELECT 2
TP4	8.5V_A
TP5	+10V_B CNTRL
TP6	+15V
TP7	+5.1V
TP8	+15V
TP9	+15V
TP10	8.5V_B
TP11	8.5V_C
TP12	-15V
TP13	-8.5V



As viewed from the front of the Hypertronic connector

HYPERTRONICS CONNECTOR
(ROW A PINOUTS)
HEAD AND SURFACE COIL TO SYSTEM INTERCONNECT

Pin Out		Description	Style	PreAmp on Coil Connection Requirements	System PreAmp Connection Requirements
A	1	Head Coil Receive w/Transmit Load	Co–Axial	N/A	
A	2	N/A		N/A	
A	3	N/A		N/A	
A	4	Head Coil Transmit w/Transmit Load	Co–Axial	N/A	

G3 & 3T94 EXCITE HD (12.x), HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
HYPERTRONIC CONNECTOR PINOUTS
(12.x – Forward Production & Upgrades)
(14.x – Upgrades)
SHEET 9E OF 18

NOTE: For Pin–outs B10 thru B17 below, MC Bias is 17–24 on 14.x systems.

HYPERTRONICS CONNECTOR
(ROW B PINOUTS)
DC, COIL ID, AND MULTICOIL BIAS TO SYSTEM INTERCONNECT

Pin Out		Description	Style	PreAmp on Coil Connection Requirements	System PreAmp Connection Requirements
B	1	+ 15 V DC	Pin		
B	2	Ground	Pin	Ground	Ground
B	3	+10 V_BPreAmp DC	Pin	+10 Volt 1A for PreAmp coil.	N/C
B	4	Ground	Pin	Ground	Ground
B	5	Spare	Pin		
B	6	Spare	Pin		
B	7	Spare	Pin		
B	8	Coil ID	Pin		
B	9	Ground	Pin		
B	10	T/R Switch Bias 1 (17)	Pin	T/R Switch Bias	T/R Switch Bias
B	11	T/R Switch Bias 2 (18)	Pin	T/R Switch Bias	T/R Switch Bias
B	12	T/R Switch Bias 3 (19)	Pin	T/R Switch Bias	T/R Switch Bias
B	13	T/R Switch Bias 4 (20)	Pin	T/R Switch Bias	T/R Switch Bias
B	14	T/R Switch Bias 5 (21)	Pin	T/R Switch Bias	T/R Switch Bias
B	15	T/R Switch Bias 6 (22)	Pin	T/R Switch Bias	T/R Switch Bias
B	16	T/R Switch Bias 7 (23)	Pin	T/R Switch Bias	T/R Switch Bias
B	17	T/R Switch Bias 8 (24)	Pin	T/R Switch Bias	T/R Switch Bias
B	18	Integrated Preamp (System) Switch Control	Pin	For Preamp on Coil: Connect B18 with C17	For Non–Preamp on Coil: Leave this pin as Non–Connect
B	19	Spare	Pin		
B	20	Coil Present	Pin	Must connect B20 to C6	Must connect B20 to C6
B	21	Spare	Pin		
B	22	+15 VDC	Pin	1A max	1A max
B	23	Spare	Pin		
B	24	Spare	Pin		

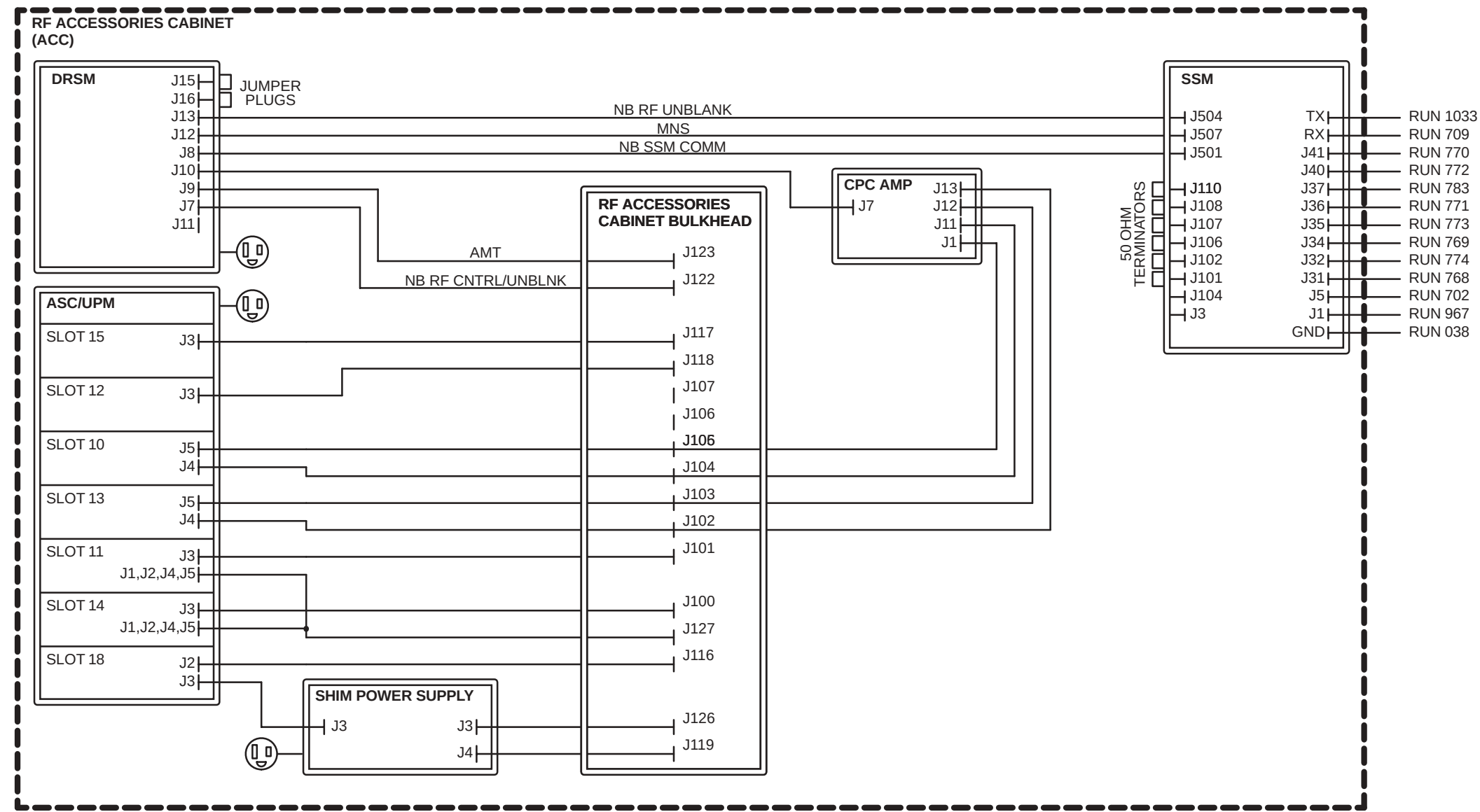
HYPERTRONICS CONNECTOR (ROW C PINOUTS) DC, COIL ID, AND MULTICOIL TO SYSTEM INTERCONNECT					
Pin Out		Description	Style	PreAmp on Coil Connection Requirements	System PreAmp Connection Re-quirements
C	1	Spare	Pin		
C	2	Spare	Pin		
C	3	Spare	Pin		
C	4	Spare	Pin		
C	5	Spare	Pin		
C	6	Coil Present Return	Pin	Must connect B20 to C6	Must connect B20 to C6
C	7	Coil ID Return	Pin		
C	8	Ground	Pin	Ground	Ground
C	9	Reserved	Pin	Reserved	Reserved
C	10	Reserved	Pin	Reserved	Reserved
C	11	Reserved	Pin	Reserved	Reserved
C	12	Reserved	Pin	Reserved	Reserved
C	13	Reserved	Pin	Reserved	Reserved
C	14	Reserved	Pin	Reserved	Reserved
C	15	Reserved	Pin	Reserved	Reserved
C	16	Reserved	Pin	Reserved	Reserved
C	17	Inegrated Preamp Switch Control RTN	Pin	For Preamp on Coil: Connect B18 with C17	For Non–Preamp on coil, leave this pin as Non–Connect
C	18	Ground	Pin	Ground	Ground
C	19	Spare	Pin		
C	20	Ground	Pin	Ground	Ground
C	21	+10V_BPreAmp DC	Pin	+10 Volt 1A for Preamp Coil	N/C
C	22	Ground	Pin	Ground	Ground
C	23	Spare	Pin		
C	24	Spare	Pin		

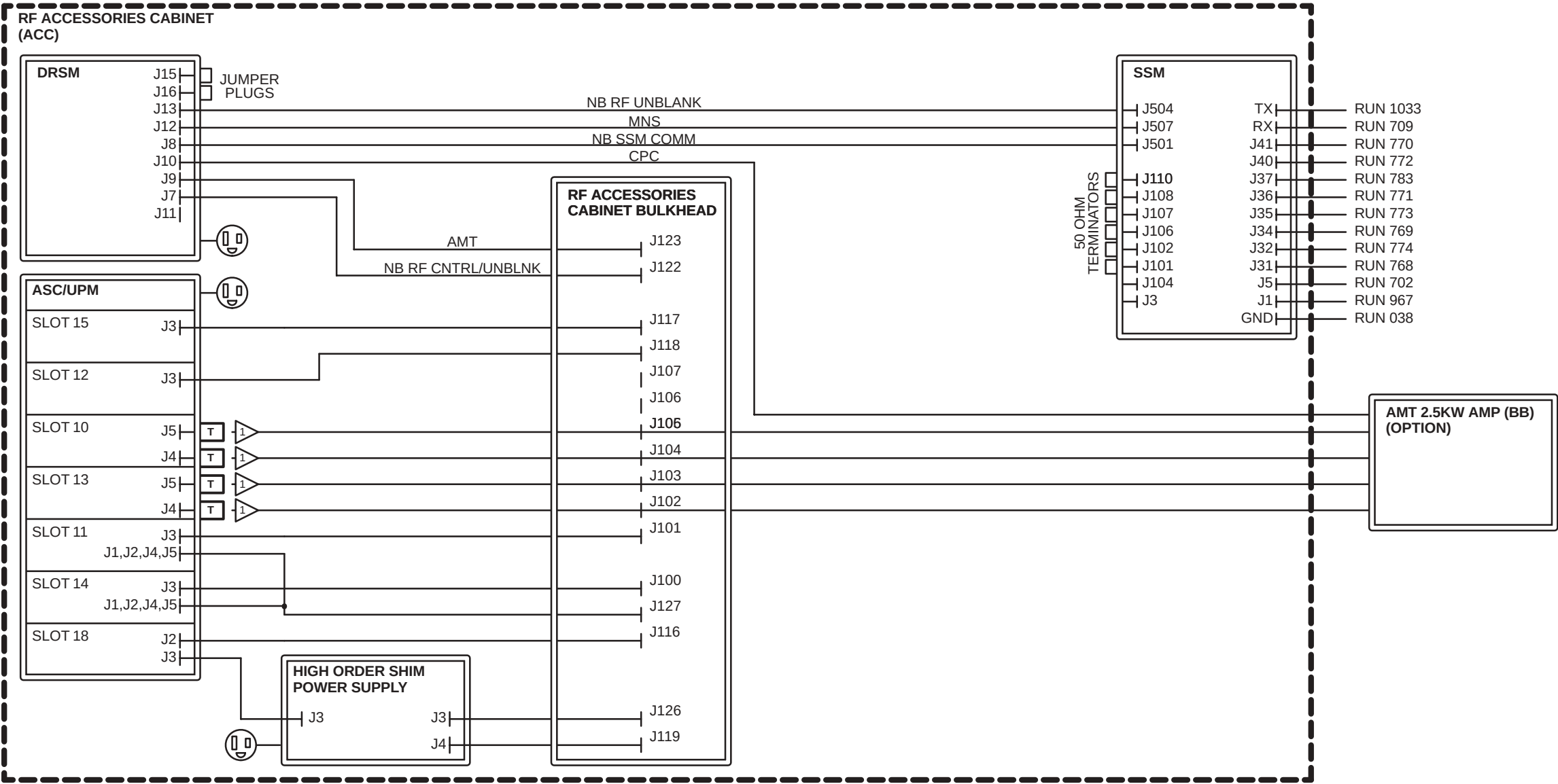
HYPERTRONICS CONNECTOR (ROW C PINOUTS)
DC, COIL ID, AND MULTICOIL TO SYSTEM INTERCONNECT

Pin Out		Description	Style	PreAmp on Coil Connection Requirements	System PreAmp Connection Requirements
D	1	Coil Integrated Preamp RF1 In	Co–Axial	PreAmp on Coil RF in / No Bias	
D	2	Coil Integrated Preamp RF2 In	Co–Axial	PreAmp on Coil RF in / No Bias	
D	3	Coil Integrated Preamp RF3 In	Co–Axial	PreAmp on Coil RF in / No Bias	
D	4	Coil Integrated Preamp RF4 In	Co–Axial	PreAmp on Coil RF in / No Bias	
D	5	Coil Integrated Preamp RF5 In	Co–Axial	PreAmp on Coil RF in / No Bias	
D	6	Coil Integrated Preamp RF6 In	Co–Axial	PreAmp on Coil RF in / No Bias	
D	7	Coil Integrated Preamp RF7 In	Co–Axial	PreAmp on Coil RF in / No Bias	
D	8	Coil Integrated Preamp RF8 In	Co–Axial	PreAmp on Coil RF in / No Bias	

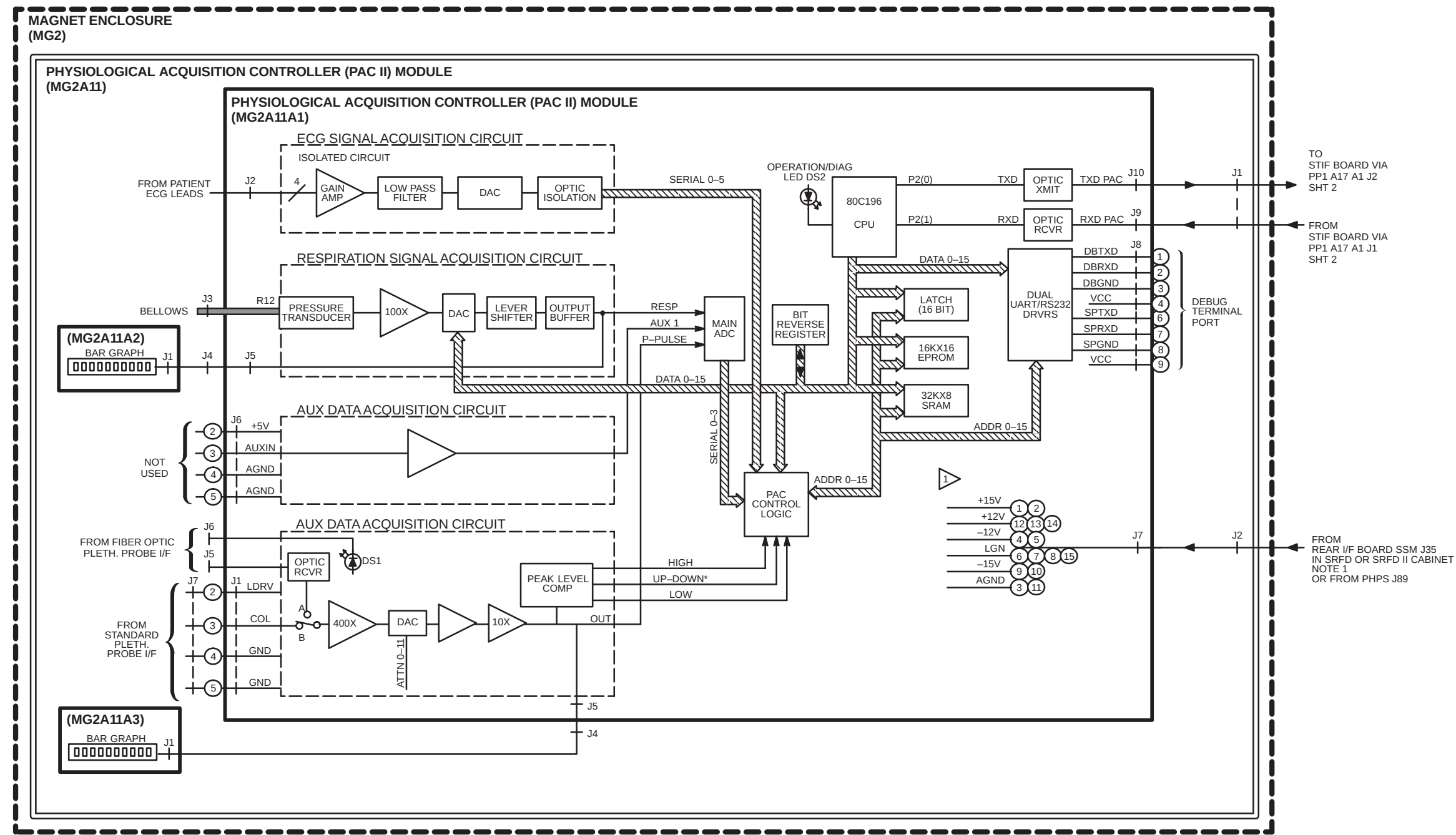
HYPERTRONICS CONNECTOR (ROW E PINOUTS)
SYSTEM PREAMPLIFIER MULTICOIL TO SYSTEM INTERCONNECT

Pin Out		Description	Style	PreAmp on Coil Connection Requirements	System PreAmp Connection Requirements
E	1	RF1 In to System PreAmp	Co–Axial		Non–PreAmp on Coil RF in with / Bias
E	2	RF2 In to System PreAmp	Co–Axial		Non–PreAmp on Coil RF in with / Bias
E	3	RF3 In to System PreAmp	Co–Axial		Non–PreAmp on Coil RF in with / Bias
E	4	RF4 In to System PreAmp	Co–Axial		Non–PreAmp on Coil RF in with / Bias
E	5	RF5 In to System PreAmp	Co–Axial		Non–PreAmp on Coil RF in with / Bias
E	6	RF6 In to System PreAmp	Co–Axial		Non–PreAmp on Coil RF in with / Bias
E	7	RF7 In to System PreAmp	Co–Axial		Non–PreAmp on Coil RF in with / Bias
E	8	RF8 In to System PreAmp	Co–Axial		Non–PreAmp on Coil RF in with / Bias





NOTES:
1 IF SITE HAS MNS, THESE CONNECTIONS WILL GO TO MNS AMPLIFIER CABINET. IF MNS NOT PART OF SYSTEM, THEN UPM BOARDS MUST HAVE 50 OHM TERMINATORS AT J4 AND J5 AT SLOTS 10 AND 13.



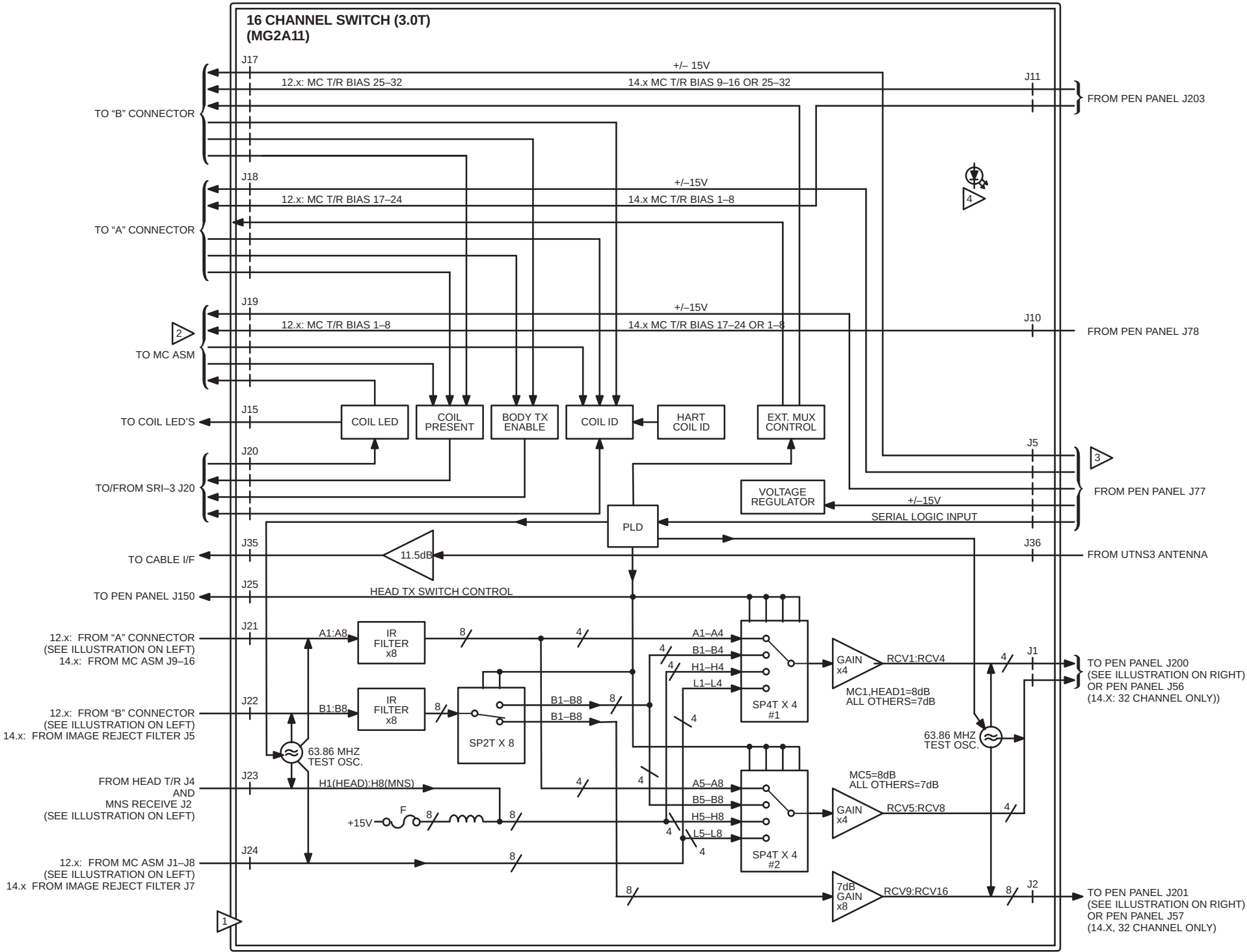
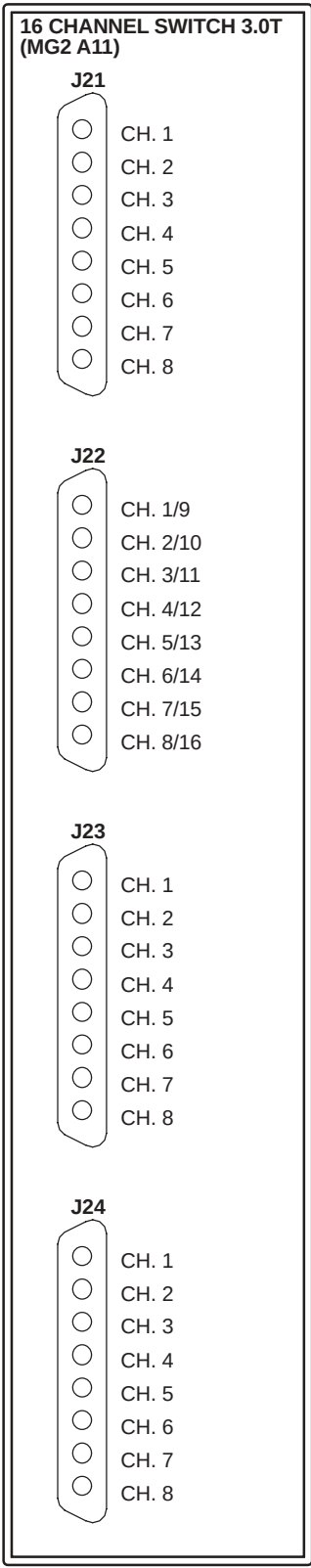
NOTES:

1) MULTIPLE WIRES/PINS WITH SAME SIGNAL ARE NOT SHOWN

REFER TO THE FOLLOWING FUNCTIONAL BLOCK DIAGRAM FOR ADDITIONAL INFORMATION:

NOTE:

1) PATIENT HANDLING



NOTES:

- 1 Switch positions shown in 16 Channel Receive mode.
- 2 More connector pinouts present on sheet 9A to 9C.

J5 PINS . SIGNALS

- 1 XPT_DATA_POS
- 2 XPT_STROBE_POS
- 3 XPT_OE_POS
- 4, 5 OPEN
- 6, 7, 8 ... +15V
- 9, 10 OPEN
- 11, 12 ... GND
- 13 OPEN
- 14 XPT_DATA_N
- 15 XPT_STROBE_N
- 16 XPT_OE_N_N
- 17, 18 ... OPEN
- 19, 20, 21 GND
- 22, 23 ... OPEN
- 24, 25 ... -15V

LED'S

- COIL ID A GREEN
- COIL ID A RED
- COIL ID B GREEN
- COIL ID B RED
- COIL ID LEGACY GREEN
- COIL ID LEGACY RED
- COIL PRESENT A
- COIL PRESENT B
- COIL PRESENT 1
- COIL PRESENT 2
- TX ENABLE A
- TX ENABLE B
- DIAG OUT
- DIAG IN
- -15V POWER
- +15V POWER
- +24V POWER
- J21 ON CH. 1-4
- J22 ON CH. 1-4
- J23 ON CH. 1-4
- J24 ON CH. 1-4
- J21 ON CH. 5-8
- J22 ON CH. 5-8
- J23 ON CH. 5-8
- J24 ON CH. 5-8
- J22 ON CH. 9-12
- J22 ON CH. 13-16

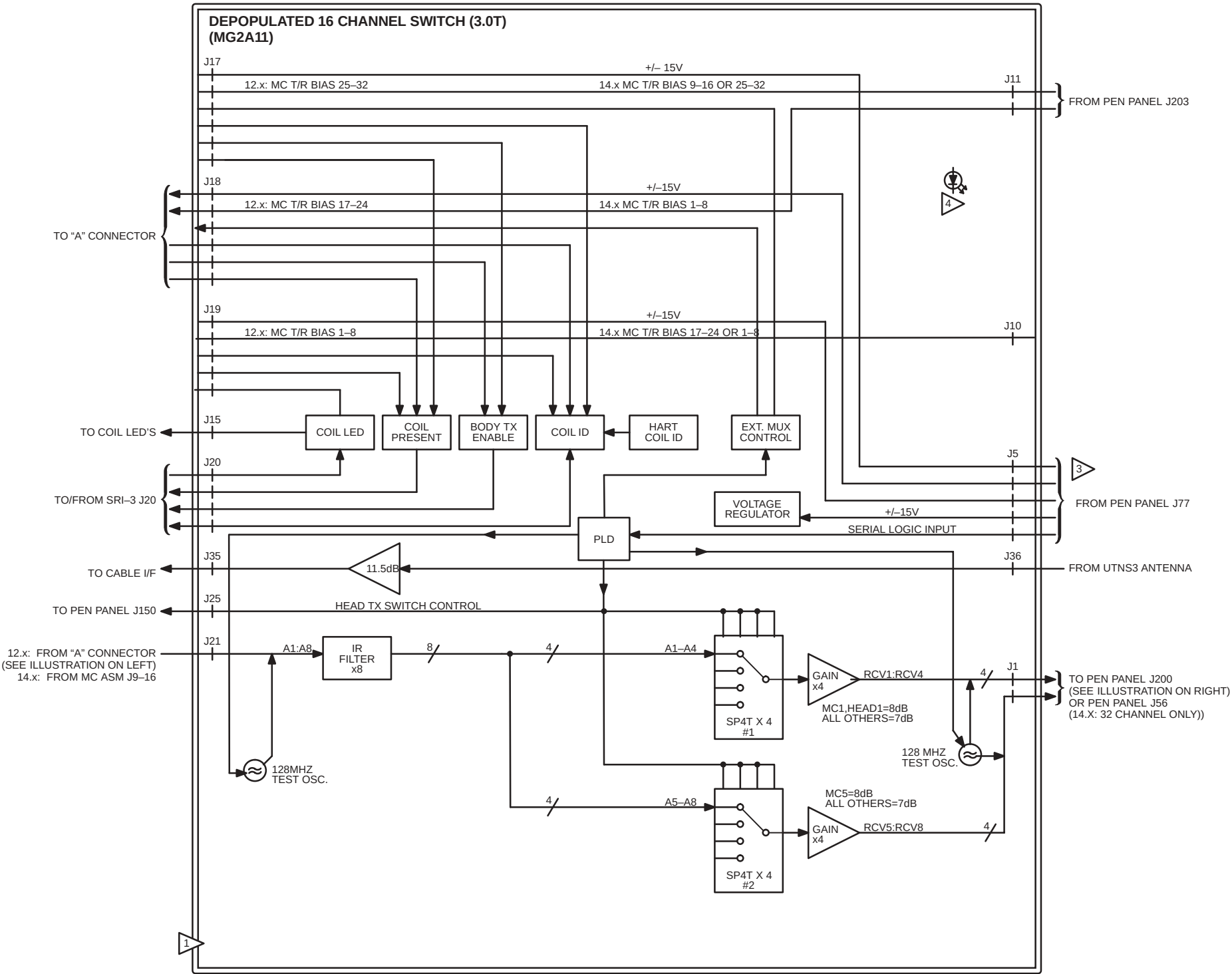
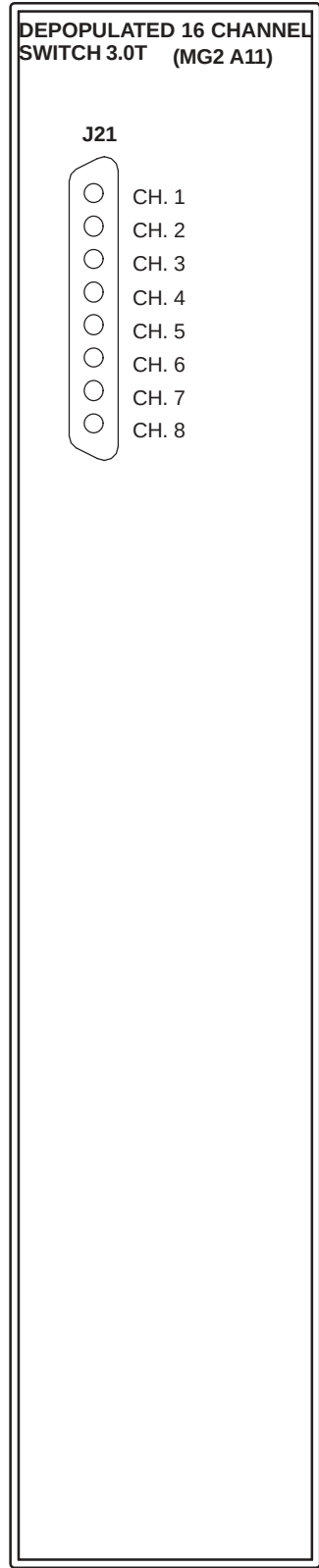
16 CHANNEL SWITCH 1.5T (MG2 A11)

J1

- CH. 1
- CH. 2
- CH. 3
- CH. 4
- CH. 5
- CH. 6
- CH. 7
- CH. 8

J2

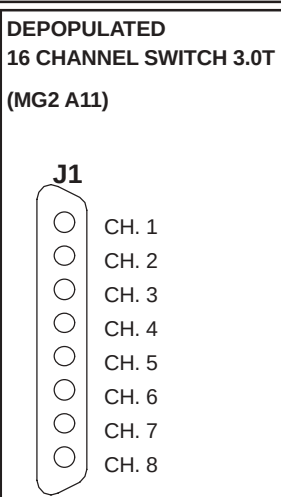
- CH. 9
- CH. 10
- CH. 11
- CH. 12
- CH. 13
- CH. 14
- CH. 15
- CH. 16

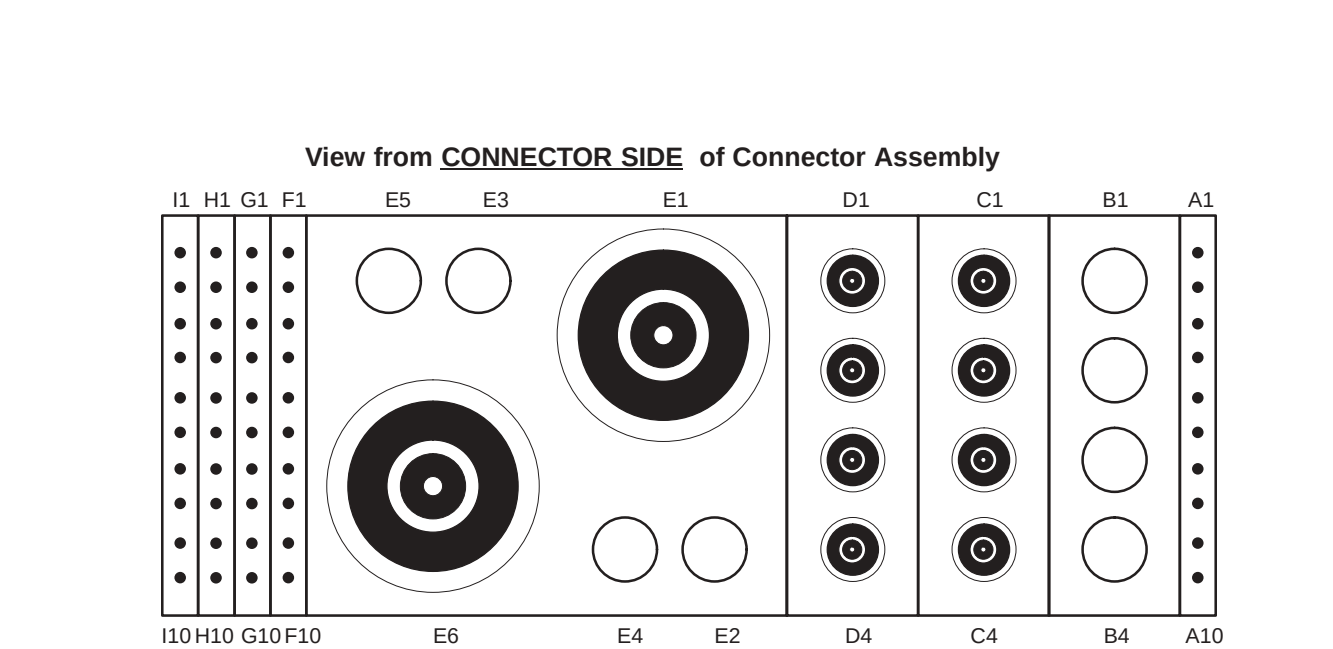


- NOTES:**
- 1 Switch positions shown in 8 Channel Receive mode.
 - 2 More connector pinouts present on sheet 9A.

- 3 J5 PINS . SIGNALS**
- 1 XPT_DATA_POS
 - 2 XPT_STROBE_POS
 - 3 XPT_OE_POS
 - 4, 5 OPEN
 - 6, 7, 8 ... +15V
 - 9, 10 ... OPEN
 - 11, 12 ... GND
 - 13 OPEN
 - 14 XPT_DATA_N
 - 15 XPT_STROBE_N
 - 16 XPT_OE_N_N
 - 17, 18 ... OPEN
 - 19, 20, 21 GND
 - 22, 23 ... OPEN
 - 24, 25 ... -15V

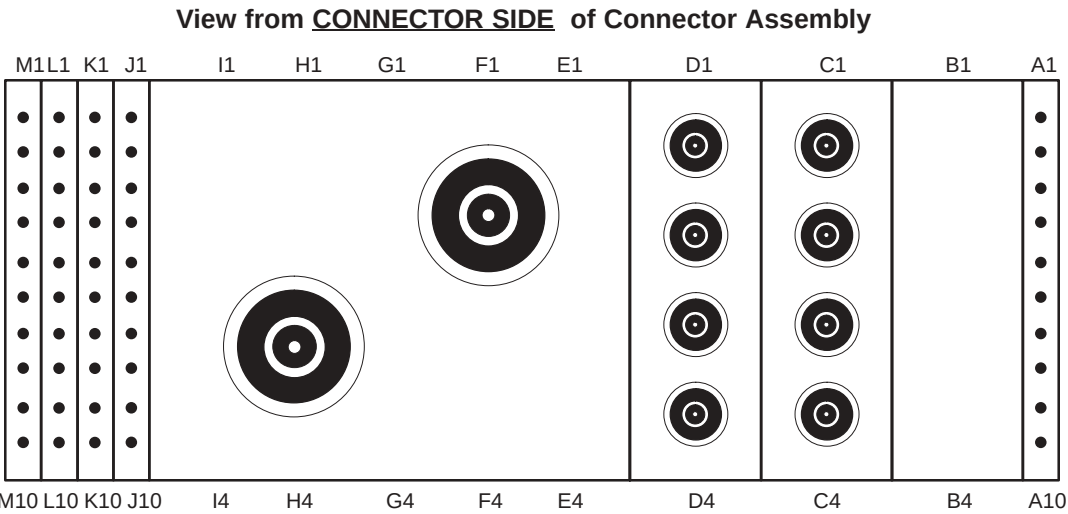
- 4 LED's**
- COIL ID A GREEN
 - COIL ID A RED
 - COIL PRESENT A
 - TX ENABLE A
 - -15V POWER
 - +15V POWER
 - +24V POWER
 - J21 ON CH. 1-4





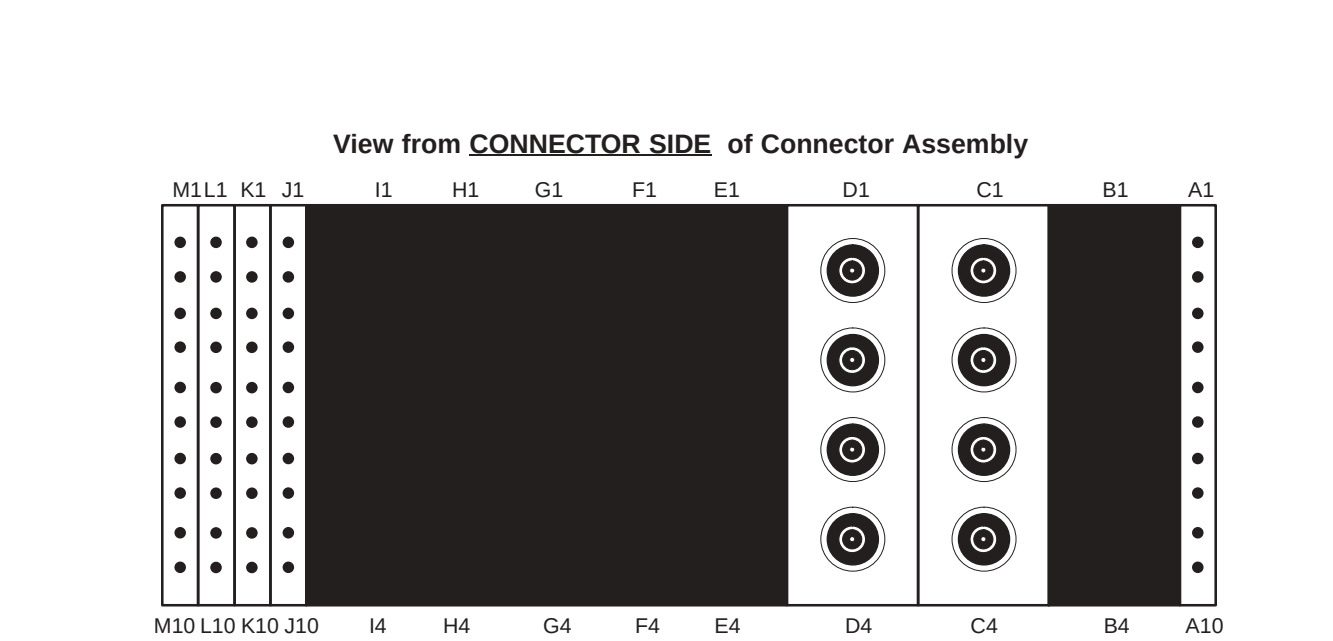
Contact Function Allocation			
Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
B1	n/a	Key blocked in receptacle E	
B2	n/a	Key blocked in receptacle F	
B3	n/a	Key blocked in receptacle G	
B4	n/a	Key blocked in receptacle H	
C1	n/a	MC-1A,B or MC-9B	Preamp out 1, no bias, DC blocked
C2	n/a	MC-2A,B or MC-10B	Preamp out 2, no bias, DC blocked
C3	n/a	MC-3A,B or MC-11B	Preamp out 3, no bias, DC blocked
C4	n/a	MC-4A,B or MC-12B	Preamp out 4, no bias, DC blocked
D1	n/a	MC-5A,B or MC-13B	Preamp out 5, no bias, DC blocked
D2	n/a	MC-6A,B or MC-14B	Preamp out 6, no bias, DC blocked
D3	n/a	MC-7A,B or MC-15B	Preamp out 7, no bias, DC blocked
D4	n/a	MC-8A,B or MC-16B	Preamp out 8, no bias, DC blocked

Contact Function Allocation			
Pin	Twisted with	Description	Comments
E1	n/a	Transmit signal up to 2kw (1.5T) or 4kw (3.0T) PEP.	“A” Connector only. Not present in “B” Connector
E2	n/a	Key blocked in receptacle C	
E3	n/a	Key blocked in receptacle A	Blocked for “A”
E4	n/a	Key blocked in receptacle B	Blocked for “B”
E5	n/a	Key blocked in receptacle D	
E6	n/a	Reflected power to 50 ohm load 2.5kw PEP, 50w RMS.	“A” Connector only. Not present in “B” Connector
F1	G1	MC bias 1	MC T/R Bias Line 17 for “A”, 25 for “B”
F2	G2	MC bias 2	MC T/R Bias Line 18 for “A”, 26 for “B”
F3	G3	MC bias 3	MC T/R Bias Line 19 for “A”, 27 for “B”
F4	G4	MC bias 4	MC T/R Bias Line 20 for “A”, 28 for “B”
F5	G5	MC bias 5	MC T/R Bias Line 21 for “A”, 29 for “B”
F6	G6	MC bias 6	MC T/R Bias Line 22 for “A”, 30 for “B”
F7	G7	MC bias 7	MC T/R Bias Line 23 for “A”, 31 for “B”
F8	G8	MC bias 8	MC T/R Bias Line 24 for “A”, 32 for “B”
F9	F10	body transmit enable	
F10	F9	body transmit enable return	
G1	F1	MC bias return 1	MC T/R Bias Line 17 for “A”, 25 for “B”
G2	F2	MC bias return 2	MC T/R Bias Line 18 for “A”, 26 for “B”
G3	F3	MC bias return 3	MC T/R Bias Line 19 for “A”, 27 for “B”
G4	F4	MC bias return 4	MC T/R Bias Line 20 for “A”, 28 for “B”
G5	F5	MC bias return 5	MC T/R Bias Line 21 for “A”, 29 for “B”
G6	F6	MC bias return 6	MC T/R Bias Line 22 for “A”, 30 for “B”
G7	F7	MC bias return 7	MC T/R Bias Line 23 for “A”, 31 for “B”
G8	F8	MC bias return 8	MC T/R Bias Line 24 for “A”, 32 for “B”
G9	n/a	Reserved	
G10	n/a	Reserved	
H1	n/a	Reserved	
H2	n/a	Reserved	
H3	n/a	Reserved	
H4	n/a	Reserved	
H5	n/a	Reserved	
H6	n/a	Reserved	
H7	n/a	Reserved	
H8	n/a	Reserved	
H9	n/a	Reserved	
H10	n/a	Reserved	
I1	n/a	Reserved	
I2	I5	+15Vdc (auxiliary & preamp power)	
I3	I6	+10Vdc (auxiliary & preamp power)	
I4	n/a	-15Vdc (auxiliary power)	
I5	I2	Return for aux power	
I6	I3	Return for aux power	
I7	I8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
I8	I7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
I9	I10	coil present'	
I10	I9	coil present return'	



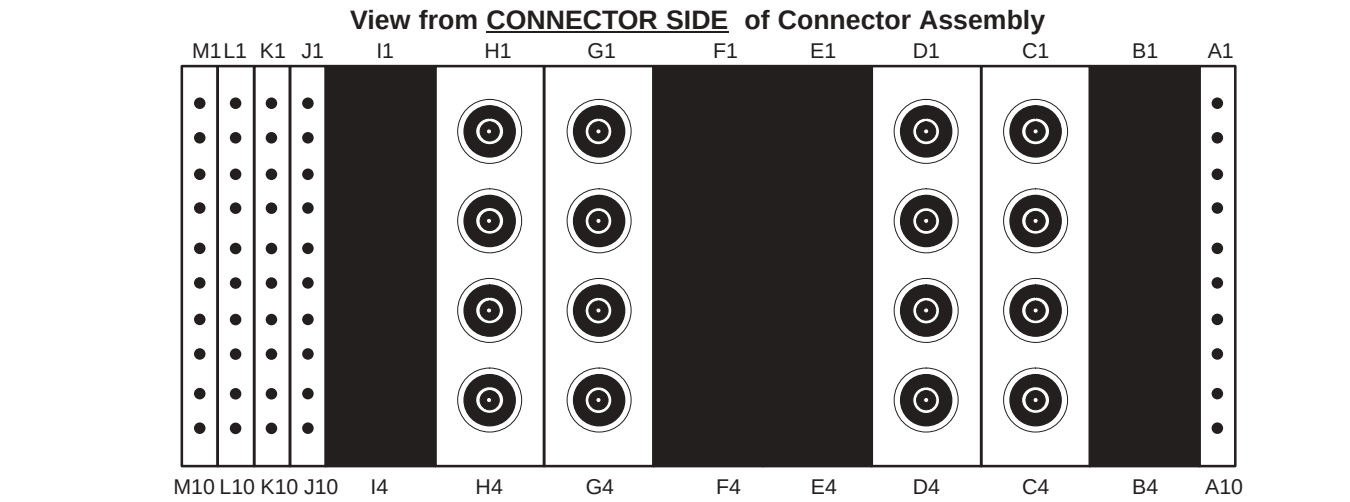
Contact Function Allocation			
Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
B1	n/a	Key blocked in receptacle E	
B2	n/a	Key blocked in receptacle F	
B3	n/a	Key blocked in receptacle G	
B4	n/a	Key blocked in receptacle H	
C1	n/a	MC-A1	Preamp out 1, no bias, DC blocked
C2	n/a	MC-A2	Preamp out 2, no bias, DC blocked
C3	n/a	MC-A3	Preamp out 3, no bias, DC blocked
C4	n/a	MC-A4	Preamp out 4, no bias, DC blocked
D1	n/a	MC-A5	Preamp out 5, no bias, DC blocked
D2	n/a	MC-A6	Preamp out 6, no bias, DC blocked
D3	n/a	MC-A7	Preamp out 7, no bias, DC blocked
D4	n/a	MC-A8	Preamp out 8, no bias, DC blocked

Contact Function Allocation			
Pin	Twisted with	Description	Comments
F2	n/a	Transmit Signal up to 8kw PEP	
H3	n/a	Reflected power to 50 ohm load 2.5kw PEP, 50w RMS	"A" connector only
J1	K1	MC bias 1	
J2	K2	MC bias 2	
J3	K3	MC bias 3	
J4	K4	MC bias 4	
J5	K5	MC bias 5	
J6	K6	MC bias 6	
J7	K7	MC bias 7	
J8	K8	MC bias 8	
J9	J10	body transmit enable	
J10	J9	body transmit enable return	
K1	J1	MC bias return 1	
K2	J2	MC bias return 2	
K3	J3	MC bias return 3	
K4	J4	MC bias return 4	
K5	J5	MC bias return 5	
K6	J6	MC bias return 6	
K7	J7	MC bias return 7	
K8	J8	MC bias return 8	
K9	K10	System preamp enable return	
K10	K9	System preamp enable	
L1	n/a	Reserved	
L2	n/a	Reserved	
L3	n/a	Reserved	
L4	n/a	Reserved	
L5	n/a	Reserved	
L6	n/a	Reserved	
L7	n/a	Reserved	
L8	n/a	Reserved	
L9	n/a	Reserved	
L10	n/a	Reserved	
M1	n/a	Reserved	
M2	M5	+15Vdc (auxiliary & preamp power)	
M3	M6	+10Vdc (auxiliary & preamp power)	
M4	n/a	–15Vdc (auxiliary power)	
M5	M2	Return for aux power	
M6	M3	Return for aux power	
M7	M8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
M8	M7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
M9	M10	coil present'	
M10	M9	coil present return'	



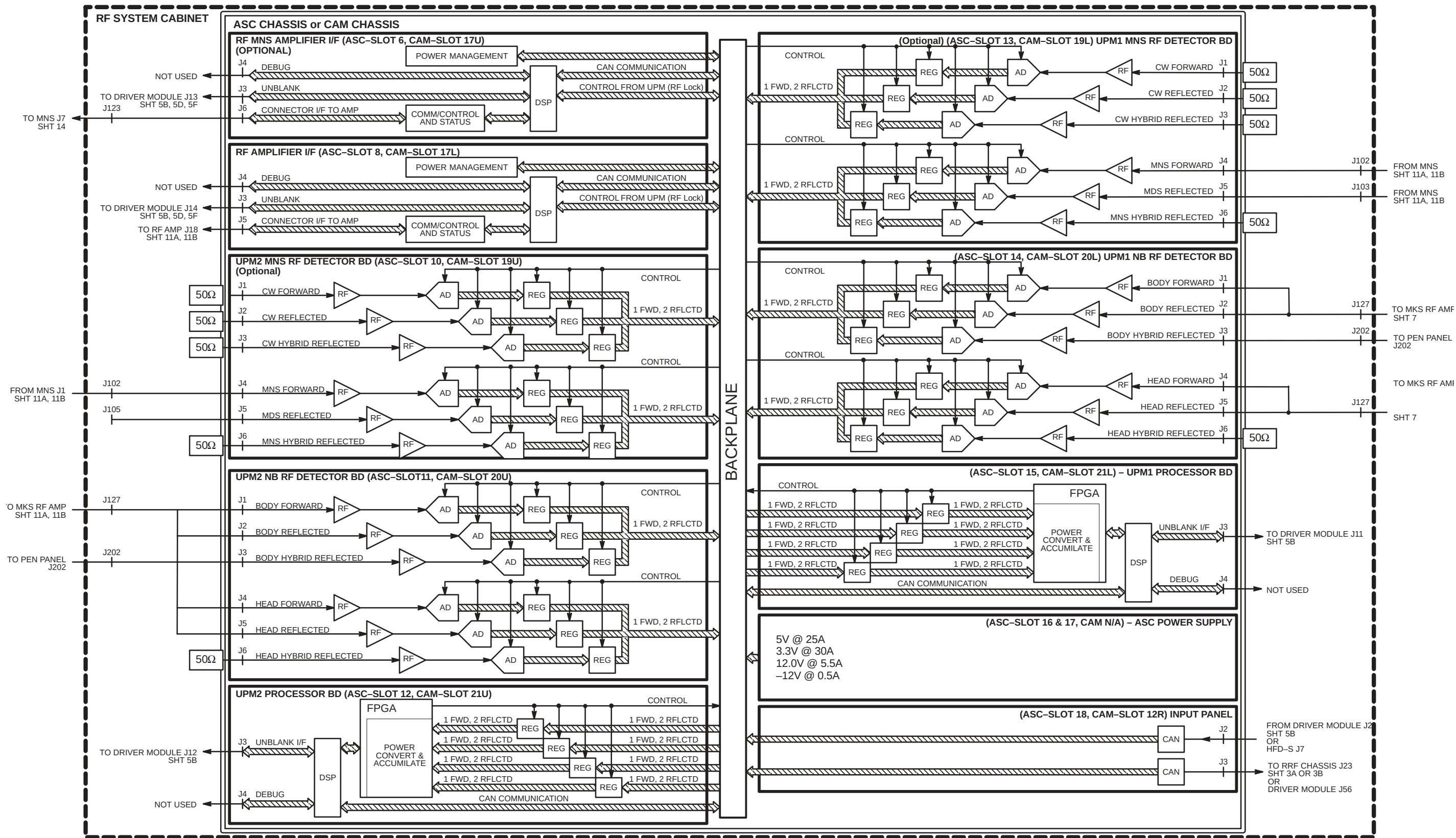
Contact Function Allocation			
Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
B1	n/a		
B2	n/a		
B3	n/a		
B4	n/a		
C1	n/a	MC-B1	Preamp out 1, MC bias 1 (9)
C2	n/a	MC-B2	Preamp out 2, MC bias 2 (10)
C3	n/a	MC-B3	Preamp out 3, MC bias 3 (11)
C4	n/a	MC-B4	Preamp out 4, MC bias 4 (12)
D1	n/a	MC-B5	Preamp out 5, MC bias 5 (13)
D2	n/a	MC-B6	Preamp out 6, MC bias 6 (14)
D3	n/a	MC-B7	Preamp out 7, MC bias 7 (15)
D4	n/a	MC-B8	Preamp out 8, MC bias 8 (16)

Contact Function Allocation			
Pin	Twisted with	Description	Comments
F2	n/a		
J1	K1	MC bias 1 (9)	
J2	K2	MC bias 2 (10)	
J3	K3	MC bias 3 (11)	
J4	K4	MC bias 4 (12)	
J5	K5	MC bias 5 (13)	
J6	K6	MC bias 6 (14)	
J7	K7	MC bias 7 (15)	
J8	K8	MC bias 8 (16)	
J9	J10	body transmit enable	
J10	J9	body transmit enable return	
K1	J1	MC bias return 1 (9)	
K2	J2	MC bias return 2 (10)	
K3	J3	MC bias return 3 (11)	
K4	J4	MC bias return 4 (12)	
K5	J5	MC bias return 5 (13)	
K6	J6	MC bias return 6 (14)	
K7	J7	MC bias return 7 (15)	
K8	J8	MC bias return 8 (16)	
K9	K10	System preamp enable return	
K10	K9	System preamp enable	
L1	n/a	Reserved	
L2	n/a	Reserved	
L3	n/a	Reserved	
L4	n/a	Reserved	
L5	n/a	Reserved	
L6	n/a	Reserved	
L7	n/a	Reserved	
L8	n/a	Reserved	
L9	n/a	Reserved	
L10	n/a	Reserved	
M1	n/a	Reserved	
M2	M5	+15Vdc (auxiliary & preamp power)	
M3	M6	+10Vdc (auxiliary & preamp power)	
M4	n/a	-15Vdc (auxiliary power)	
M5	M2	Return for aux power	
M6	M3	Return for aux power	
M7	M8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
M8	M7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
M9	M10	coil present'	
M10	M9	coil present return'	

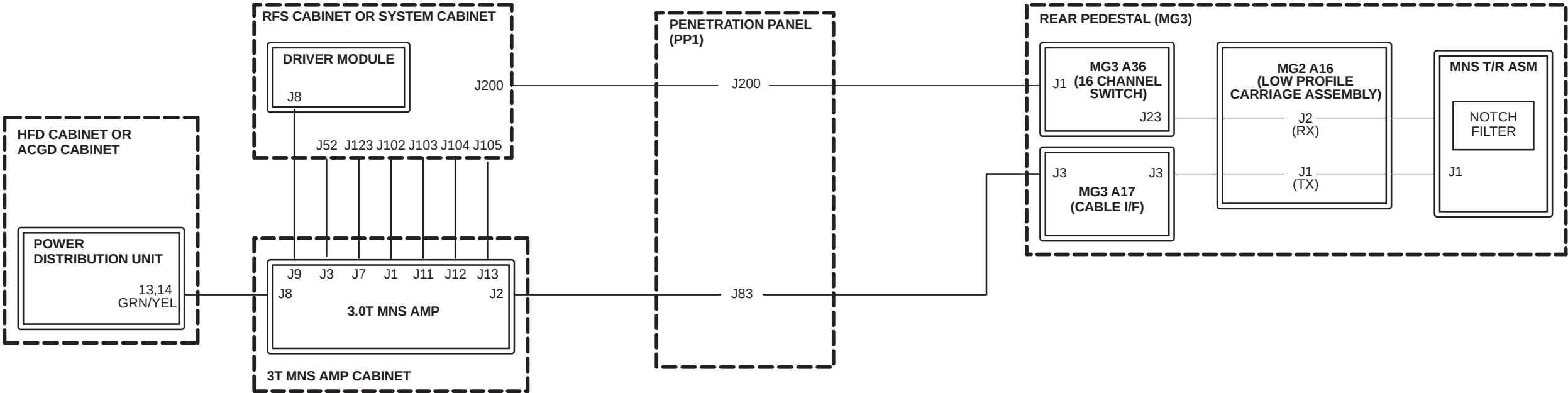


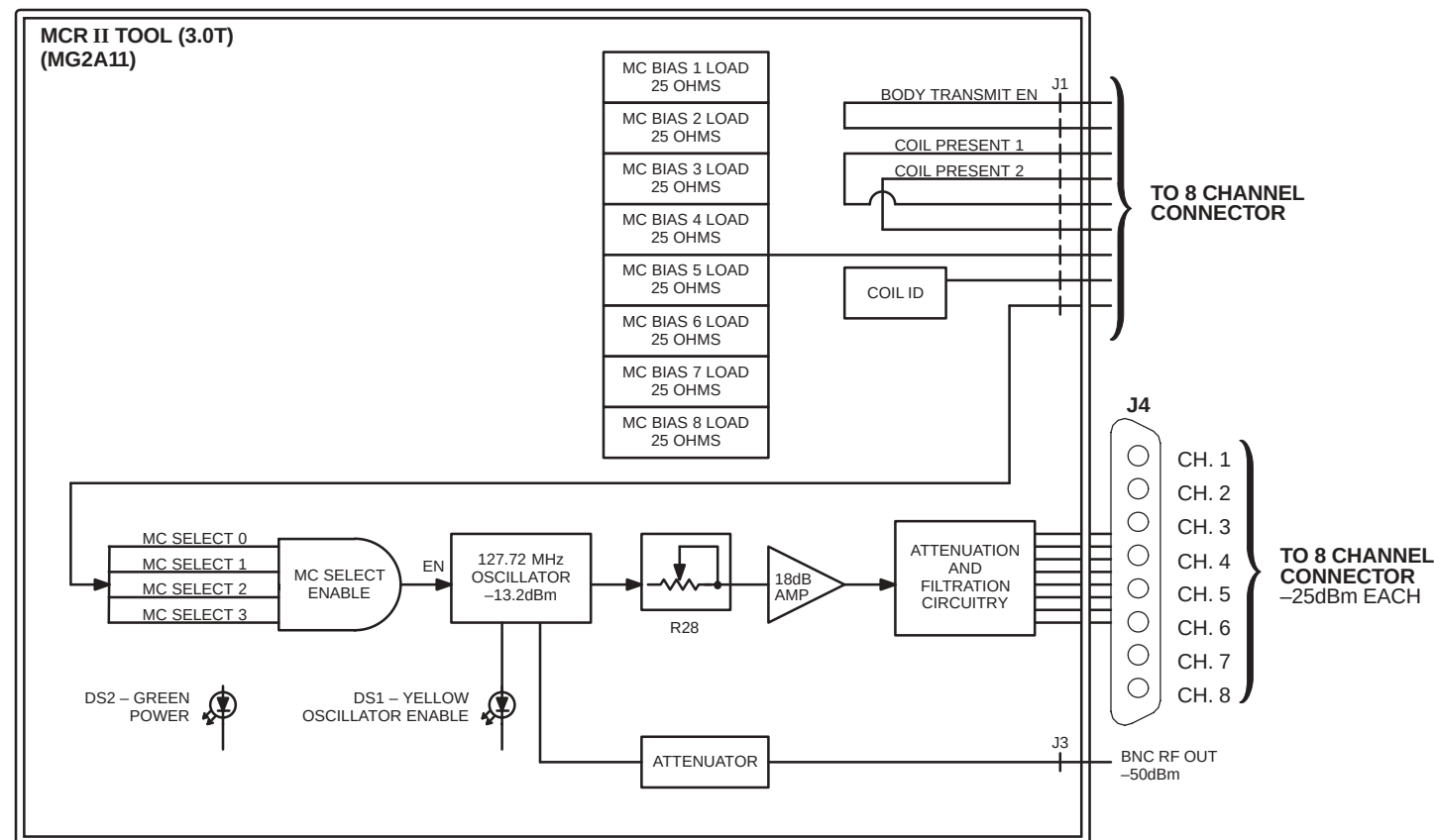
Contact Function Allocation			
Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
C1	n/a	MC-B1 or MC-C1, MC bias 1B (9), 1C (25)	Preamp out 1, MC bias 1B (9), 1C (25)
C2	n/a	MC-B2 or MC-C2, MC bias 2B (10), 2C (26)	Preamp out 2, MC bias 2B (10), 2C (26)
C3	n/a	MC-B3 or MC-C3, MC bias 3B (11), 3C (27)	Preamp out 3, MC bias 3B (11), 3C (27)
C4	n/a	MC-B4 or MC-C4, MC bias 4B (12), 4C (28)	Preamp out 4, MC bias 4B (12), 4C (28)
D1	n/a	MC-B5 or MC-C5, MC bias 5B (13), 5C (29)	Preamp out 5, MC bias 5B (13), 5C (29)
D2	n/a	MC-B6 or MC-C6, MC bias 6B (14), 6C (30)	Preamp out 6, MC bias 6B (14), 6C (30)
D3	n/a	MC-B7 or MC-C7, MC bias 7B (15), 7C (31)	Preamp out 7, MC bias 7B (15), 7C (31)
D4	n/a	MC-B8 or MC-C8, MC bias 8B (16), 8C (32)	Preamp out 8, MC bias 8B (16), 8C (32)
G1	n/a	MC-B9 or MC-C9, MC bias 9B (25), 9C (1)	Preamp out 1, MC bias 9B (25), 9C (1)
G2	n/a	MC-B10 or MC-102, MC bias 10B (26), 10C (2)	Preamp out 2, MC bias 10B (26), 10C (2)
G3	n/a	MC-B11 or MC-C11, MC bias 11B (27), 11C (3)	Preamp out 3, MC bias 11B (27), 11C (3)
G4	n/a	MC-B12 or MC-C12, MC bias 12B (28), 12C (4)	Preamp out 4, MC bias 12B (28), 12C (4)
H1	n/a	MC-B13 or MC-C13, MC bias 13B (29), 13C (5)	Preamp out 5, MC bias 13B (29), 13C (5)
H2	n/a	MC-B14 or MC-C14, MC bias 14B (30), 14C (6)	Preamp out 6, MC bias 14B (30), 14C (6)
H3	n/a	MC-B15 or MC-C15, MC bias 15B (31), 15C (7)	Preamp out 7, MC bias 15B (31), 15C (7)
H4	n/a	MC-B16 or MC-C16, MC bias 16B (32), 16C (8)	Preamp out 8, MC bias 16B (32), 16C (8)

Contact Function Allocation			
Pin	Twisted with	Description	Comments
J1	K1	MC bias 1 (9)-B / MC bias 1 (25)-C	
J2	K2	MC bias 2 (10)-B / MC bias 2 (26)-C	
J3	K3	MC bias 3 (11)-B / MC bias 3 (27)-C	
J4	K4	MC bias 4 (12)-B / MC bias 4 (28)-C	
J5	K5	MC bias 5 (13)-B / MC bias 5 (29)-C	
J6	K6	MC bias 6 (14)-B / MC bias 6 (30)-C	
J7	K7	MC bias 7 (15)-B / MC bias 7 (31)-C	
J8	K8	MC bias 8 (16)-B / MC bias 8 (32)-C	
J9	J10	body transmit enable	
J10	J9	body transmit enable return	
K1	J1	MC bias return 1 (25)-B / MC bias return 1 (1)-C	
K2	J2	MC bias return 2 (26)-B / MC bias return 2 (2)-C	
K3	J3	MC bias return 3 (27)-B / MC bias return 3 (3)-C	
K4	J4	MC bias return 4 (28)-B / MC bias return 4 (4)-C	
K5	J5	MC bias return 5 (29)-B / MC bias return 5 (5)-C	
K6	J6	MC bias return 6 (30)-B / MC bias return 6 (6)-C	
K7	J7	MC bias return 7 (31)-B / MC bias return 7 (7)-C	
K8	J8	MC bias return 8 (32)-B / MC bias return 8 (8)-C	
K9	K10	System preamp enable return	
K10	K9	System preamp enable	
L1	n/a	Reserved	
L2	n/a	Reserved	
L3	n/a	Reserved	
L4	n/a	Reserved	
L5	n/a	Reserved	
L6	n/a	Reserved	
L7	n/a	Reserved	
L8	n/a	Reserved	
L9	n/a	Reserved	
L10	n/a	Reserved	
M1	n/a	Reserved	
M2	M5	+15Vdc (auxiliary & preamp power)	
M3	M6	+10Vdc (auxiliary & preamp power)	
M4	n/a	-15Vdc (auxiliary power)	
M5	M2	Return for aux power	
M6	M3	Return for aux power	
M7	M8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
M8	M7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
M9	M10	coil present'	
M10	M9	coil present return'	

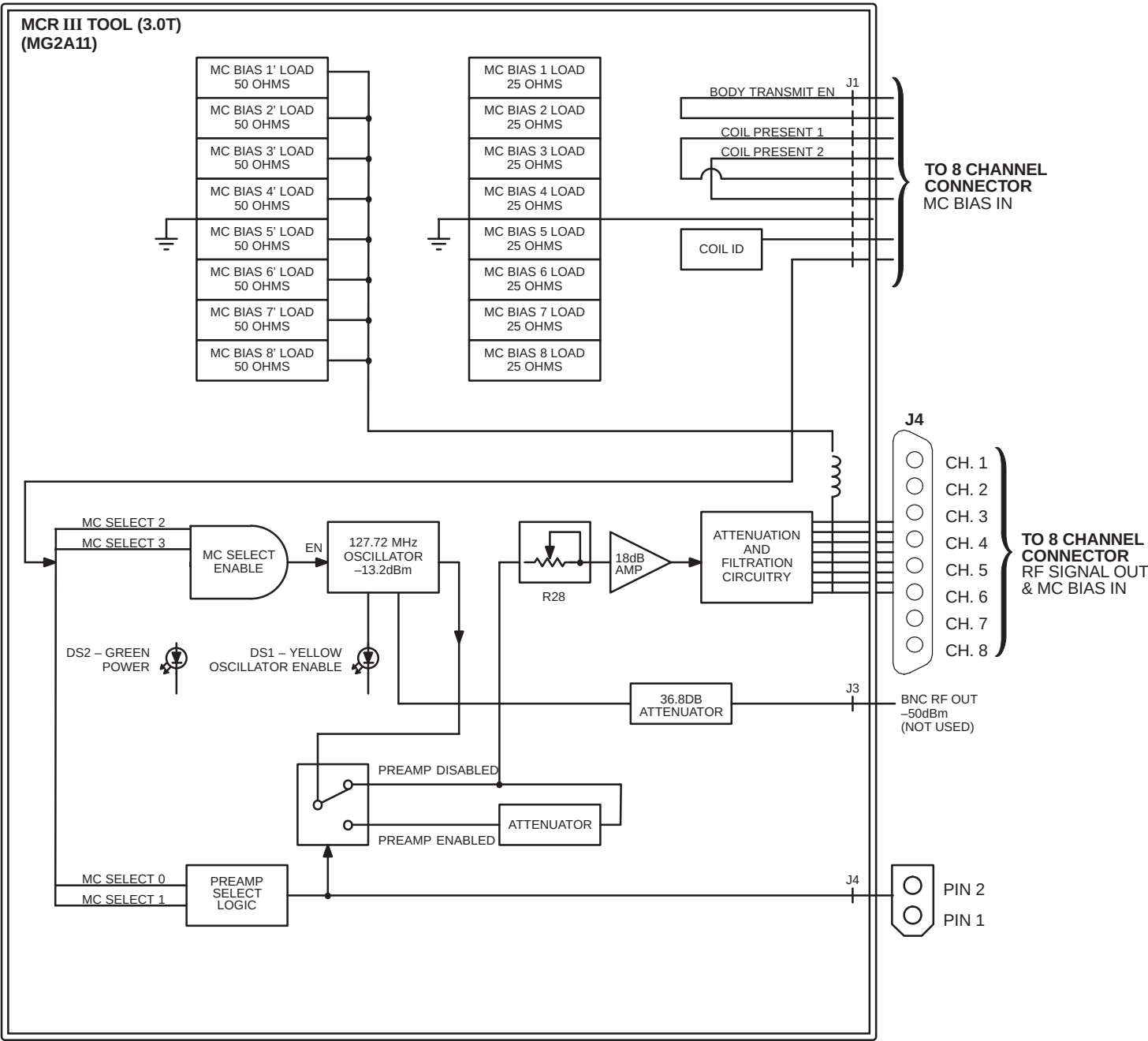


G3 & 3T94 EXCITE (12.x) and HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
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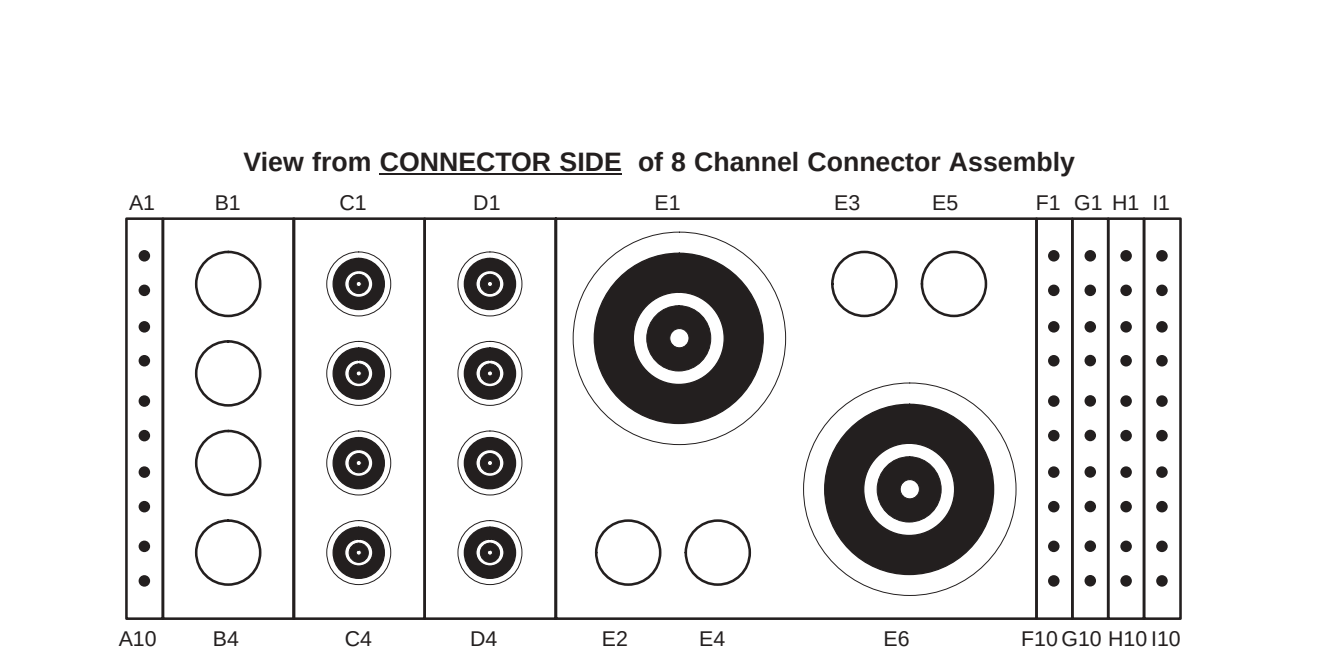


J1	
PINS	SIGNAL
1	MC Bias 1
2	MC Bias 2
3	MC Bias 3
4	MC Bias 4
5	MC Bias 5
6	MC Bias 6
7	MC Bias 7
8	MC Bias 8
9	MC Select 0
10	MC Select 1
11	MC Select 2
12	MC Select 3
13	Body Transmit Enable
14	Coil ID
15	Coil Present 15
16	Coil Present 2
17	+15VDC (Auxiliary Power)
18	+10VDC (Auxiliary Power)
19	–15VDC (Auxiliary Power)
20	MC Bias Return 1
21	MC Bias Return 2
22	MC Bias Return 3
23	MC Bias Return 4
24	MC Bias Return 5
25	MC Bias Return 6
26	MC Bias Return 7
27	MC Bias Return 8
28	MC Select 0 Return
29	MC Select 1 Return
30	MC Select 2 Return
31	MC Select 3 Return
32	Body Transmit Enable Return
33	Coil ID Return
34	Coil Present Return 1
35	Coil Present Return 2
36	Auxiliary Power Return
37	Auxiliary Power Return



J1	
PINS	SIGNAL
1	MC Bias 1
2	MC Bias 2
3	MC Bias 3
4	MC Bias 4
5	MC Bias 5
6	MC Bias 6
7	MC Bias 7
8	MC Bias 8
9	MC Select 0
10	MC Select 1
11	MC Select 2
12	MC Select 3
13	Body Transmit Enable
14	Coil ID
15	Coil Present 15
16	Coil Present 2
17	+15VDC (Auxiliary Power)
18	+10VDC (Auxiliary Power)
19	-15VDC (Auxiliary Power)
20	MC Bias Return 1
21	MC Bias Return 2
22	MC Bias Return 3
23	MC Bias Return 4
24	MC Bias Return 5
25	MC Bias Return 6
26	MC Bias Return 7
27	MC Bias Return 8
28	MC Select 0 Return
29	MC Select 1 Return
30	MC Select 2 Return
31	MC Select 3 Return
32	Body Transmit Enable Return
33	Coil ID Return
34	Coil Present Return 1
35	Coil Present Return 2
36	Auxiliary Power Return
37	Auxiliary Power Return

J4	
PINS	SIGNAL
1	Preamp Enable
2	Preamp Enable RTN

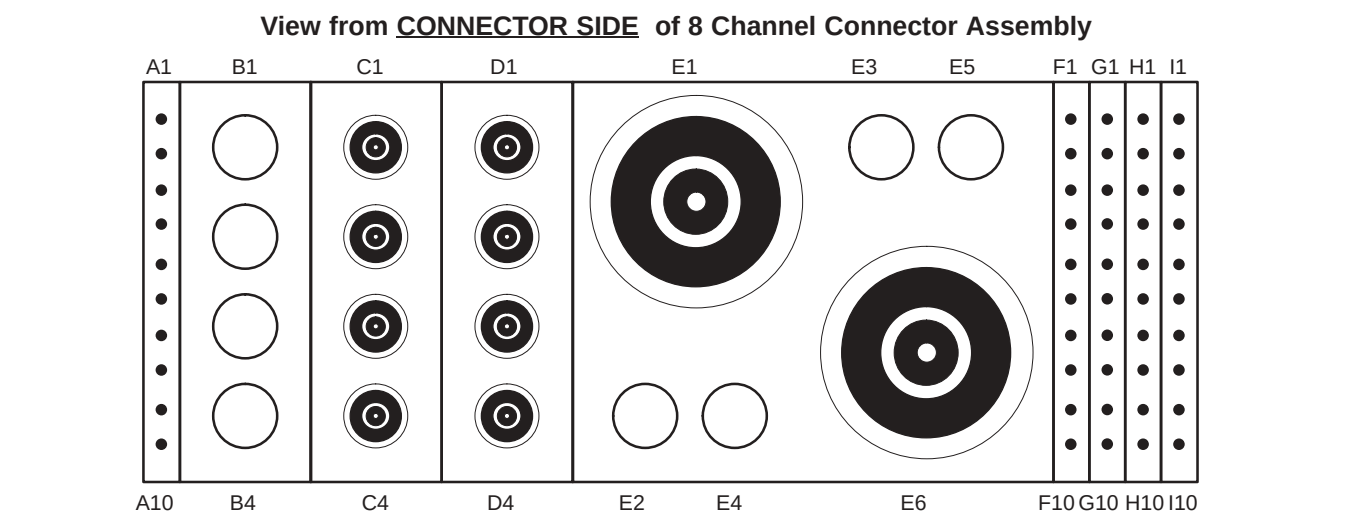


Contact Function Allocation			
Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
B1	n/a		Not used
B2	n/a		Not used
B3	n/a		Not used
B4	n/a		Not used
C1	n/a	MC-1A,B or MC-9B	Preamp out 1, no bias, DC blocked
C2	n/a	MC-2A,B or MC-10B	Preamp out 2, no bias, DC blocked
C3	n/a	MC-3A,B or MC-11B	Preamp out 3, no bias, DC blocked
C4	n/a	MC-4A,B or MC-12B	Preamp out 4, no bias, DC blocked
D1	n/a	MC-5A,B or MC-13B	Preamp out 5, no bias, DC blocked
D2	n/a	MC-6A,B or MC-14B	Preamp out 6, no bias, DC blocked
D3	n/a	MC-7A,B or MC-15B	Preamp out 7, no bias, DC blocked
D4	n/a	MC-8A,B or MC-16B	Preamp out 8, no bias, DC blocked

G3 & 3T94 EXCITE HD (12.x) and HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
MCR II Tool – 8 Channel Connector
(12.x – Forward Production & Upgrades)
(14.x – Upgrades)

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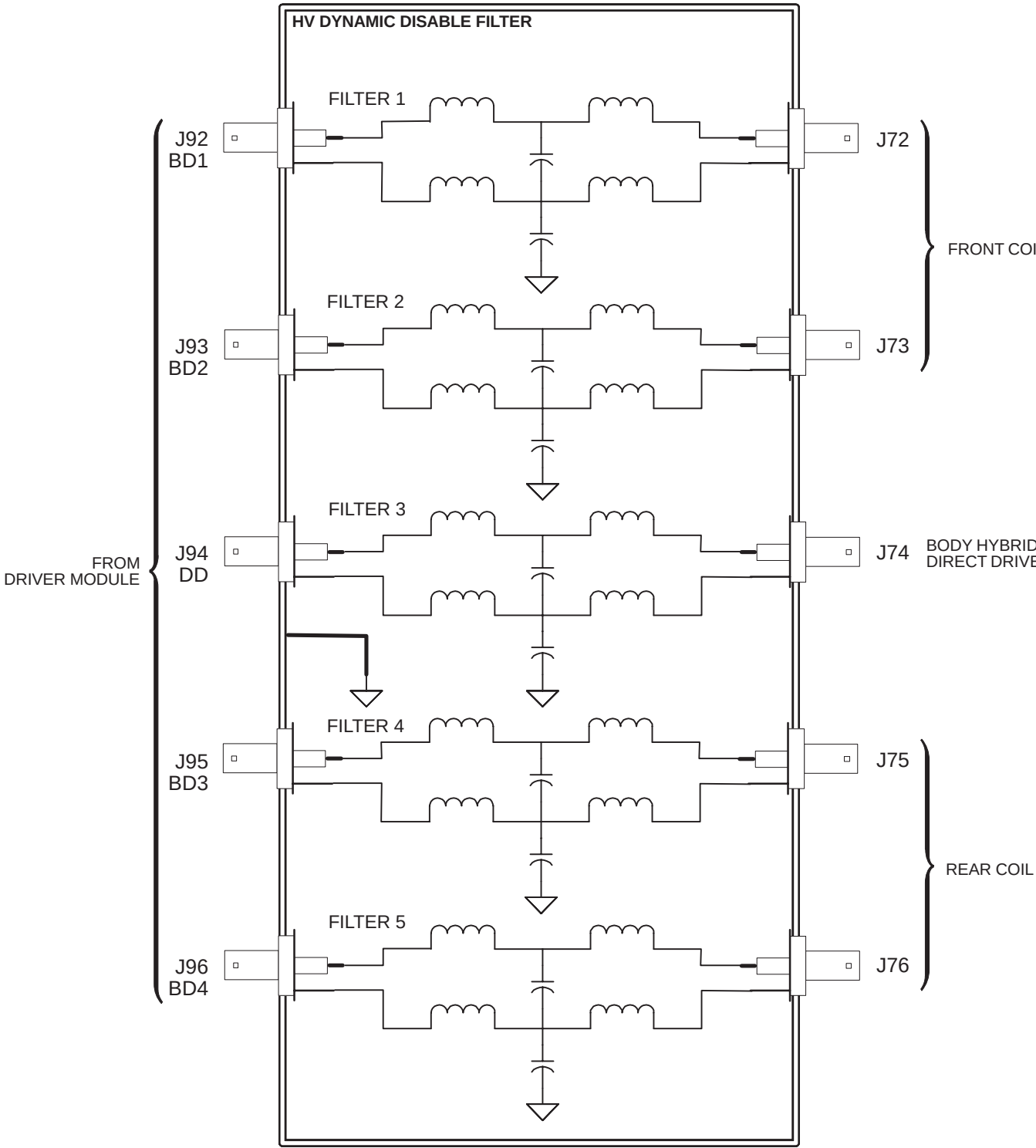
Contact Function Allocation			
Pin	Twisted with	Description	Comments
E1	n/a	Open	Not used
E2	n/a	Open	Not used
E3	n/a	Open	Not used
E4	n/a	Open	Not used
E5	n/a	Open	Not used
E6	n/a	Open	Not used
F1	G1	MC bias 1	MC T/R Bias Line 17 for "A", 25 for "B"
F2	G2	MC bias 2	MC T/R Bias Line 18 for "A", 26 for "B"
F3	G3	MC bias 3	MC T/R Bias Line 19 for "A", 27 for "B"
F4	G4	MC bias 4	MC T/R Bias Line 20 for "A", 28 for "B"
F5	G5	MC bias 5	MC T/R Bias Line 21 for "A", 29 for "B"
F6	G6	MC bias 6	MC T/R Bias Line 22 for "A", 30 for "B"
F7	G7	MC bias 7	MC T/R Bias Line 23 for "A", 31 for "B"
F8	G8	MC bias 8	MC T/R Bias Line 24 for "A", 32 for "B"
F9	F10	body transmit enable	
F10	F9	body transmit enable return	
G1	F1	MC bias return 1	MC T/R Bias Line 17 for "A", 25 for "B"
G2	F2	MC bias return 2	MC T/R Bias Line 18 for "A", 26 for "B"
G3	F3	MC bias return 3	MC T/R Bias Line 19 for "A", 27 for "B"
G4	F4	MC bias return 4	MC T/R Bias Line 20 for "A", 28 for "B"
G5	F5	MC bias return 5	MC T/R Bias Line 21 for "A", 29 for "B"
G6	F6	MC bias return 6	MC T/R Bias Line 22 for "A", 30 for "B"
G7	F7	MC bias return 7	MC T/R Bias Line 23 for "A", 31 for "B"
G8	F8	MC bias return 8	MC T/R Bias Line 24 for "A", 32 for "B"
G9	n/a	Reserved	
G10	n/a	Reserved	
H1	n/a	Reserved	
H2	n/a	Reserved	
H3	n/a	Reserved	
H4	n/a	Reserved	
H5	n/a	Reserved	
H6	n/a	Reserved	
H7	n/a	Reserved	
H8	n/a	Reserved	
H9	n/a	Reserved	
H10	n/a	Reserved	
I1	n/a	Reserved	
I2	I5	+15Vdc (auxiliary & preamp power)	
I3	I6	+10Vdc (auxiliary & preamp power)	
I4	n/a	-15Vdc (auxiliary power)	
I5	I2	Return for aux power	
I6	I3	Return for aux power	
I7	I8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
I8	I7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
I9	I10	coil present'	
I10	I9	coil present return'	



Contact Function Allocation			
Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
B1	n/a		Not used
B2	n/a		Not used
B3	n/a		Not used
B4	n/a		Not used
C1	n/a	MC-1A,B or MC-9B	Preamp out 1, no bias, DC blocked
C2	n/a	MC-2A,B or MC-10B	Preamp out 2, no bias, DC blocked
C3	n/a	MC-3A,B or MC-11B	Preamp out 3, no bias, DC blocked
C4	n/a	MC-4A,B or MC-12B	Preamp out 4, no bias, DC blocked
D1	n/a	MC-5A,B or MC-13B	Preamp out 5, no bias, DC blocked
D2	n/a	MC-6A,B or MC-14B	Preamp out 6, no bias, DC blocked
D3	n/a	MC-7A,B or MC-15B	Preamp out 7, no bias, DC blocked
D4	n/a	MC-8A,B or MC-16B	Preamp out 8, no bias, DC blocked

G3 HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
MCR III Tool – 8 Channel connector
(14.x – Forward Production)
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Contact Function Allocation			
Pin	Twisted with	Description	Comments
E1	n/a	Open	Not used
E2	n/a	Open	Not used
E3	n/a	Open	Not used
E4	n/a	Open	Not used
E5	n/a	Open	Not used
E6	n/a	Open	Not used
F1	G1	MC bias 1	MC T/R Bias Line 17 for “A”, 25 for “B”
F2	G2	MC bias 2	MC T/R Bias Line 18 for “A”, 26 for “B”
F3	G3	MC bias 3	MC T/R Bias Line 19 for “A”, 27 for “B”
F4	G4	MC bias 4	MC T/R Bias Line 20 for “A”, 28 for “B”
F5	G5	MC bias 5	MC T/R Bias Line 21 for “A”, 29 for “B”
F6	G6	MC bias 6	MC T/R Bias Line 22 for “A”, 30 for “B”
F7	G7	MC bias 7	MC T/R Bias Line 23 for “A”, 31 for “B”
F8	G8	MC bias 8	MC T/R Bias Line 24 for “A”, 32 for “B”
F9	F10	body transmit enable	
F10	F9	body transmit enable return	
G1	F1	MC bias return 1	MC T/R Bias Line 17 for “A”, 25 for “B”
G2	F2	MC bias return 2	MC T/R Bias Line 18 for “A”, 26 for “B”
G3	F3	MC bias return 3	MC T/R Bias Line 19 for “A”, 27 for “B”
G4	F4	MC bias return 4	MC T/R Bias Line 20 for “A”, 28 for “B”
G5	F5	MC bias return 5	MC T/R Bias Line 21 for “A”, 29 for “B”
G6	F6	MC bias return 6	MC T/R Bias Line 22 for “A”, 30 for “B”
G7	F7	MC bias return 7	MC T/R Bias Line 23 for “A”, 31 for “B”
G8	F8	MC bias return 8	MC T/R Bias Line 24 for “A”, 32 for “B”
G9	G10	Preamp Enable	
G10	G9	Preamp Enable Return	
H1	n/a	Reserved	
H2	n/a	Reserved	
H3	n/a	Reserved	
H4	n/a	Reserved	
H5	n/a	Reserved	
H6	n/a	Reserved	
H7	n/a	Reserved	
H8	n/a	Reserved	
H9	n/a	Reserved	
H10	n/a	Reserved	
I1	n/a	Reserved	
I2	I5	+15Vdc (auxiliary & preamp power)	
I3	I6	+10Vdc (auxiliary & preamp power)	
I4	n/a	-15Vdc (auxiliary power)	
I5	I2	Return for aux power	
I6	I3	Return for aux power	
I7	I8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
I8	I7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
I9	I10	coil present'	
I10	I9	coil present return'	



NOTE: Connector shields are electrically isolated from enclosure housing.

COOLING SYSTEM

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10 OUTDOOR MRCC	
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Note:

For all Chillers and Cooling System variations (SEE Vendor Manuals on MR Service Methods CDROM)

